

A Statistical Framework for Analysis of Trial-Level Temporal Dynamics in Fiber Photometry Experiments

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Abstract

Fiber photometry has become a popular technique to measure neural activity in vivo, but common analysis strategies can reduce detection of effects because they condense *within-trial* signals into summary measures, and discard trial-level information by averaging *across-trials*. We propose a novel photometry statistical framework based on functional linear mixed modeling, which enables hypothesis testing of variable effects at *every trial time-point*, and uses trial-level signals without averaging. This makes it possible to compare the timing and magnitude of signals across conditions while accounting for between-animal differences. Our framework produces a series of plots that illustrate covariate effect estimates and statistical significance at each trial time-point. By exploiting signal autocorrelation, our methodology yields *joint* 95% confidence intervals that account for inspecting effects across the entire trial and improve the detection of event-related signal changes over common multiple comparisons correction strategies. We reanalyze data from a recent study proposing a theory for the role of mesolimbic dopamine in reward learning, and show the capability of our framework to reveal significant effects obscured by standard analysis approaches. Our method identifies two dopamine components with distinct temporal dynamics that may be hard to explain under currently competing learning theories. In simulation experiments, our methodology yields improved statistical power over common analysis approaches. Finally, we provide an open-source package implementing our framework.

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This **important** work presents a new methodology for the statistical analysis of fiber photometry data, improving statistical power while avoiding the bias inherent in the choices that are necessarily made when summarizing photometry data. The reanalysis of two recent photometry data sets, the simulations, and the mathematical detail provide **convincing** evidence for the utility of the method and the main conclusions, however, the discussion of the re-analyzed data is **incomplete** and would be improved by a deeper consideration of the limitations of the original data. In addition, consideration of other data sets and photometry methodologies including non-linear analysis tools, as well as a discussion of the importance of the data normalization are needed.

1 Introduction

Fiber photometry is a photonic technique used to measure neural activity *in vivo*. The assay quantifies bulk fluorescence emitted from fluorescent biosensors that detect neurotransmitters or physiological processes (e.g., calcium influx) with high neurochemical and cell-type specificity (Wang et al., 2021 [↗](#); Cui et al., 2013 [↗](#); Gunaydin et al., 2014 [↗](#)). The popularity of photometry has increased nearly exponentially since its development (Cui et al., 2013 [↗](#); Gunaydin et al., 2014 [↗](#)), with roughly 1,500 references to it in the last year alone.¹ [↗](#) Although photometry is an invaluable tool, there is little consensus on analysis strategies for the data produced. Many common analysis procedures were not designed for photometry, specifically, but rather grew organically out of adapting approaches historically applied in the cyclic voltammetry (Phillips et al., 2003 [↗](#); Heien et al., 2005 [↗](#)), EEG (Adrian and Matthews, 1934 [↗](#)), and electrophysiology communities (Fatt and Katz, 1952 [↗](#)). Arguably the most common photometry analysis strategy proceeds by: 1) averaging event-aligned signals across trials (“trial-averaging”) and animals for comparison of different conditions (e.g., treatment/control), 2) graphing each condition’s average signal (“trace”), 3) calculating a signal summary measure (e.g., Area-Under-the-Curve (AUC)), and 4) conducting hypothesis tests (e.g., ANOVA) on that summary statistic. Although these analysis conventions are parsimonious, they may dilute important patterns in the data related to, for example, individual animal differences in the timing and magnitude of signals, and the evolution of signals across trials. Part of the appeal of photometry is that probes can be implanted chronically, thereby enabling its application in sophisticated multi-session (“longitudinal”) experiments. Such designs yield, however, complex datasets in which associations between the signal and experimental variables can vary across trials (e.g., due to learning) and animals. To illustrate this, we present a typical analysis on real photometry data (Coddington et al., 2023 [↗](#)) in **Figure 1** [↗](#). These measurements were collected on mesolimbic dopamine neurons in well trained, head-fixed animals performing a Pavlovian reward learning task. **Figure 1A** [↗](#) shows that the signals exhibit considerable heterogeneity across animals, suggesting that it can be difficult to identify one summary measure that captures the target effect in all subjects. Even within-animal, traces are highly variable across conditions (**Figure 1B** [↗](#)), trials within-session (**Figure 1C** [↗](#)), and sessions (**Figure 1D** [↗](#)). These figures illustrate how averaging the signal *across trials* can obscure behavior–signal associations, and how summarizing *within trial* signals (e.g., with AUC) can reduce one’s ability to distinguish between trial-level signals that differ in dynamics, but yield similar summary values.

Despite the complexity of the data, there are few analysis methods developed specifically for photometry. *Encoding models* of point-by-point signal–behavior relationships have been used for predicting the signal values from behavioral variables (Markowitz et al., 2023 [↗](#); Willmore et al., 2022 [↗](#); Choi et al., 2020 [↗](#)). When used for inference, however, these approaches only test whether or not there is an overall behavioral effect on the signal in the analyzed time-window. By not testing the association at each time-point, it makes it difficult to determine when the temporal dynamics of associations are meaningful. Moreover, one model is fit per animal and thus data is not pooled across subjects, which can substantially reduce statistical power. In Jean-Richard-dit-Bressel et al. (2020) [↗](#), the authors propose to compare photometry signals through the combination of *permutation testing* and non-parametric (cluster/subject-level) bootstraps to construct confidence intervals (CIs). This is restricted to comparisons between two conditions, however, which precludes testing for continuous variables, multi-level factors, or multivariate analyses. Investigators have analyzed data from techniques like photometry and calcium imaging by fitting *pointwise generalized linear models (GLMs)* (Pinto and Dan, 2015 [↗](#)), or Pearson correlations (Markowitz et al., 2023 [↗](#)) to assess associations between variables and the signal at each trial time-point. However, standard GLMs and Pearson correlations do not yield valid inference when applied to multi-animal repeated-measures datasets. This therefore requires one to analyze a single trial-averaged signal per animal, discarding trial-level information. Moreover,

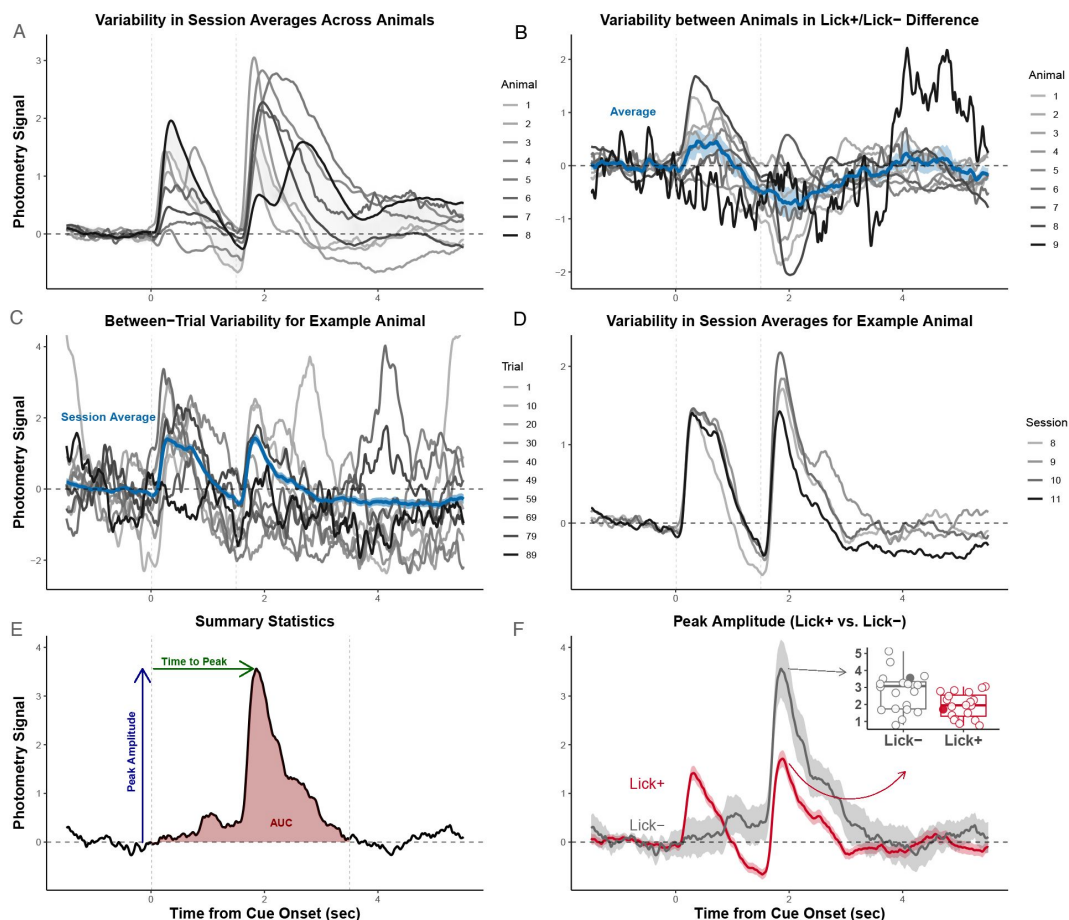


Figure 1.

Variability in photometry signals highlights the need for trial-level analyses.

Signals were recorded from a Pavlovian task in which reward-delivery (sweetened water) followed a stimulus-presentation (0.5 sec auditory cue) after a 1 sec delay. Signals are aligned to cue-onset. **(A)** Signals exhibit heterogeneity across animals. Each trace is a trial-averaged signal on one session for one animal. **(B)** Signals exhibit heterogeneity across animals in the effect of condition. Each trace is from one animal on the same session as in (A). Signals were separately averaged across trials in which animals did (Lick+) or did not (Lick-) engage in anticipatory licking. Each trace represents the pointwise *difference* between average Lick+ and Lick-signals. **(C)** Signals exhibit heterogeneity across trials within animal. Each trace is a randomly selected trial from the same animal on the same session. **(D)** Signals exhibit heterogeneity across sessions. Each trace plotted is the trial-averaged signal for one session for one subject. **(E)** Illustration of common summary measures. Depending on the authors, Area-Under-the-Curve (AUC) can be the area of the shaded region or the average signal amplitude. **(F)** Example hypothesis test of Lick+/Lick-differences using peak amplitude as the summary measure. All signals are measurements of calcium dynamics from axons of mesolimbic dopamine neurons recorded from fibers in the nucleus accumbens (Coddington et al., 2023 [DOI](#)).

the methods do not adjust for multiple comparisons of testing at different time-points, which can inflate Type I error. Lee et al. (2019) [fit pointwise linear mixed models](#) and then apply Benjamini Hochberg correction. However, this method does not yield *joint* CIs and thus does not exploit the correlation between signal values across time-points. The method therefore requires one to adjust for a different test at each sample in a trial's signal. Thus, two analyses of the same data, down-sampled with different sampling rates, could yield significant results in one analysis and not the other, simply because higher sampling rates require one to correct for more comparisons. More generally, this method can be very conservative and dramatically hinder the detection of effects.

In sum, we argue that existing photometry analysis approaches reduce the ability to detect effects. Summary measure analyses coarsen information by 1) condensing the photometry signal into a single statistic (e.g., AUC) that summarizes across time-points *within-trial*, and/or 2) averaging *across-trials* for each animal before conducting hypothesis tests. For methods that estimate associations at each trial time-point, effects can be obscured because current approaches do not exploit the correlation across time-points, do not provide *joint* CIs, and thus yield very conservative inference.

We present an analysis framework that fills this gap and extracts more nuanced information from photometry data by 1) enabling hypothesis testing of the effects of experimental variables on signals at *every trial time-point*, and 2) using the signal from every trial and animal in the analysis. Our proposed approach is based on functional linear mixed models and allows one to compare the temporal evolution (“temporal dynamics”) of the signal between conditions – in timing and magnitude – while accounting for between-animal differences. The statistical procedure uses 1) mixed effects modeling to enable the analysis of sophisticated nested experiments (e.g., designs that include multiple conditions, sessions, and trials), and 2) functional regression to exploit autocorrelation in the signal to calculate *joint* 95% CIs. These *joint* CIs account for examining effects throughout the entire trial, but are not overly conservative CIs. Our framework outputs a plot for each covariate in the model (e.g., behavior, cue-type), which shows whether that covariate is significantly associated with the photometry signal at each time-point. The framework therefore unifies the stages of plotting signals and then conducting hypothesis tests into a joint analysis and visualization procedure.

2 Results

In this section, we introduce our photometry analysis framework based on functional linear mixed models. We focus on explaining the implementation steps and analysis outputs. We then demonstrate how the approach can be used to formulate the scientific questions posed in a recent paper, by re-analyzing their datasets and expanding their results. Finally, we conduct realistic data-driven simulations to show that our approach has desirable statistical properties for photometry analyses.

2.1 Functional Linear Mixed Models (FLMM)

Linear Mixed Models (LMM) are a class of methods for testing the association between covariates (“independent” variables) and outcomes (“dependent” variables) for repeated-measures data (see Methods section 4.1 for a brief introduction to LMM). They can be used to analyze trial-level summary measures (e.g., AUCs) pooled across all trials and animals, preventing the loss of information from trial-averaging. However, this still requires condensing signals into scalar summary measures, which coarsens within-trial information across time-points. In contrast, functional regression methods can be used to model a photometry time series as a “function”, which makes it possible to test the association between covariates of interest and the signal value

at each time-point in the trial (see Methods section 4.2 for a brief introduction to functional regression). However, most functional regression methods require trial averaging of the signals prior to analysis, which discards trial-level information.

The photometry analysis framework we are introducing is based on Functional Linear Mixed Models (FLMM), which combines the benefits of LMM and functional regression to extract the information in the signal both *across*- and *within*- trials (Cui et al., 2021 [DOI](#); Scheipl et al., 2012 [DOI](#); Davidson, 2009 [DOI](#); Morris and Carroll, 2006 [DOI](#)). By modeling covariates that can vary between 1) trial (e.g., cue-type, latency-to-press), 2) session (e.g., training stage), and 3) animal (e.g., treatment group), *FLMM* estimates the effects (termed *functional fixed-effects*) and statistical significance of those covariates for longitudinal designs. By further including *functional random-effects*, the framework can also estimate and adjust for between-animal differences in 1) photometry signal dynamics, and 2) effects of a covariate on the signal, within- and across-trials. In essence, this enables one to model between-animal and between-trial variability in both the “shape” of the signal, and the evolution of covariate effects across trial time-points. The result is a *single plot of the coefficient estimates for each covariate*, which visualizes when (and to what extent) the covariate has a statistically significant (fixed-effect) association with the photometry signal during a trial.

In **Figure 2A** [DOI](#), we illustrate the steps in *FLMM* parameter estimation, which are implemented by our software (see Methods section 4.3 for details). The first input is a matrix $\mathbf{Y}$, where column s , $\mathbf{Y}(s)$, contains the photometry signal value at trial time-point s from every trial in every session and animal, as shown in **A1**. The other inputs are covariate matrices, $\mathbf{X}$ and $\mathbf{Z}$, associated with the fixed-effects and random-effects regression coefficients, respectively. Then, for each time-point, s , we fit a LMM model to $\mathbf{Y}(s)$ as the outcome variable, with the conditional mean of animal i modeled as:

$$\mathbb{E}[\mathbf{Y}_i(s) | \mathbf{X}_i, \mathbf{Z}_i, \gamma_i(s)] = \mathbf{X}_i \beta(s) + \mathbf{Z}_i \gamma_i(s). \quad (1)$$

This yields estimates for fixed-effect regression coefficients ($\hat{\beta}(s)$, common across animals), random-effect coefficients ($\hat{\gamma}_i(s)$, animal-specific), and $\hat{\beta}(s)$ *pointwise* 95% confidence intervals (CIs), as shown in the last row of **Figure 2A1** [DOI](#). We then smooth the $\hat{\beta}_k$ across trial time-points for each covariate k , as illustrated in **Figure 2A2** [DOI](#). The smoothed *pointwise* 95% CI (dark gray) is constructed using a closed form covariance expression when $\mathbf{Y}_i(s) \propto \mathbf{Z}_i \gamma_i(s)$ is modeled as Gaussian, or through a bootstrap-based procedure for other distributions. To account for the multiple comparisons of inspecting coefficients across the entire trial, we construct a *joint* 95% CI (light gray), yielding the plots in **Figure 2A3** [DOI](#). We detail the *joint* 95% CI estimation procedure, the subject of our statistical methodology contribution, in Sections 4.3 and 6. By treating the signal as a “function”, *FLMM* exploits the correlation across trial time-points to construct narrower *joint* 95% CI, thereby enabling one to identify more significant effects.

Figure 2B [DOI](#) illustrates the *FLMM* output of an example analysis. The inset in **Figure 2B** [DOI](#) [bottom left] shows how the effect at a given time-point is estimated and interpreted: when correlating (with an LMM) the signal values (pooled across trials and animals) measured at trial time-point $s = 1.7$ sec with the covariate, Latency, the slope of that line is $\hat{\beta}_1(0.1) = 0.82$. More generally, the interpretation of the *FLMM* plot for covariate k at time-point s , $\hat{\beta}_k(s)$, is the “average change in the photometry signal at trial timepoint s associated with a one unit increase in covariate k , holding other covariates constant.” **Figure 2B** [DOI](#) [bottom right] shows estimated functional random-effects for an example animal and illustrates how these random-effects model individual differences in the Latency–signal association. Each *period* for which the *joint* 95% CI does not contain 0 anywhere denotes that the fixed-effect coefficients are *statistically significantly* different

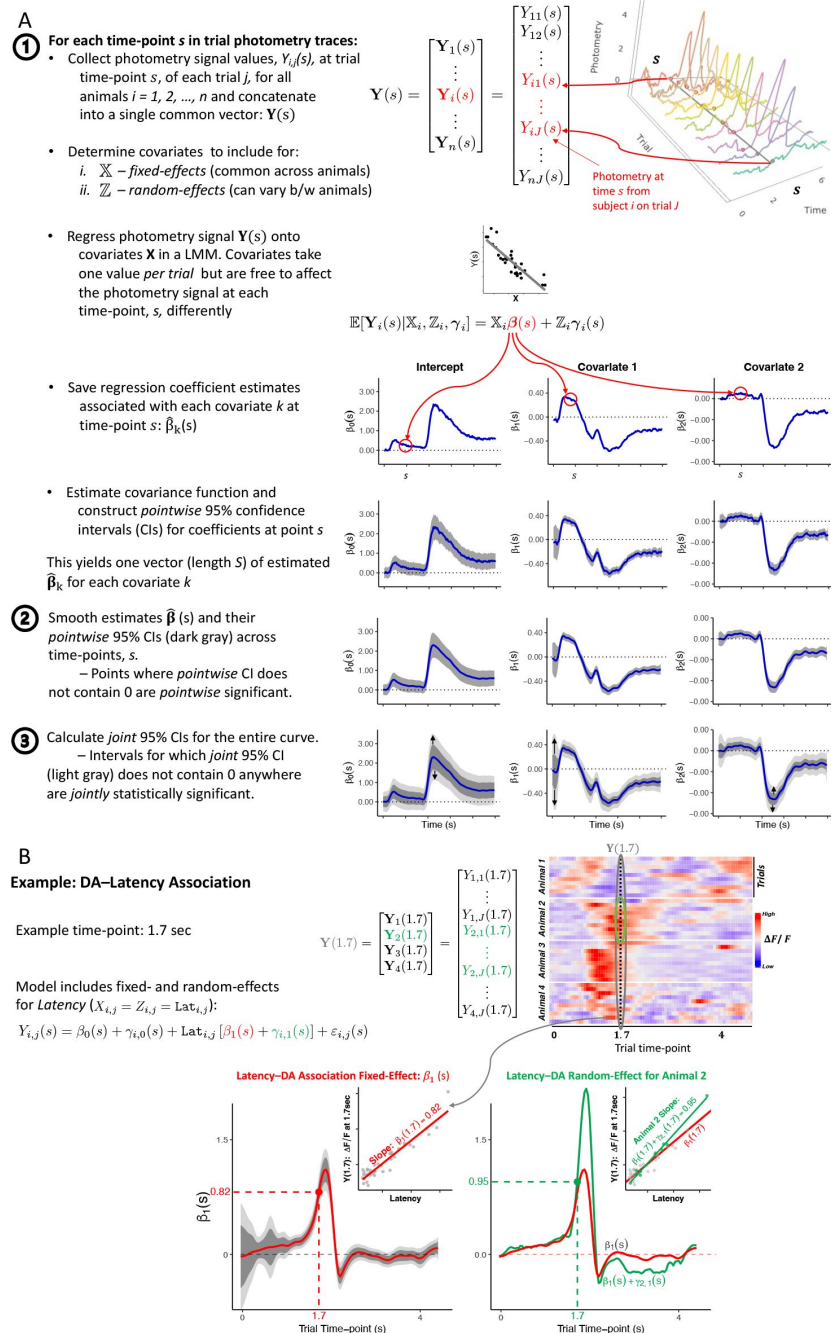


Figure 2.

Functional Linear Mixed Models estimation.

(A) General procedure. **(B)** Example analysis of Latency-signal (Latency-to-lick) association. To illustrate how the plots are constructed, we show the procedure at an example trial time-point ($s = 1.7$ sec), corresponding to values in the heatmap [Top left]. Each point in the FLMM $\beta_1(s)$ coefficient plot [Bottom Left] can be conceptualized as pooling signal values at time s across trials/animals (a slice of the heatmap) and correlating that pooled vector, $\mathbf{Y}(1.7)$, against Latency, via an LMM. [Bottom Right] shows how functional random-effects can be used to model variability in the Latency-DA slope across animals. The inset shows how at $s = 1.7$, the model treats an example animal's slope, $\beta_1(1.7) + \gamma_{2,1}(1.7) = 0.95$, to differ from the shared/common fixed-effect, $\beta_1(1.7) = 0.82$.

from 0 throughout the entire period. The plot conveys: 1) where effects are statistically significant; 2) the estimated effect magnitudes; and 3) *joint* 95% CIs, thereby providing a *complete*, interpretable, and simplified presentation of statistical results.

2.2 A photometry study of the role of mesolimbic dopamine in learning

To demonstrate how our method can be used to answer scientific questions in photometry experiments, we reanalyzed data from a recent article proposing a new model for mesolimbic dopamine's role in learning (Jeong et al., 2022²). We used this study for two reasons. First, the dataset exhibits many common characteristics of photometry data that can dilute effects, or even invalidate results if left unaccounted for. Second, the dataset contains data from multiple experiments, which allows us to illustrate how *FLMM* can be used to test hypotheses across a range of experimental designs. In this section, we discuss how *FLMM* handles those characteristics; in subsequent sections, we show how those hypotheses can be posed in our framework.

One of the most important characteristics of these data is that the dopamine (DA) measurements were collected in within-subject nested longitudinal experiments. For example, in the first behavioral task we discuss, mice were trained across multiple sessions. Each session involved delivery of a sequence of 100 sucrose rewards. The authors analyzed average dopamine (AUC) during the “reward period” time-window (aligned to the first lick after reward-delivery) as a trial (Figure 3A). Thus trials were nested in session, which were nested in animal (Figure 3B). Photometry experiments often exhibit this type of nested longitudinal structure, which can induce correlation patterns in the data within-animal, and obscure effects if not accounted for statistically. This structure can occur in both within-subjects and between-subjects designs, as illustrated in Figure 3D. For example, these data exhibited high within-animal correlations across sessions (Figure 3E), and across trials within-session (Figure 3F).

FLMM can model effect heterogeneity, as well as correlation patterns within- and across trials, through the inclusion of functional random-effects. These allow one to estimate what is common across all animals, and what is unique to each. This is critical because photometry datasets often exhibit between-animal variability in covariate–signal associations. For example, in the task described above, the time between rewards, (IRI: “inter-reward interval”) varied between trials unpredictably (Figure 3C), and the authors reported that “reward period” AUC was correlated with IRI. Figure 3G shows that the magnitude of this correlation varied considerably across animals and sessions. *FLMM* can model this variability in IRI–DA correlation magnitude through the inclusion of animal- and session-specific random effects. More broadly, by varying which covariates and random-effects are included in the model, *FLMM* can analyze data from experimental designs that include within- and between-subject contrasts. For example, one can implement functional versions of methods like correlations, or repeated-measures ANOVA, and, more generally, model a wide range of dependence structures. Finally, *FLMM* can accommodate missing data and subjects with different sample sizes, which are often unavoidable characteristics of photometry experiments. In Figure 3H we show how this dataset included photometry recordings on behavioral sessions at various stages of training that differed across animals. If not accounted for statistically, this can obscure associations between the signal and covariates. For example, average reward period AUC levels were reported to increase across sessions. Thus animals with data collected only on early sessions may appear to have lower AUC levels than other animals, which can increase uncertainty in estimates of covariate effects.

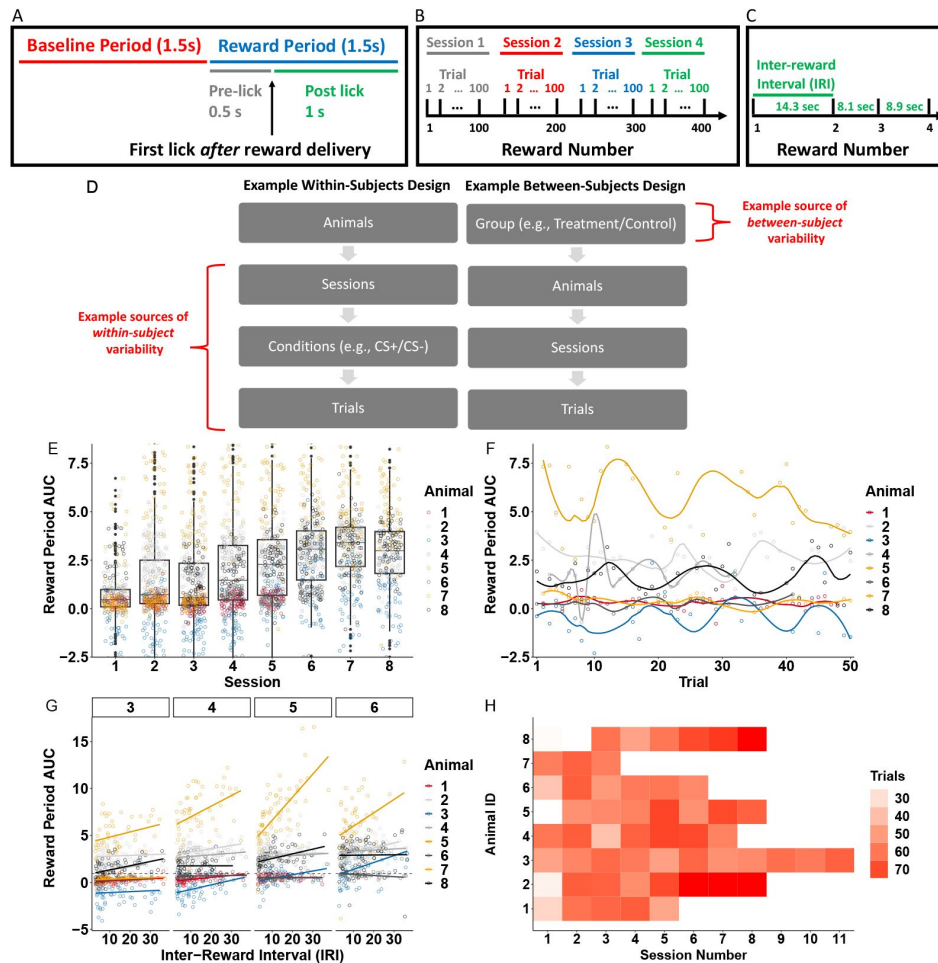


Figure 3.

Nested longitudinal designs in photometry experiments can result in correlation patterns and missing data that dilute effects if not unaccounted for statistically.

Descriptive statistics and figures pertain to data from Jeong et al. (2022), reanalyzed in Section 2.3. **(A)** Experiment trial time-windows used to construct photometry signal summary measures (AUC). The reward was delivered at random times and signals were aligned to the first lick following reward delivery. Reward delivery may occur during the Baseline Period or Reward Period, depending on the lick time. **(B)** The Reward Number is defined as the cumulative number of rewards (interchangeably referred to as “trials”) pooled across sessions. Each session involved delivery of 100 trials. **(C)** The time between two rewards (inter-reward interval or IRI) was a random draw from an exponential distribution (mean 14). **(D)** Examples of experimental designs that exhibit hierarchical nesting structure. Trials/sessions and conditions such as cue-type (e.g., CS+/CS-) contribute to variability within-animal. Between-subject variability can arise from, for example, experimental groups, photometry probe placement, or natural between-animal differences. **(E)** Reward Period AUC values are correlated across sessions. Each dot indicates the average reward period AUC value of one trial. Between-session correlation in AUC values can be seen within-subject since reward period AUC values are similar within-animal on adjacent sessions. Between-session correlation can be seen on average across animals: session boxplot medians are similar on adjacent sessions. **(F)** Temporal correlation within-subject on session 3, chosen because it is the only session common to all animals. Reward period AUC on each trial for any animal is similar on adjacent trials. **(G)** Lines show association (OLS) between IRI and reward period AUC for each animal and session, revealing individual differences in association magnitude. The heterogeneity in line slopes highlights the need for random-effects to account for between-animal and between-session variability. **(H)** Number of sessions and trials per session (that meet inclusion criteria) included varies considerably between animals. For example, one animal’s data was collected on sessions 1-11 while another’s was collected on sessions 1-3.

2.3 Using *FLMM* to test associations between signal and covariates throughout the trial

We first recreate an analysis of the experiment described in the previous section. Jeong et al. (2022) [reported](#), in Section “Tests 1 and 2,” that the trial-level “reward period” AUC was positively correlated with IRI. The authors fit separate Pearson correlations to data from each animal. To illustrate how to test this question in our framework, we show a recreation of their analysis in **Figure 4A-B** [and](#) the *FLMM* analysis estimates of the IRI–DA association in **Figure 4C-D** [and](#). While Jeong et al. (2022) [assessed](#) the IRI–AUC correlation in each animal separately, the *FLMM* can test the IRI–DA association at each trial time-point and in all animals jointly, thereby improving our ability to detect effects. To implement a test most similar to the Pearson correlation, we fit an *FLMM* model with IRI as the (fixed-effect) covariate. Given the between-animal and between-session heterogeneity in signal profiles highlighted in **Figure 3** [and](#), we included nested random-effects to account for the variability across both subjects and sessions. The relationship is significantly positive throughout the time-window [-0.5, 1.5] sec.

These results corroborate the paper’s findings, and provide finer-grained details inaccessible with standard analyses. For example, the temporal dynamics, revealed by *FLMM*, suggest that the neural signal associated with IRI may be composed of two distinct components. The first component rises rapidly starting at around -0.75 sec and decreases quickly after lick-onset (the first lick after reward-delivery). Since the association reaches its peak before lick-onset, the signal may reflect motivation, movement, or reward detection. The second component begins after lick-onset and rises and falls slower than the first component. The timing suggests it may track sucrose consumption. Importantly, the reward period AUC in the paper averages across these putative components. In contrast, *FLMM* is able to partially disentangle them through their distinct temporal dynamics, and offers insight into the role the components play in behavior.

FLMM can also identify results completely obscured by standard methods. We show this by recreating an analysis in which Jeong et al. (2022) [report](#) that reward period AUC was positively correlated with that trial’s Reward Number. **Figure 3B** [illustrates](#) the definition of Reward Number, the cumulative trials/rewards received across sessions. The finding that the Reward Number–AUC correlation was positive was controversial because it conflicted with the prediction of the Reward Prediction Error (“RPE”) model, a prevailing hypothesis for the role of DA in reward learning. The authors argued that RPE predicts a negative Reward Number–DA association, while their model (“ANCCR”) predicts a positive association. Using *FLMM* on the same data, we estimate that the Reward Number–DA association is in fact *negative* within-session. The discrepancy in findings arises because the analysis in Jeong et al. (2022) [pools](#) trials *across* sessions in a manner that does not consider the session number that a reward was delivered on. *FLMM* finds the effect above by accounting for the nested session/trial task structure, and suggests why it occurs by estimating changes at each trial time-point.

The source of the conflicting results above is most clearly illustrated by fitting Reward Number–AUC correlations in each animal and session separately. In **Figure 5A** [and](#) Appendix Figure 10 we show session-specific linear regression fits overlaid on the fits from the session-pooled analysis conducted by Jeong et al. (2022) [and](#). We parameterized session-specific models to yield intercepts (highlighted as large black circles) with the interpretation “the expected AUC value on the first trial of the session.” These intercepts tend to increase across sessions, ultimately resulting in a positive overall Reward Number–AUC correlation in the session-pooled analysis (see Appendix Figure 10). However, the majority of the within-session *slopes* are actually negative, indicating that the AUC decreases across trials *within-session*. The apparent disagreement between the positive Reward Number–AUC correlation estimated in the session-pooled analysis, and the negative Trial Number–AUC association identified in the within-session analysis, is an example of *Simpson’s paradox*. This is caused by pooling data without taking into account its hierarchical structure.

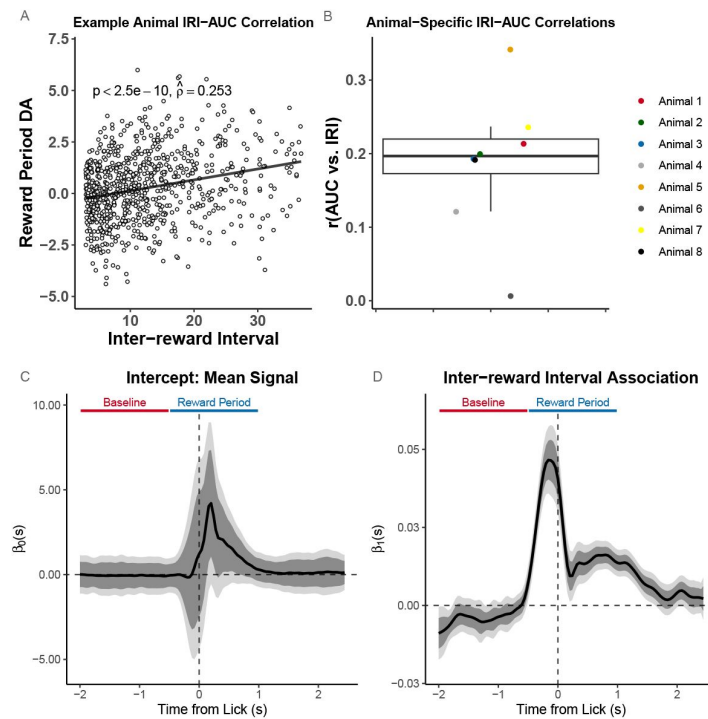


Figure 4.

***FLMM* reveals distinct components obscured by summary measure analyses.**

(A)-(B) show a recreation of statistical analyses conducted in Jeong et al. (2022) on the random inter-trial interval (IRI) reward delivery experiment, and (C)-(D) show our analyses. (A) Analysis of IRI-AUC correlation on all trials in an example animal, as presented in (Jeong et al., 2022). (B) Recreation of boxplot summarizing IRI-AUC correlation coefficients from each animal. (C)-(D) Coefficient estimates from *FLMM* analysis of IRI-DA association: functional intercept estimate (C), and functional IRI slope (D). Although we do not use AUC in this analysis, we indicate the trial periods, “Baseline” and “Reward Period,” that Jeong et al. (2022) used to calculate the AUC. They quantified DA by a measure of normalized AUC of $\Delta F/F$ during a window ranging 0.5 s before to 1 s after the first lick following reward delivery. All plots are aligned to this first lick after reward delivery. The IRI-DA association is statistically significantly positive in the time interval $\sim [-0.5, 1.75]$ sec.

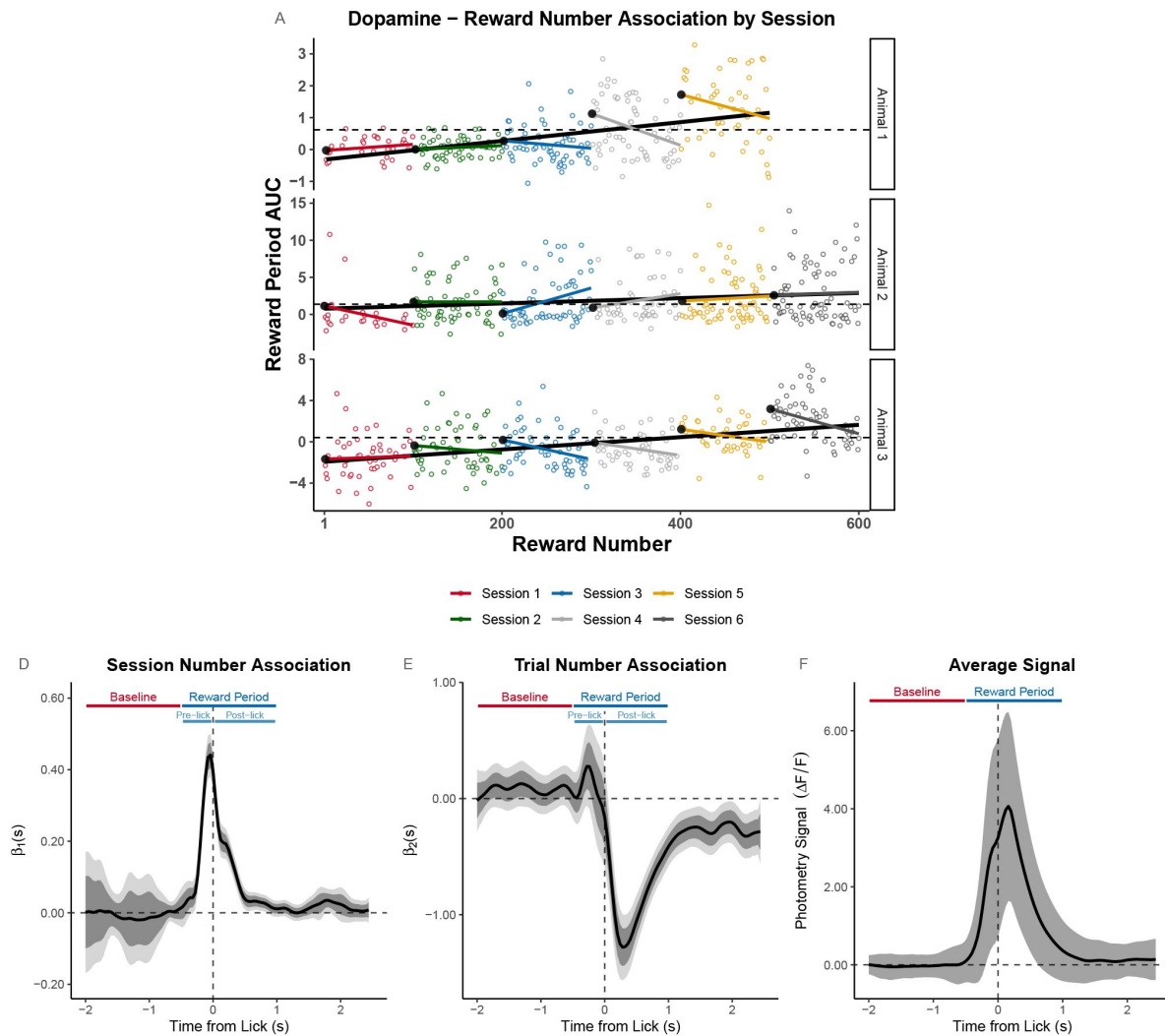


Figure 5.

FLMM identifies within-session signal decreases obscured by standard analyses.

(A) Visualization of the Simpson's paradox: the AUC decreases within-session, but increases across sessions. Plot shows Reward Number–AUC linear regressions fit to data pooled across sessions (black lines), or fit to each session separately (colored lines) in three example animals. Each colored dot is the AUC value for that animal on the corresponding session and trial. The black dots at the left of each color line indicate the intercept value of the session- and animal-specific linear regression model. Intercepts were parameterized to yield the interpretation as the “expected AUC value on the first trial of the session for that animal.” The dotted lines indicate the animal-specific median of the intercepts (across sessions) and are included to visualize that the intercepts increase over sessions. (B)–(C) Coefficient estimates from FLMM analysis of the Reward Number–DA association that models Reward Number with Session Number and Trial Number (linear) effects to capture between-session and within-session effects, respectively. The plots are aligned to the first lick after reward-delivery. The Baseline and Reward Period show the time-windows used to construct AUCs in the summary measure analysis from Jeong et al. (2022) [\[14\]](#). Pre-lick and post-lick time-windows indicate the portions of the Reward Period that occur before and after the lick, respectively. The Session Number effect is *jointly* significantly positive roughly in the interval $[-0.25, 0.5]$ sec, and peaks before lick-onset. This suggests DA increases across sessions during that interval. The Trial Number effect is briefly *pointwise* significantly positive around ~ -0.3 sec and *jointly* significantly negative in the interval $[0, 2.5]$ sec. This suggests DA decreases across trials within-session during the interval $[0, 2.5]$ sec. (D) Average signal pooled across sessions and animals. Shaded region shows standard error of the mean.

To resolve this “paradox,” we fit an *FLMM* that models the nested design by including *both* between-session and within-session fixed-effects, as well as nested random-effects. Estimating the Session Number and Trial Number functional coefficients reveals that *within-session* and *between-session* effects have distinct temporal dynamics. For example, **Figure 5D-E** shows that the *FLMM* estimates that the mean signal decreases significantly *within-session* beginning immediately post-lick, but *increases across-sessions* for a brief interval around the lick. Importantly, the Session Number–DA association peaks and begins falling before the lick, while the Trial Number–DA correlation becomes significant only after the lick. Together these results suggest that DA increases across sessions in response to reward *anticipation*, but decreases within-session during reward *consumption*.

An alternative interpretation of the opposing within- and between-session effects is that DA rises within-session, but appears to decrease due to photobleaching. We assess this in Appendix Section 7.3. We repeat the above analyses on the signal aligned to reward-delivery, and analyze two other behavioral experiments, collected on the same animals. These show that, *within-session*, DA responses increase to reward-predictive cues, but decrease to reward-delivery (predicted and unpredicted). That we can detect both within-session increases and decreases to separate events (cue-onset and reward-delivery) occurring on the same trials, suggests that photobleaching is not hiding within-session DA increases. A parsimonious interpretation of these findings is that there is a DA reduction to unpredicted rewards within-session, independent of photobleaching.

2.4 Using *FLMM* to compare signal “temporal dynamics” across conditions

We next describe an example comparing the signal “temporal dynamics” across conditions, from Section “Tests 3 to 7” of Jeong et al. (2022). In this experiment, each trial consisted of a presentation of a 2 sec cue, followed by reward 3 sec after cue-onset. After many training sessions, the delay was lengthened to 9 sec (**Figure 6A-B**). On the first 9 sec delay session, the authors report that “dopaminergic cue responses [(Cue Period AUC)] showed no significant change,” relative to trials from the last short-delay session. This finding conflicts with what an RPE hypothesis of dopamine coding would anticipate (Kobayashi and Schultz, 2008). Like the authors, we compared the average signals between the last short-delay and the first long-delay sessions. However, we sought to test whether the signal “dynamics” over trial time-points changed after lengthening the delay. To directly compare the difference in signal magnitudes between short- and long-delay sessions, at each time-point, we fit a *FLMM* model with a single binary covariate representing short/long-delays (similar to a functional paired t-test). The *FLMM* estimated mean DA was significantly lower in the latter parts of the initial cue-elicited DA response (~ 1-2.5 sec after cue-onset) on long-compared to short-delay trials (shown in **Figure 6C**). This effect may have been occluded because the AUC analyzed in Jeong et al. (2022) was constructed by averaging signal values in a time window that contains opposing effects: the significant (relative) *reduction* in mean DA ~ 0.5-2.5 sec after cue-onset (marked as interval (2) in **Figure 6H**), identified by *FLMM*, is potentially diluted in the AUC by the (non-significant) *increase* in the cue response in the first ~ 0.5 sec (marked as interval (1) in **Figure 6H**). In our reanalyses we identified other experiments where effects were obscured by the use of AUCs averaging over opposite effects (see Appendix Section 7.2).

This example shows how *FLMM* can detect subtle effects that are difficult to identify by eye, and thus hard to construct appropriate summary measures for. For example, the delay-change effect is significant only during the falling-portion of the cue-elicited transient (0.5-2.5s after cue-onset). This small *relative* reduction during a later portion of the cue-response is overshadowed by the much larger *overall* DA response immediately following (0-0.5 sec) cue-onset. Finally, we show animal-specific functional random-effect estimates in **Figure 6E-F**. These provide intuition about how *FLMM* adjusts for between-animal differences in the “temporal dynamics” of covariates effects.

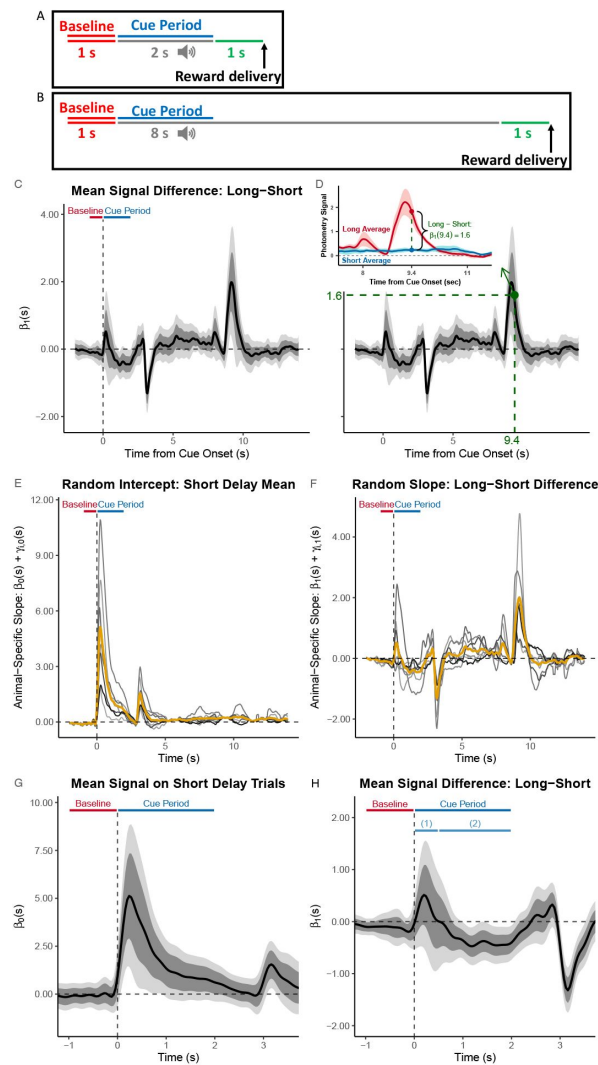


Figure 6.

FLMM identifies significant temporal dynamics effects missed by summary measure analyses.

The analysis of the Delay Length change experiment in Jeong et al. (2022) [used the following summary measure: the average Cue Period AUC – Baseline AUC \(AUCs in the windows \[0,2\] and \[-1,0\] sec, respectively, relative to cue onset\). \(A\)-\(B\)](https://doi.org/10.7554/eLife.95802.1) Behavioral task design and Baseline/Cue Period are illustrated for short-delay (A) and long-delay (B) sessions. (C)-(H) These plots show coefficient estimates from FLMM re-analysis of the experiment. (C) The coefficient value at time-point s on the plot is interpreted as the mean *change* in average DA signal at time-point s between long- and short-delay trials (i.e., positive values indicate a larger signal on long-delay trials), aligned to cue onset. (D) Same Figure as in (C) but the inset shows the interpretation of an example time-point ($s = 9.4$): the difference in magnitude between the average traces (pooled across animals and trials) of long- and short-delay sessions. (E)-(F) Gold lines indicate the fixed-effect estimates and grey lines indicate animal-specific estimates (calculated as the sum of functional fixed-effect and random-effect estimates (Best Linear Unbiased Predictor)) for the random intercept, and random slope, respectively. (G)-(H) Fixed-effect coefficient estimates shown with expanded time axis. In (H), it is clear that long-delay trials exhibit average (relative) increases (sub-interval (1)) and decreases (sub-interval (2)) in signal that would likely cancel out and dilute the effect, if analyzing with a summary measure (AUC) that averages the signal over the entire Cue Period.

2.5 Simulation experiments

We conducted experiments to assess how *FLMM* performs on synthetic data based on the Delay-Length experiment data from Jeong et al. (2022) [introduced in Section 2.4](#). We simulated data in the small sample sizes typical in photometry experiments. We incorporated the key characteristics discussed earlier, namely trial-to-trial and session-to-session correlation, as well as animal-to-animal variability in photometry signal i) magnitudes and ii) levels of association with the covariates. We did this by treating the parameter estimates from a *FLMM* model, fit to the Delay-Length dataset, as the “true” parameter values. This ensured values fell within a realistic range. As the simulated data contained a single binary covariate, this allowed comparison with other statistical approaches discussed in Section 1, namely the permutation method (Jean-Richard-dit Bressel et al., 2020 [introduced in Section 2.4](#)), a paired-samples t-test, and a (non-functional) LMM. The latter two only work with summary measures, so we applied them to the reward period AUC summary measure used in Jeong et al. (2022) [introduced in Section 2.4](#). **Figure 7A** [shows average traces of the real data that the simulations are based on](#). **Figure 7B** [presents simulated trials from one “animal,”](#) and **Figure 7C** [presents simulated session averages from seven “animals.”](#)

The quantitative measures for comparison are *pointwise* and *joint* 95% confidence interval (CI) coverage, and statistical power. **Figure 7D** [shows that the *FLMM* achieves *joint* 95% CI coverage at roughly the nominal level for all sample sizes tested](#). The permutation method, *Perm*, (Jean-Richard-dit Bressel et al., 2020 [introduced in Section 2.4](#)) provides only *pointwise* CIs and thus yields low *joint* coverage. We next analyzed the cue-period to allow comparison between the average performance of *FLMM*, evaluated at time-points in that time-window, with the performance of standard methods applied to the AUC summary measure. *FLMM* achieved roughly the nominal *pointwise* coverage (**Figure 7E** [introduced in Section 2.4](#)), while *Perm* does not, likely because the cluster-bootstrap often performs poorly in small sample settings (Ju, 2013 [introduced in Section 2.4](#)). *FLMM* yields substantially better statistical power than the other methods (**Figure 7F** [introduced in Section 2.4](#)). The LMM and t-test exhibit lower power, likely because effects are diluted by analyzing summary measures. In Appendix 8.2, we present additional simulation results that demonstrate that the performance of summary measure methods is highly sensitive to minor changes in the length of the time-interval that the AUC summarizes. These results show how *FLMM* improves *pointwise* coverage and statistical power compared to standard methods and provides reasonable *joint* coverage.

3 Discussion

We introduced a Functional Linear Mixed Modeling framework for photometry data analysis that: 1) enables hypothesis testing at every trial time-point; 2) accounts for individual differences and trial-to-trial variability; and 3) allows modeling of longitudinal experimental designs. We extend the *FLMM* framework in (Cui et al., 2021 [introduced in Section 2.4](#)) to enable parameter estimation with *pointwise* and *joint* 95% CIs for general random-effect specifications. Previous work provided *FLMM* implementations for either *joint* CIs for simple random-effects models (Cui et al., 2021 [introduced in Section 2.4](#); Sergazinov et al., 2023 [introduced in Section 2.4](#); Crainiceanu et al., 2024 [introduced in Section 2.4](#)), or *pointwise* CIs for nested models (Scheipl et al., 2016 [introduced in Section 2.4](#)). However, they do not allow computation of *joint* 95% CIs in the presence of general random-effects specifications, which is necessary for modeling many characteristics of photometry datasets. When conducting the analyses in this paper, we found that nested random-effects specifications, enabled by our method, substantially improved model fits compared to simpler models (e.g., where the only random-effects were animal-specific random-intercepts).

We demonstrated the capabilities of *FLMM* by reanalyzing datasets from a recently published study Jeong et al. (2022) [introduced in Section 2.4](#). First, we showed that the main scientific questions asked by the authors can easily be answered by specifying covariates of interest, fitting an *FLMM* model, and referring to the coefficient estimate plots for results. Furthermore, we showed that *FLMM* allows direct comparison – and testing – of differences in signal time-courses between conditions,

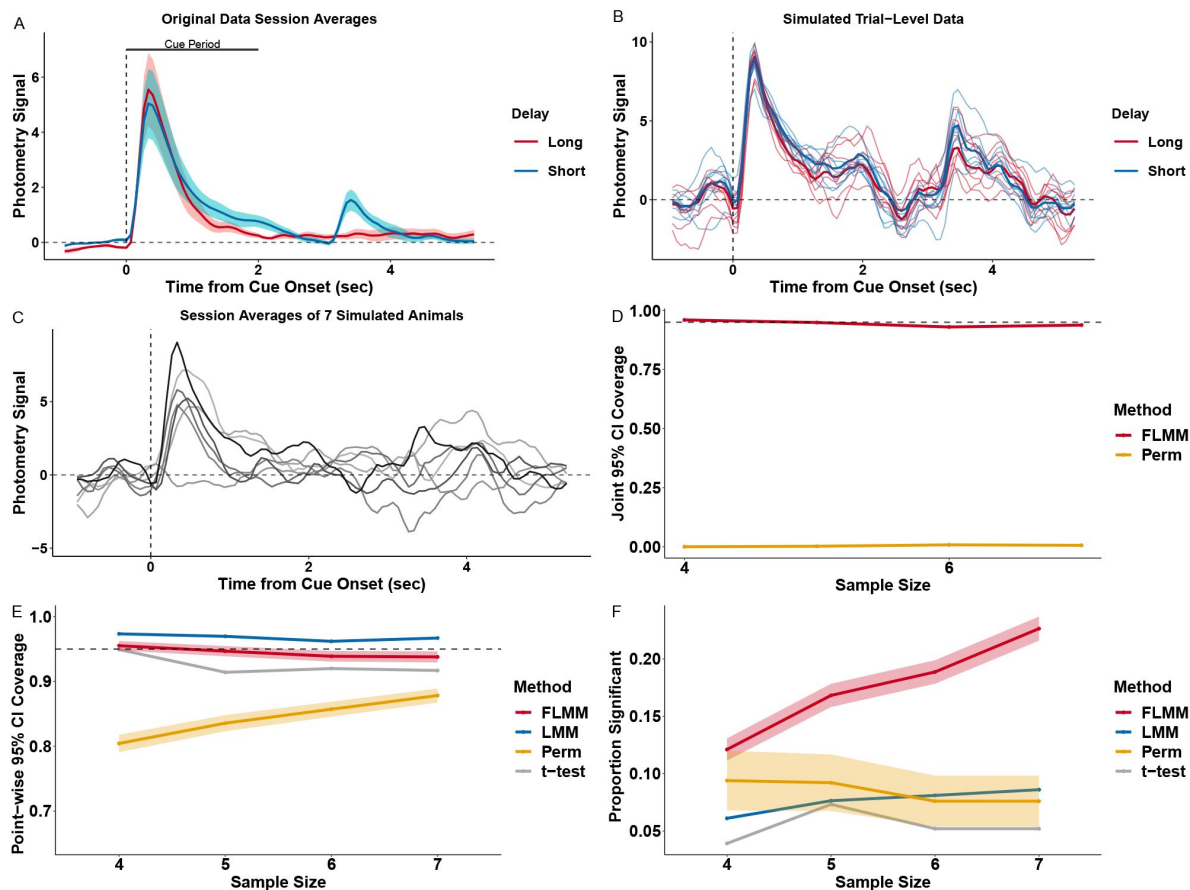


Figure 7.

Realistic simulation experiments show that *FLMM* exhibits desirable statistical properties for photometry analyses.

The simulation produces synthetic photometry data similar to (Jeong et al., 2022), with the same sources of variability across trials and animals. **(A)** Lines show average traces from the original photometry data. The traces are averaged across trials and animals from the last short- and the first long-delay session. Bar shows the cue-period analyzed in the paper and in our experiments. **(B)** Each thin line is the signal from a single simulated trial from the same “animal.”; bold lines show the average trace for each trial type. **(C)** Each line is the trial-averaged trace from one session for seven simulated “animals.” **(D)** *FLMM* exhibits approximately correct *joint* 95% CI coverage. *Perm* does not provide *joint* CIs and thus its *joint* coverage is low, as expected. **(E)** *FLMM* exhibits approximately correct *pointwise* 95% CI coverage. **(F)** *FLMM* improves statistical power during the cue-period compared to standard methods at each sample size tested. Power is calculated for *Perm* based on the full consecutive threshold criteria. For figures (E) and (F) the LMM and t-test were fit on the cue-period AUC and thus each replicate yields one indicator of CI inclusion and statistical significance. We represent the corresponding proportions with a line plot. For other methods, estimates are provided at each time-point. We therefore average performance across the cue period and then summarize variability of these replicate-specific averages with a 95% confidence band.

something that is difficult with standard methods relying on summary measure tests. Finally, we showed that *FLMM* can reveal significant effects that were tested by the authors, but obscured by summary measure analyses.

The specific effects identified with *FLMM* bear on one of the key questions in Jeong et al. (2022) [\[1\]](#), namely which of two competing models – RPE and ANCCR – better describe changes in DA responses to unpredicted rewards over trials. For example, our finding that DA increases *across-sessions* in the random IRI task may be inconsistent with RPE, while the *within-session* decreases conflict with ANCCR predictions. The within-session post-lick DA decreases might be predicted by RPE because, with sufficient training, earlier states within a session will come to predict later states. This will result in two outcomes: 1) DA responses to reward-consumption will decrease on later states, since they are better predicted than earlier states; and 2) by virtue of predicting future reward states, earlier states in the session will acquire further value. Importantly, satiation or habituation may also contribute. They appear not to be the sole cause, though, as the temporal dynamics of within-session reductions vary between-session (Appendix Figure 11), despite equal levels of food-restriction. RPE may also partially explain between-session increases in pre-lick DA. The authors suggested that the reward-delivery mechanism produces stimuli that could act as reward-predictive cues. As learning progresses, DA should increase in response to these cues. However, RPE would then also predict that DA during reward-consumption would decrease across sessions as cue-elicited DA increases, an effect which we do not observe. In personal correspondence, the authors of Jeong et al. (2022) [\[1\]](#) offered the following possible explanations for the within-session DA decreases within the ANCCR framework. Specifically, while the ANCCR simulations described in Jeong et al. (2022) [\[1\]](#) did not consider the session a trial occurred on, there may be session boundary-effects in animal learning. For instance, animals may estimate the rate of reward to be lower at the start of a session due to a degraded memory from prior sessions, or to generalization between the high reward rate within the head-fixed context and the low rate outside it. Then the systematic increase in animals' estimate of reward rate during the session would result in a corresponding decrease in the learning rate of the reward-reward predecessor representation (Burke et al., 2023 [\[2\]](#)). If so, the magnitude of the reward-reward predecessor representation increase at reward-delivery would reduce across sessions, and result in a negative Session Number–DA correlation.

Beyond these findings, the results of other experiments in Jeong et al. (2022) [\[1\]](#) may not support a purely RPE-based explanation. For example, in Appendix Section 7.5, we reanalyze data in which the authors tested whether, across several sessions of Pavlovian learning, DA activity “backpropagates” from reward delivery (3 sec after cue-onset) to the presentation of reward-predictive cues. We argue that if the specific backpropagation effect defined in Jeong et al. (2022) [\[1\]](#) were present, our method would be ideally suited to identify it, because it tests how the signal *timing* evolves both within- and across-sessions. However, our results align with those reported by Jeong et al. (2022) [\[1\]](#). Even in the delay-length change experiment, in which *FLMM* identified significant changes obscured by AUC analyses, it is not clear to us whether proposed versions of RPE would predict the specific change in dynamics observed. More generally, modeling trial-level mesolimbic DA as a vector (e.g., see Wörnberg and Kumar (2023) [\[3\]](#)) may improve theories for its role in reward learning. While adjudicating between ANCCR and RPE is beyond the scope of this paper, our analyses produced a set of findings that would be hard to identify using existing analysis methods. Those findings also suggest future experiments that control trial-by-trial reward rates, and would allow quantitative testing of the explanations proposed. By estimating trial-level temporal dynamics, *FLMM* should make it possible to further refine these theories.

We describe *FLMM* further in the Methods section 4.3 and provide a full derivation of our methodology in Appendix 6 [\[4\]](#). For further evaluation, we reanalyzed a second study (Coddington et al. (2023) [\[5\]](#), see Appendix 9 [\[6\]](#)) that measures calcium dynamics, showing the applicability of the method to different photometry sensors. There, we demonstrate additional capabilities to: 1) model how signal “dynamics” early in training predict behavior later in training; and 2) estimate

covariate interactions across both trial time-points and sessions. Finally, we carried out simulation experiments on realistic synthetic data to verify that, in small sample settings, the *FLMM* framework yields *joint* and *pointwise* CIs that achieve approximately 95% coverage, and improves statistical power versus the alternative approaches introduced. As *FLMM* commits type I errors at roughly nominal levels and outperforms methods that analyze trial-averaged data, we verify *FLMM* is not overly susceptible to trial-level noise or spurious correlations.

Beyond the technical capabilities of *FLMM* method, we believe there are a number of ways in which it can enhance scientific practice. First, it may reduce biases arising from summary statistic analyses. Photometry measures aggregate activity from a collection of neurons, which may contain sub-populations with different functions. By coarsening the temporal resolution, summary measures may obscure the heterogeneity of the target neural population, and bias the study of systems towards sub-populations that exhibit larger signals. Thus through analyzing each time-point, *FLMM* may more effectively characterize neuronal heterogeneity. Second, functional random-effects can be used in place of some photometry pre-processing steps, thereby improving reproducibility. Longitudinal experiments can induce correlation patterns across trials that standard techniques (e.g., repeated-measures ANOVA) may not adequately account for. For these reasons, analysts sometimes normalize data with, for example, trial-specific baselines. This can reduce reproducibility, however, because normalization approaches can differ across labs. Moreover, common strategies that use *one* statistic to standardize *all* trial time-points may drown out effects, if different time-points require different normalization approaches. *FLMM* instead allows one to adjust each time-point with a different value by, for example, including trial- and animal-specific functional random-effects. This makes assumptions about the signal explicit through the model specification, and provides an objective way to select an analysis strategy (e.g., model-fit criteria). Finally, we found that *FLMM* made it easier to identify model misspecification, as estimate plots often provided obvious visual indications (e.g., CI magnitude varying enormously across sub-intervals of the trial). We recommend fitting a few models and selecting one based on fit criteria (e.g., AIC/BIC), without considering statistical significance.

FLMM is applicable in a wide range of experimental settings. We provide an open source package, *fastFMM*³, that is the first, to our knowledge, to provide *joint* CIs for functional mixed models with nested random-effects. For demonstration, we selected a dataset collected on a behavioral paradigm, photometry sensor, and signaling pathway that are well characterized, making it easier to evaluate the framework. Although selecting summary measures and time-windows to quantify is relatively easy here, this may not be the case when collecting data under new conditions. In such cases, *FLMM* can characterize the association between neural activity and covariates without strong assumptions about how to summarize signals. The *FLMM* framework may also be applicable to techniques like electrophysiology and calcium imaging. For example, our package can fit functional generalized LMMs with a count distribution (e.g., Poisson). Additionally, our method can be extended to model time-varying covariates. This would enable one to estimate how the level of association between signals, simultaneously recorded from different brain regions, fluctuates across trial time-points.

To conclude, we believe research in statistical methods research for fiber photometry is needed, given the widespread use of the technique and the variability of analysis procedures across labs. We hope that our proposed *FLMM* framework and software further enable researchers to extract the rich information contained in photometry signals.

4 Methods

The framework we introduced is based on functional linear mixed models (*FLMM*), which is a combination of linear mixed modeling and functional regression. In this section, we introduce both of these prior to describing the *FLMM* approach. We then describe the analyses and modeling

methods we used to reanalyze data from Jeong et al. (2022) (the results of which were presented in Section 2) as well as details of the simulation scheme.

We describe our strategy for analysis of data after application of standard preprocessing steps such as calculation of $\Delta F/F$

4.1 Linear Mixed Models

Linear mixed modeling (LMM, also known as multilevel or hierarchical modeling) is a method for testing the association between covariates (i.e., “predictor” or “independent” variables) and an outcome (“dependent”) variable for repeated-measures data (or multivariate/longitudinal). We provide a brief description of mixed models below, but see Aarts et al. (2014); Yu et al. (2022); Magezi (2015); Barr et al. (2013); Barr (2013); Baayen et al. (2008) for more extensive descriptions of their use in neuroscience and psychology.

LMM enable hypothesis testing in repeated measures designs, and can account for and characterize individual differences through inclusion of “random-effects.” Conceptually, random-effects serve as animal-specific regression coefficients.⁴ This allows the relationship between a covariate of interest (e.g., locomotion) and the outcome (e.g., lever pressing) to vary between-animals in the model, while still pooling data from all animals to estimate parameters. For example, they can be used to model between-animal differences in photometry signal magnitudes on baseline conditions (e.g., on a control condition). This can allow the model to capture how variables of interest (e.g., behavior) are associated with the signal after “adjusting” for the fact that some animals may, on average, exhibit lower signal magnitudes on all conditions. While LMMs may not feel intuitive initially, they actually share connections to many familiar hypothesis testing procedures. Just as ANOVAs, t-tests, and correlations can be cast as special cases of linear models, repeated measures versions of these tests (e.g., paired-sample t-tests, repeated-measures ANOVAs, and MANOVAs) possess similar connections to linear *mixed* models. We recommend section 1.2-1.3 of Crowder (2009) for more explanation of these connections. This reference also includes descriptions of the capacity of LMM (and thus FLMM) to accommodate: 1) unbalanced designs, 2) varying sample sizes across individuals, and 3) more complicated correlation structures than methods like repeated-measures ANOVA.

In order to provide better intuition about LMM in the photometry context, we illustrate the role of random-effects in **Figure 8**. In panel A, we plot data from 5 animals⁵ on multiple trials, to examine the association between the signal and behavior on a test session. Averaging the signal across trials and correlating its peak amplitude with average session behavior results in only five observations, shown in the inset. In **Figure 8B**, we show the alternative approach that correlates a summary of *each trial* (e.g., peak amplitude) with the trial-level behavioral variable. The linear regression is fit to many more observations than in the session average strategy, but the standard regression hypothesis testing procedure may now be invalid, as we violated the independence assumption by pooling repeated observations within animals. Moreover, it ignores that the numbers of samples (trials) differs between animals. We magnify the inset from **Figure 8B** in **Figure 8C** to emphasize how within-subject correlation arises, at least partially, from animal-to-animal variability. Indeed, the linear regressions (Ordinary Least Squares or “OLS”) fit to each animal separately differ from the fit obtained if we first pool all animals’ trial-level data (“All Animals OLS Fit”). This is expected since 1) each animal may have a different “true” regression line (i.e., individual differences), and 2) the estimated lines may differ “by chance” alone due to sampling error. An LMM allows us to pool across animals and trials but still account for individual differences through “random-effects”, thereby striking a *balance* between i) the line fit to the trial-level data pooled across animals, and ii) the animal-level fits. In **Figure 8D** we show the fit estimated from the LMM (determined by $\hat{\beta}^{(LMM)}$), as well as the subject-specific fits that include the random-effects (defined by the *sum*: $\hat{\beta}^{(LMM)} + \hat{\gamma}_i$ for subject *i*). In **Figure 8C** the *All Animals OLS Fit* is “pulled” towards Animal 1’s data because we failed to account for the fact that more

trials were recorded from Animal 1. In **Figure 8D**, the LMM fit accounts for differing sample sizes across animals and is no longer unduly influenced by Animal 1: it is more similar to the majority of the other animals' subject-specific LMM fits than the OLS fit. Moreover, we see that animal 3 in **Figure 8C** has an OLS fit that differed considerably from all other animals. The LMM subject-specific fit is, however, “shrunk” towards the population fit, determined by $\hat{\beta}^{(LMM)}$.

The LMM used to estimate the fits presented in **Figure 8D** is

$$Y_{i,j} = \beta_0^{(LMM)} + \gamma_{0,i} + X_{i,j} \left(\beta_1^{(LMM)} + \gamma_{i,1} \right) + \epsilon_{i,j} \quad (2)$$

where $Y_{i,j}$ and $X_{i,j}$ are the outcome variable (e.g., photometry signal peak amplitude) and covariate, respectively, associated with subject i on trial j . For short hand, we concatenate the “fixed” effects, random-effects and covariates into vectors, $\boldsymbol{\beta}^{(LMM)} = [\beta_0 \ \beta_1]^T$, $\boldsymbol{\gamma}_i = [\gamma_{i,0} \ \gamma_{i,1}]^T$ and $\mathbf{X}_{i,j} = [1 \ X_{i,j}]^T$. $\boldsymbol{\beta}^{(LMM)}$ is the population regression coefficient estimate that is *shared* across all subjects, and $\boldsymbol{\gamma}_i$ is subject i 's random-effect, its deviation from the shared population slope $\boldsymbol{\beta}$. Thus the slope that determines the fit specific to subject i is $\boldsymbol{\beta}^{(LMM)} + \boldsymbol{\gamma}_i$. That is, the LMM estimates: i) what is common among all animals; and ii) what is unique to each animal. This imbues LMM (and its generalizations) with a flexibility that enables one to account for individual differences and to model correlated data arising from a wide range of complex experimental designs. LMM still require, however, one to analyze summary measures (e.g., AUCs) and thus their application in photometry analyses leads to discarding of within-trial signal information across time-points.⁶

4.2 Functional Linear Regression

We now describe functional-on-scalar regression, a functional data analysis method where the outcome variable is a *function*, rather than a scalar value. We treat the photometry time series from one trial as a “function,” which allows us to test the association between any trial-level experimental variables of interest (e.g., cue type, latency-to-press) and the photometry signal value *at each time-point in the trial*. For example, if photometry data for two groups is collected on one trial, one could fit a separate t-test at *every* time-point in the trial. The t-test at time s , then assesses the mean difference between the signal magnitude of group 1 at time s and the signal magnitude of group 2 at time s . Plotting the test statistics vs. time yields a *curve* representing how differences between groups change over time. This provides the intuition behind the functional version of a *t-test*, and is a special case of the *functional linear model*. This functional version of the t-test is in contrast with the common approach of comparing the groups' signals using a single summary measure (e.g. peak amplitude in the trial). In essence, the functional linear model⁷ is a linear regression with a functional coefficient $\boldsymbol{\beta}(s)$ across time-point s :

$$Y_i(s) = \mathbf{X}_i \boldsymbol{\beta}(s) + \epsilon_i(s) \quad (3)$$

where $Y_i(s)$ and $\epsilon_i(s)$ are the photometry value and error terms of animal i at time-point s in the signal. The $p \times 1$ vector, $\boldsymbol{\beta}(s)$, is the regression coefficient at time-point s , applied to p covariates. This functional linear regression framework allows for testing the effects of (scalar) variables at every time-point, adjusting for multiple covariates (continuous or discrete) and their interactions. Functional regression thus exploits *within-trial* signal information, but cannot be directly used to analyze multiple trials per animal since it assumes independence between trials and animals.

4.3 Functional Mixed Models

Our framework is based on Functional Linear Mixed Models (*FLMM*), which is a form of functional-on-scalar regression that combines the benefits of linear mixed effects modeling and functional linear regression. We provided an informal presentation of the approach in the Results section 2.1, with the goal of enabling researchers with a wide range of statistical backgrounds to understand

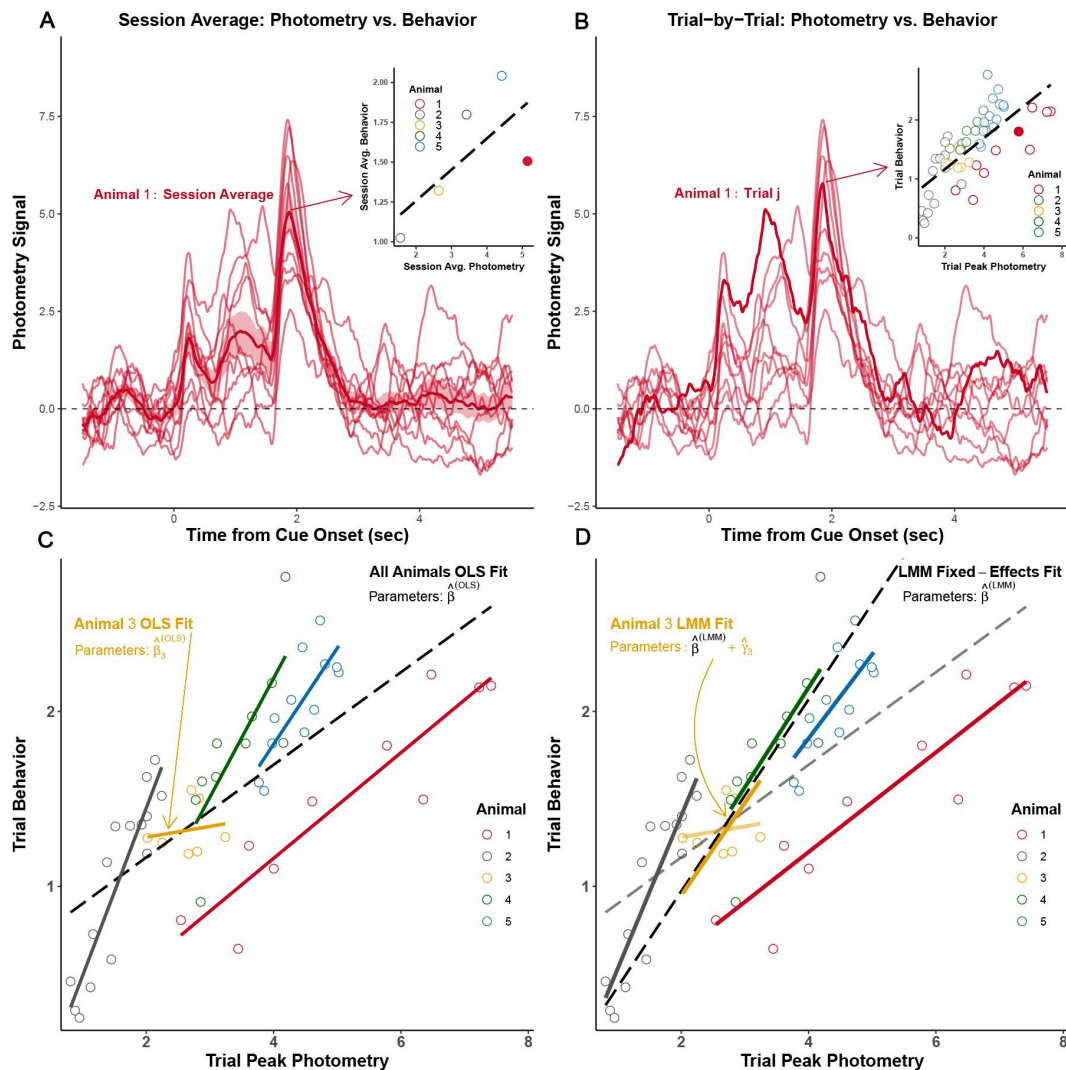


Figure 8.

Example repeated-measures data from a single test session. **(A) Session-average approach:** Photometry signals from all trials for Animal 1 are averaged across trials and summarized (peak amplitude). The session-average summary is then plotted (inset) against the session-average behavior for each of 5 animals. Example animal's session average is the filled circle. **(B) Trial-level approach:** the trial-level signal summary measure (peak amplitude) is pooled across animals and correlated with trial-level behavior. An example signal from one trial for Animal 1 is highlighted in the trace plot. That example trial is represented as the filled circle in the inset. Each dot in the inset is one trial from one animal; dot color indicates the animal ID. **(C)** Inset from (B) is magnified. Linear regressions (OLS: Ordinary Least Squares) fit separately to each animal's trial-level. A global regression fit to the trial-level data pooled across animals is displayed as the dotted black line. **(D)** Linear Mixed Modeling (LMM) strikes a balance between one model common to all animals and fitting many animal-specific models. The "global" fixed-effects fit (from $\hat{\beta}^{(LMM)}$) and the fits including the subject-level random-effect estimates (Best Linear Unbiased Predictor) are displayed. Subject-specific fit for animal i is calculated from: $\hat{\gamma}^{(LMM)} + \hat{\gamma}_i^{(LMM)}$. Note the fixed-effects, $\hat{\beta}^{(LMM)}$, and random-effects, $\hat{\gamma}_i^{(LMM)}$, are estimated in the same model.

its application and results. Here, we provide a high level overview of the methodology, with further technical details provided in **Appendix section 6**. Our *FLMM* approach is based on an estimation procedure that was first proposed in Cui et al. (2021). We selected this method because 1) it allows for calculation of *joint* 95% CIs, 2) it is readily scalable for the dataset sizes and randomeffect specifications needed for neuroscience experiments, and 3) it uses common syntax for model specification from the well-known lme4 package.

The construction of *joint* CIs in the context of functional data analysis is a very important research question; see Cui et al. (2021) and references therein. Each *point* at which the *pointwise* 95% CI does not contain 0 indicates that the coefficient is *statistically significantly* different from 0 at that point. Compared with *pointwise* CIs, *joint* CIs takes into account the autocorrelation of signal values across trial time-points (the “functional domain”). Therefore, instead of interpreting at a specific time-point, *joint* CIs enable *joint* interpretations at multiple locations along the functional domain. This aligns with interpreting covariate effects on the photometry signals across time-intervals (e.g., “cue-period”) as opposed to at a single trial time-point. Previous methodological work has provided functional mixed model implementations for either *joint* 95% CIs for simple random-effects models (Cui et al., 2021), or *pointwise* 95% CIs for nested models (Scheipl et al., 2016), but to our knowledge, do not provide explicit formulas or software for computing *joint* 95% CIs in the presence of general random-effects specifications. We found nested random-effects specifications substantially improved model fits in the neuroscience experiments we reanalyzed, and were needed to properly model the sophisticated behavioral designs (see Appendix Section 6.3 for more details). We therefore derived an extension of the estimator for the covariance of the random-effects (Cui et al., 2021), based on a Method of Moments approach described in Greven et al. (2010) to account for collinearity in the design matrix of the regression used in the Method of Moments estimation scheme.

Mixed effects modeling (and thus *FLMM*) requires one to specify the random-effects in the modeling process. In essence, this amounts to selecting which covariate effects (i.e., regression coefficients) might vary across, for example, animals or sessions (see Appendix Section 6.3 for more details on specification and interpretation of random-effects). While this may seem unfamiliar at first, in practice, most experimental designs (and the appropriate model formulations) fall under a few categories that can be modeled with extensions of familiar methods (e.g., FLMM analogues of repeated-measures ANOVA, correlations etc.). Our method is fully implemented by our package and uses the same syntax as the common mixed effects package lme4, thereby allowing one to easily fit the models and plot the results. Thus the rich literature on linear mixed model selection and model syntax can be directly applied to the FLMM case, as many of the same principles apply (e.g., see reviews of LMM and applications in neuroscience and psychology in Aarts et al. (2014); Yu et al. (2022); Magezi (2015); Barr et al. (2013); Barr (2013); Baayen et al. (2008)). Users may improve their analyses by evaluating a few candidate models and selecting the one with the best model fit criteria. To that effect, AIC, BIC, and cAIC (Säfken et al., 2018) are already provided automatically in our software implementation. We provide the code to fit models with our package below and we include all code used for this paper (e.g., data pre-processing, model-fitting) on the github page: https://github.com/gloewing/photometry_FLMM. Finally, since fiber photometry sums photon counts over many neurons (the activity of which may be modeled as only weakly dependent given model parameters), we appeal to the central limit theorem to motivate the adoption of a Gaussian-likelihood for the conditional distribution of $\mathbf{Y}_i(s) \mid \mathbb{X}_i$. However, when photon counts are low, our package can fit functional generalized LMMs with a count distribution like a Poisson (see Cui et al. (2021) for details).

4.4 Figure 1 Methods and Analyses

Figure 1 was generated from data (Coddington et al., 2023) available at (Dudman, 2023). All signals are measurements of calcium dynamics in mesolimbic dopamine cells (DAT-Cre::ai32 transgenic mice were injected with a Cre-dependent jRCaMP1b virus across the ventral midbrain)

recorded from fibers in the nucleus accumbens. In this experiment, head-fixed mice were exposed to a 0.5s stimulus, followed by reward 1s after cue-offset. Signals are aligned to cue-onset. Our github page includes code to recreate the images.

Figure 1A [↗](#) was generated from trial-averaging Lick+ trials from session 8 for control animals that received no optogenetic stimulation. To ensure the animals were sufficiently trained we selected session 8, because it is the latest session in which all control animals have recorded data available in this dataset. **Figure 1B** [↗](#) was generated from trial-averaging Lick+ and Lick- trials and then plotting the pointwise difference. **Figure 1C** [↗](#) was generated arbitrarily from the first control animal in the dataset (i.e., animal ID 1), which we selected without inspecting other animals' data to avoid biases. The data plotted are a subset of Lick+ trials (a sequence that starts from the first and ends on the last trial and takes every 10th trial) on the final training session for that animal (session 11). We selected this session to ensure the animal was as well trained as possible. **Figure 1D** [↗](#) shows session averages on Lick+ trials from sessions 8-11.

We made **Figures 1E** [↗](#) and **1F** [↗](#) for explanatory purposes, and the specific choices we made in creating these figures are not meant to reflect any specific photometry analysis procedures. We nevertheless include a description of how we generated them for completeness. **Figure 1E** [↗](#) was the trial-averaged trace from session 8 for Lick- trials for animal 1. **Figure 1F** [↗](#) was the trial-averaged trace from session 8 for Lick- and Lick+ trials separately for animal 1. The inset contains an individual point for each animal from session 8. A given point was calculated by trial-averaging Lick+ and Lick- for each animal separately, and then finding the maximum difference.

4.5 Data Reanalysis Methods

Our primary experimental results in Section 2 come from a reanalysis of the data from Jeong et al. (2022) [↗](#). Here we provide a high level description of the methods used, with additional details in Appendix Section 7 [↗](#). We describe the models we implemented and where possible, we explain the authors' hypotheses, data pre-processing, and analysis procedures. We often center all covariates to be mean zero to allow an interpretation of the intercept as the mean signal for a trial when all covariates are at their average values. Finally, the R scripts used to replicate the analyses or derive information from data (e.g., extracting consummatory lick bouts) can be found on the github page: https://github.com/gloewing/photometry_FLMM [↗](#)

4.5.1 Notation

We will denote the mean models of the FLMM as $\mathbb{E}[\mathbf{Y}_{i,j,l}(s) \mid \mathbf{X}_{i,j,l}, \mathbf{Z}_{i,j,l}]$ of animal i , on trial j of session l . We use the general notation, $\mathbf{X}_{i,j,l}$ for the covariates for the fixed-effects and $\mathbf{Z}_{i,j,l}$ for the covariates for the random-effects, because the full form of the covariates that fully specify the columns of the associated design matrices can be quite complicated (e.g., when we include animal-specific random-effects for each session). On the right hand side of the mean model, we specify the covariates by name when possible (e.g., $\text{IRI}_{i,j,l}$ as the inter-reward interval for subject i on trial j on session l), with the understanding that the vectors $\mathbf{X}_{i,j,l}$ and $\mathbf{Z}_{i,j,l}$ contain these covariates where applicable (e.g., this covariate may be included in $\mathbf{X}_{i,j,l}$, $\mathbf{Z}_{i,j,l}$ or both). As mentioned in the main text, the entry in these vectors can change from trial-to-trial, but unlike the outcome, the entries do not change within-trial and for that reason we do *not* include the notation (s): $\mathbf{X}_{i,j,l}(s)$. Finally, in some models the covariates change from session-to-session but not across trials *within* a session. We indicate how the variables change (e.g., trial-to-trial vs. session-to-session but not trial-to-trial) through outcome, covariate, and random-effect subscripts (e.g., $\mathbf{Y}_{i,j,l}(s)$, $\text{Delay}_{i,l}$, $\gamma_{i,l}(s)$).

4.5.2 Reanalysis Methods: Using *FLMM* to test associations between signal and covariates throughout the trial

We analyzed the same set of sessions that were analyzed by the authors (Jeong et al., 2022) in the section “Tests 1 and 2 (unpredicted rewards).” As noted previously, the set of sessions analyzed by the authors differed across animals, and the number of trials per session differed between sessions. The sessions analyzed and characteristics of these data were presented in **Figure 3**. All analyses are on data collected with the photometry dopamine dLight1.3b sensor (AAVDJ-CAG-dLight1.3b virus). The virus was injected and the optical fiber was implanted in the nucleus accumbens core. The authors quantified dopamine (DA), measured with normalized AUC of $\Delta F/F$ during a window 0.5 s before to 1 s after the first lick that occurred after reward-delivery. In “Data Analysis: Experiment 1” of the Supplement (p.3), the authors describe that “The [Area Under the Curve (AUC)] during 1.5 s time window before reward period was subtracted from AUC during reward period to normalize baseline activity. To test dynamics of dopamine response to reward, Pearson’s correlation was calculated between dopamine response and Reward Number, or dopamine response and inter reward interval (IRI) from the previous reward.” In all of our analyses in this section, we tested *linear effects* of all covariates of interest, given that the authors conducted Pearson correlations in their tests and we wanted to implement the most similar model in our framework (e.g., see Fig. S8 in (Jeong et al., 2022)). We followed the methods for excluding trials based on behavioral criteria, and for extracting consummatory lick bouts described in “Data Analysis: Experiment 1” of the Supplement (p.3-4) of the original manuscript. We describe this in more detail in Appendix Section 7.2.

IRI Model We tested the association between the DA response to sucrose reward and IRI ($IRI_{i,j,l}$), as shown in **Figure 4**. We selected a final model after carefully comparing models based on AIC and BIC that included, for example, subject- and session-specific random intercepts, random slopes for Trial Number, random slopes for Lick Latency (the time between reward-delivery and first lick), and random slopes for IRI. We considered this collection of models to account for trial-to-trial variability in learning (e.g., Lick Latency grows faster with learning), within-session effects such as satiation (e.g., Trial Number), between-session heterogeneity in IRI slopes, and base-line signal dynamics (e.g., the functional random-intercept). The best model included subject-specific (indexed by i) and session-specific (indexed by l) random functional intercepts and random functional slopes for Lick Latency ($Lick_{i,j,l}$):

$$\mathbb{E}[\mathbf{Y}_{i,j,l}(s) \mid \mathbf{X}_{i,j,l}, \mathbf{Z}_{i,j,l}] = \beta_0(s) + \beta_1(s)IRI_{i,j,l} + \gamma_{0,i}(s) + \gamma_{1,i,l}(s) + Lick_{i,j,l} [\gamma_{2,i}(s) + \gamma_{3,i,l}(s)].$$

As an example, we show the code from our package to fit the above model (the model specification syntax is equivalent to that used by the R package lme4):

```
model_fit = fui(photometry ~ IRI + (lick_latency | id/session), data = photometry_data, subj_ID = "id")
```

The above model assumes the random-effects $\gamma_{i,l}(s) \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \Sigma_{\gamma}(s))$, where $\mathbf{y}_{i,l}(s) = [\gamma_{0,i}(s), \gamma_{1,i,l}(s), \gamma_{2,i}(s), \gamma_{3,i,l}(s)]^T$

Reward Number Model Similarly we tested the association between the DA response to sucrose reward and Reward Number by modeling it with Session Number ($SN_{i,j,l}$), and Trial Number (indexed by j but written out as $TN_{i,j,l}$ when used as a covariate), as shown in **Figure 5**. We

selected a final model after carefully comparing models based on AIC and BIC. The best model included subject-specific (indexed by i) and session-specific (indexed by l) random functional intercepts and random functional slopes for Lick Latency ($\text{Lick}_{i,j,l}$):

$$\mathbb{E}[\mathbf{Y}_{i,j,l}(s) \mid \mathbf{X}_{i,j,l}, \mathbf{Z}_{i,j,l}] = \beta_0(s) + \beta_1(s)\text{SN}_{i,l} + \beta_2(s)\text{TN}_{i,j,l} + \gamma_{0,i}(s) + \gamma_{1,i,l}(s) + \text{Lick}_{i,j,l} [\gamma_{2,i}(s) + \gamma_{3,i,l}(s)].$$

We fit the above model with the code:

```
model_fit = fui(photometry ~ trial + session + (lick_latency | id/session), data = photometry_data,
subj_ID = "id")
```

4.5.3 Reanalysis Methods: Using *FLMM* to compare signal “temporal dynamics” across conditions

We analyzed the data described in section “Experiment 3: Tests 4-5” (Jeong et al., 2022 [\[1\]](#)). The authors noted that “[W]hen a learned delay between cue onset and reward (3 seconds) is extended permanently to a new, longer delay (9 seconds), [Reward Prediction Error] predicts that as animals learn the longer delay...there will be a concomitant reduction in the dopamine cue response due to temporal discounting (46). By contrast, [their model] predicts little to no change in the dopamine cue response as the structure of the task is largely unchanged...Experimentally, [they] observed that although animals learned the new delay rapidly, dopaminergic cue responses showed no significant change” (Jeong et al., 2022 [\[1\]](#)). Their analysis of DA cue-response was based on a baseline subtracted AUC (AUC [0,2s] [-1,0s] AUC relative to cue onset). From the supplement (p. 4): “Experiments 2-5: [T]o analyze learning-dependent dynamics of dopamine response, AUC of $\Delta F/F$ during 2 s from CS+ onset was normalized by AUC during baseline period.” See Appendix 7.4 for more details.

We compared a number of models and selected the one that exhibited the best model fit criteria (AIC and BIC). We fit a random slope model for the Long Delay (9s) indicator ($\text{Delay}_{i,l}$), where $\text{Delay}_{i,l} = 1$ on Long Delay trials and $\text{Delay}_{i,l} = 0$ on Short Delay trials:

$$\mathbb{E}[\mathbf{Y}_{i,j,l}(s) \mid \mathbf{X}_{i,l}, \mathbf{Z}_{i,l}] = \beta_0(s) + \gamma_{0,i}(s) + \text{Delay}_{i,l} [\beta_1(s) + \gamma_{1,i}(s)].$$

Note that because the experimental design included the delay length switches *between* sessions, the covariate $\text{Delay}_{i,l}$ does not include a trial index j (i.e., all trials on session l have the same delay length).

As an example, we show the code from our package to fit the above model:

```
model_fit = fui(photometry ~ delay + (delay | id), data = photometry_data)
```

The above model assumes the random-effects $\gamma_i(s) \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \Sigma_\gamma(s))$, where $\mathbf{y}_i(s) = [\gamma_{0,i}(s) \gamma_{1,i}(s)]^T$

4.6 Functional Mixed Models R Package

We provide the fastFMM package that implements our proposed methods. This can be downloaded on the github page: <https://github.com/gloewing/fastFMM> [\[1\]](#). We refer analysts to our documentation for directions, and recommend lme4’s thorough package resources, since our package is based on the lme4 software and model syntax (e.g., see **Table 2** [\[2\]](#) in (Bates et al., 2014 [\[3\]](#)), **Table 1** [\[4\]](#) in (Barr et al., 2013 [\[5\]](#)) and (Bates, 2010 [\[6\]](#))). Appendix 6.4 contains more information about functional data analysis modeling approaches.

	Longitudinal	Cross-Sectional
Functional	Functional Mixed Models	Functional Regression
Scalar	Generalized Linear Mixed-Effects (GLMM)	Generalized Linear Models (GLM)

Table 1

Regression model classes based on functional vs. scalar response variables and for longitudinal (repeated-measures) vs. cross-sectional data. We use “functional mixed models” as a short-hand for function-on-scalar mixed models. We use “functional regression” as a short-hand for single level function-on-scalar regression.

	Functional Outcome	Scalar Outcome
Functional Predictors	Function-on-Function Regression (FoFR)	Scalar-on-Function Regression (SoFR)
Scalar Predictors	Function-on-Scalar Regression (FoSR)	Generalized Linear Models (GLM)

Table 2

Cross-sectional regression model classes based on functional vs. scalar predictor variables (i.e., covariates) and functional vs. scalar outcome variables. We take the FoFR, FoSR and SoFR to be the single level (non-longitudinal) versions of these methods.

4.7 Simulation Experiments

For the simulation experiments reported in Section 2.4, we fit a random slope model to the Delay Length data described in 2.4 from (Jeong et al., 2022 [DOI](#)). We selected the model that achieved the best AIC on the data, out of a collection of candidate models. We used all trials on the last session of short-delay (2s) and all trials on the first session of the long-delay (8s). We simulated data from the functional linear mixed effects model:

$$Y_{i,k}(s) = \beta_0(s) + \gamma_{0,i}(s) + \text{Delay}_{i,k} [\beta_1(s) + \gamma_{1,i}(s)] + \epsilon_{i,k}(s)$$

where $Y_{i,k}(s)$ is the observed photometry signal of subject i at time s of trial k , $\beta_0(s)$ is the (functional) intercept at time s on all trials, $\gamma_{0,i}$ is the random (functional) intercept of subject i at time s on all trials. $\text{Delay}_{i,k} = 1$ on long-delay trials (8s) and $\text{Delay}_{i,k} = 0$ on short-delay trials (2s). $\beta_1(s)$ is the (functional) coefficient representing the average change in the mean at trial time-point s on long-delay trials (compared to short-delay trials). $\epsilon_{i,k}(s)$ is the error term and $\gamma(s)_{i,j,k} \propto \epsilon_{i,k}(s)$ for $i = 1, 2, n, j = 1, 2, n_j, k = 1, 2, q$ and $s = 1, 2, T$

For each simulation replicate, we drew $\gamma_{j,i} \sim \mathcal{N}_T(0, \hat{\Sigma}_j^\gamma)$ where $\hat{\Sigma}_j^\gamma$ was the $T \times T$ estimated variance covariance matrix of random-effect j across time. We set $\beta = \hat{\beta}$, where $\hat{\beta}$ was the estimated coefficient vector from the photometry dataset. Finally, we drew $\epsilon_{i,k} \sim \mathcal{N}_T(0, \Sigma_i^\epsilon)$ where $\epsilon_{i,k}$ represents the $T \times 1$ vector of errors across the trial. The errors were drawn independently across different trials and subjects. To induce additional correlation across time-points within a trial, we set $\Sigma_i^\epsilon = 5 * \hat{\Sigma}_i^\epsilon$, where $\hat{\Sigma}_i^\epsilon$ was the estimated covariance matrix of the model residuals fit to the observed photometry data for subject i . The index i was drawn uniformly from the observed animal indices $\{1, 7\}$ without replacement when the simulated sample size $n \leq 7$. Otherwise, we set the first 7 to the indices $\{1, 7\}$ and the remaining $n - 7$ were drawn without replacement from the indices $\{1, 7\}$. This induced a small correlation between subjects for $n > 7$, which we noticed had a small but predictably deleterious effect on method performances.

For comparisons in the cue-period window, we assessed the average performance of *FLMM* evaluated at time-points in that window, against the performance of standard methods that analyzed a summary measure of the window (cue-period average-signal, which we denote as AUC) as the outcome variable. The LMM was applied to trial-level AUCs, the paired t-test was applied to subject- and condition-level AUCs. The performances of the *Perm* and *FLMM* were quantified according to the average pointwise performance during the cue-period. We present the full consecutive threshold criteria of *Perm* in [Figure 7](#) [DOI](#) and show performance of the 1/2 consecutive threshold criteria in Appendix section 8.2 for completeness. We implemented the consecutive threshold criteria method, adapted to our simulated sampling rate of 15Hz.

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5 Appendix Photometry Citations

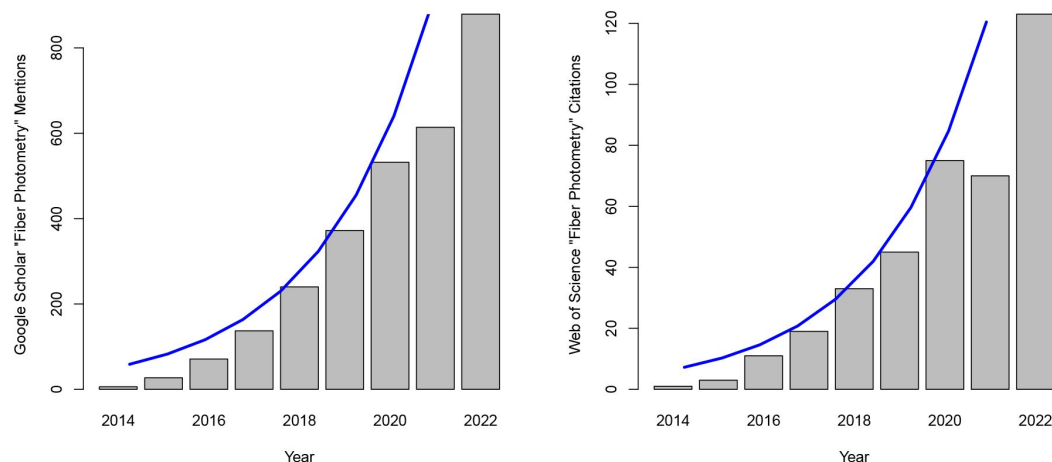


Figure 9.

[Left] Google scholar mentions of the string "Fiber Photometry" by year. There were 549 mentions between January 1, 2023 – June 22, 2023. [Right] Web of Science citations of papers that include the string "Fiber Photometry" by year. There were 50 citations between January 1, 2023 – June 22, 2023. Blue lines indicate fitted values from an exponential fit to the data: $\text{Citations}_i = \alpha * \exp[\beta * \text{Year}_i]$, where α and β were estimated with the nls package in R. The 1,500 references to photometry described in the main text refers to Google Scholar mentions in the previous 12 months.

6 Appendix Functional Mixed Models Methods

For a thorough introduction to the Functional Mixed Model fitting framework presented in our manuscript, please refer to Cui et al. (2021) [\[1\]](#). However, for completeness, we provide a short description of the estimation details and then provide a brief derivation of our proposed estimator. The majority of the details presented here can be found in greater depth in section 3.1. *Analytic Inference for Gaussian Functional Data* of Cui et al. (2021) [\[1\]](#) but we believe we have included sufficient description of the statistical estimation scheme to explain the derivation of our proposed estimator (presented in Section 6.2). For clarity, we focus on functional *linear* mixed models. However, our package also provides the capability to fit the wider class of functional generalized linear mixed models for the distributions supported by the R lme4 package.

6.1 Functional Linear Mixed Models

We focus on the functional linear mixed model,

$$\mathbf{Y}_i(s) = \mathbf{X}_i\boldsymbol{\beta}(s) + \mathbf{Z}_i\boldsymbol{\gamma}_i(s) + \boldsymbol{\epsilon}_i(s), \quad (4)$$

where $\mathbf{Y}_i(s) \in \mathbb{R}^{n_i}$ is the vector of observations of functional outcomes at time-point s for subject i , n_i is the number of functional observations of subject i , and $s \in \{1, 2, L\}$, where L is the number of points on the grid of the functional domain (i.e., the number of trial time-points or trial photometry samples; we use S in the main text to refer to L). We take N to be the number of

subjects (e.g., animals) and $n^* = \sum_{i=1}^N n_i$. We denote $\beta(s) \in \mathbb{R}^p$ to be the vector of functional fixed-effects at point s . $\gamma_i(s) \in \mathbb{R}^{q_i}$ and $\epsilon_i(s) \in \mathbb{R}^{n_i}$ are vectors of mutually independent random-effects and error terms for time-point s , with multivariate Gaussian distributions. We include a subject-specific index for q_i because, depending on the random-effect specification and n_i , the dimension of the random-effects vector, $\gamma_i(s)$, can vary across subjects (but q_i is fixed within-subjects across trial time-points, s). We denote $\mathbb{X}_i \in \mathbb{R}^{n_i \times p}$ and $\mathbb{Z}_i \in \mathbb{R}^{n_i \times q_i}$ as (design) matrices of subject i containing the covariates for the fixed- and random-effects terms, respectively. Denote $\mathbb{Z} \in \mathbb{R}^{n^* \times \sum_i q_i}$ to be a block diagonal matrix where the i^{th} block entry is $\mathbb{Z}_i$ and let $\mathbb{X} \in \mathbb{R}^{n^* \times p}$ refer to the design matrix from row concatenating the subject-specific matrices $\mathbb{X}_i$. Estimates for the fixed-effects $\beta(s)$ are correlated across trial time-points (i.e., the functional domain), which is incorporated into *joint* inference through assuming $\text{Cov}(\gamma(s_1), \gamma(s_2)) = \mathbb{G}(s_1, s_2) \in \mathbb{R}^{\sum_i q_i \times \sum_i q_i}$ and $\text{Cov}(\epsilon(s_1), \epsilon(s_2)) = 0$ for all $s_1 \neq s_2$, where $\gamma(s) = [\gamma_1^T(s) \gamma_2^T(s) \dots \gamma_N^T(s)]^T \in \mathbb{R}^{\sum_i q_i}$ and $\epsilon(s) = [\epsilon_1^T(s) \epsilon_2^T(s) \dots \epsilon_N^T(s)]^T \in \mathbb{R}^{n^*}$.

We used a two-stage approach for fitting functional linear mixed models. In the first step, we fit pointwise linear mixed models at each time-point s , yielding fixed-effect estimates which we denote as $\tilde{\beta}(s)$. The pointwise estimator admits the closed form expression,

$\tilde{\beta}(s) = [\mathbb{X}^T \mathbb{V}^{-1}(s) \mathbb{X}]^{-1} \mathbb{X}^T \mathbb{V}^{-1}(s) \mathbf{Y}(s)$, where $\mathbb{V}(s) = \mathbb{Z} \mathbb{H}(s) \mathbb{Z}^T + \tilde{\mathbb{R}}(s)$, and $\mathbb{H}(s)$ and $\tilde{\mathbb{R}}(s)$ are the covariance matrices of $\gamma(s)$ and $\epsilon(s)$, respectively. The variance of the pointwise estimator is $\text{Var}(\tilde{\beta}(s)) = [\mathbb{X}^T \mathbb{V}^{-1}(s) \mathbb{X}]^{-1}$.

The correlation across trial time-points can be seen more explicitly through the following expression:

$$\text{Cov}(\tilde{\beta}(s_1), \tilde{\beta}(s_2)) = [\mathbb{X}^T \mathbb{V}^{-1}(s_1) \mathbb{X}]^{-1} \mathbb{X}^T \mathbb{V}^{-1}(s_1) \mathbb{W}(s_1, s_2) \mathbb{V}^{-1}(s_2) \mathbb{X} [\mathbb{X}^T \mathbb{V}^{-1}(s_2) \mathbb{X}]^{-1}$$

where $\mathbb{W}(s_1, s_2) = \mathbb{Z} \mathbb{G}(s_1, s_2) \mathbb{Z}^T$. While estimates of $\mathbb{H}(s)$ and $\tilde{\mathbb{R}}(s)$ are provided by standard mixed modeling software, $\mathbb{G}(s_1, s_2)$ must be estimated separately. We detail the estimation scheme below as it plays a critical role in *joint* inference and constitutes the main topic of our statistical contribution. Note that the estimates of $\mathbb{G}(s_1, s_2)$, $\mathbb{H}(s)$, and $\tilde{\mathbb{R}}(s)$ can be smoothed using, for example, fast bivariate P-splines (Xiao et al., 2013) along the functional domain (that is, trial time-points) to reduce variability; and then any negative eigenvalues of the smoothed matrices can be trimmed at 0 to ensure the resulting covariance matrix estimates are positive semi-definite (Cui et al., 2021; Greven et al., 2010).

Our package provides the option to use an array of smoothing approaches for the regression coefficient estimates, $\tilde{\beta}(s)$, but defaults to using penalized splines (Ruppert et al., 2003). When using penalized splines to smooth the pointwise estimates, $\tilde{\beta}(s)$, the covariance matrix for the final fixed-effects estimates admits a closed form expression, and allows for fast calculation of *joint* confidence band estimates. To see this, denote $\hat{\beta}(s)$ as the final fixed-effect estimates after smoothing the raw pointwise estimates, $\tilde{\beta}(s)$, directly provided by mixed model software. Then a smoothed estimator for the t^{th} fixed-effect is $\hat{\beta}^{(t)} = \mathbb{S}_t \tilde{\beta}^{(t)}$, where

$$\tilde{\beta}^{(t)} = [\tilde{\beta}^{(t)}(s_1), \dots, \tilde{\beta}^{(t)}(s_L)]^T \in \mathbb{R}^L. \text{ Thus, } \text{Cov}(\hat{\beta}^{(t)}) = \mathbb{S}_t \text{Cov}(\tilde{\beta}^{(t)}) \mathbb{S}_t^T, \text{ where}$$

$\mathbb{S}_t = \mathbb{B}_t (\mathbb{B}_t^T \mathbb{B}_t + \lambda_t \mathbb{P}_t)^{-1} \mathbb{B}_t^T$, $\mathbb{B}_t$ is a K -dimensional spline basis matrix, λ_t is a smoothing parameter, and $\mathbb{P}_t$ is a penalty matrix. Typically the number of knots KL , but as long as a sufficient number of knots are specified, the smoother and exact K selected appear to have little impact on the final coefficient estimates or 95% CIs in practice. To be conservative, we used a high

number of knots ($K = L/2$), which we set as the default in our package. In practice, the specific number could be altered depending on the sampling rate of the photometry data analyzed and the specific sensor (since the kinetics of neurochemical signaling systems can vary widely). The results presented in the paper were calculated based upon implementation of our method using thin-plate splines since these performed well in practice.

Our package leverages the well-known lme4 package in R for fitting the pointwise mixed effects models (Bates, 2010; Bates et al., 2014). This precludes, however, the specification of structure on the covariance matrix of the errors, $\epsilon_i(s)$ across trials (for a fixed trial time-point s) in Gaussian models, functionality provided by, for example, the nlme in R (Pinheiro et al., 2023). Future work could explore functional linear mixed models that allow for this functionality. However, in our experimentation analyzing photometry data summary measures (e.g., AUCs) from a range of experiments that observed multiple trials and sessions per animal, we found that model fit criteria (e.g., AIC, BIC, cAIC (Säfken et al., 2018)) were usually much better when accounting for correlation across trials within-animal through random-effect specifications as opposed to placing structure on the covariance of the errors within-subject across trials. Moreover, we found that models that specified the types of error covariance structures applicable to photometry experiments were often slow to fit which is a drawback in our approach to fitting functional mixed models given that it requires fitting many pointwise models. For those reasons, we set our package's default to use the lme4 package thereby providing both speed and modeling flexibility to users.

We now provide a brief description of our proposed estimator for $\mathbb{G}(s_1, s_2)$. The critical role that this plays in inference and in neuroscience applications arises from the nested designs common in the sophisticated behavioral experiments commonly used alongside fiber photometry. We briefly describe this in Appendix Section 6.3.

6.2 Appendix: Covariance Estimator

We begin with a high-level description of the estimator for the covariance matrix, $\mathbb{G}(s_1, s_2)$, and include details below. Denote $Y_{i,k}(s_1) \in \mathbb{R}$ as the functional outcome at time-point s_1 , for subject i on trial k . The method of moments estimator proposed in (Greven et al., 2010) and applied to our functional mixed model estimation procedure was presented in Section 3.1, equation (4) from Cui et al. (2021),

$$E \left[\{Y_{i,k}(s_1) - \mathbf{X}_{i,k}^T \beta(s_1)\} \{Y_{i,j}(s_2) - \mathbf{X}_{i,j}^T \beta(s_2)\} \right] = \sum_{t=1}^q \sum_{v=1}^q Z_{i,k,t} Z_{i,j,v} \text{Cov}(\gamma_{i,t}(s_1), \gamma_{i,v}(s_2)). \quad (5)$$

This expression suggests an estimator that regresses the residual products $[Y_{i,k}(s_1) - \mathbf{X}_{i,k}^T \beta(s_1)][Y_{i,j}(s_2) - \mathbf{X}_{i,j}^T \beta(s_2)]$ onto the random-effect covariate products $\{Z_{i,k,t} Z_{i,j,v} j, k = 1, 2, \dots, J_i\}$, where J_i indicates the number of trials for subject i , and t, v are random-effect covariate indices. Concatenating these covariate products into a design matrix, $\tilde{\mathbf{Z}}$, and concatenating the residual products into an “outcome vector”, $\tilde{\mathbf{Y}}(s_1, s_2) \in \mathbb{R}^{n^*}$, allows us to express the estimator as a solution to the least squares problem,

$$\hat{\alpha}(s_1, s_2) \in \underset{\alpha(s_1, s_2)}{\text{argmin}} \left\| \tilde{\mathbf{Y}}(s_1, s_2) - \tilde{\mathbf{Z}} \alpha(s_1, s_2) \right\|_2^2. \quad (6)$$

After calculating $\hat{\alpha}(s_1, s_2)$, $\hat{\mathbb{G}}(s_1, s_2)$ is obtained by re-organizing the elements of the vector $\hat{\alpha}(s_1, s_2)$ into the entries of the matrix $\hat{\mathbb{G}}(s_1, s_2)$, which we describe in greater detail below. This estimator is flexible and in principle is agnostic to the random-effect specification, but in practice requires modification when including general random-effects specifications. A common scenario requiring these modifications is when random-effects are included for binary covariates.

Such settings arise anytime one includes, for example, nested random-effect structures. This is a substantial drawback because these types of random-effect specifications are critical to properly model the sophisticated behavioral designs common in neuroscience which often exhibit multiple layers of nesting. For instance, suppose a collection of animals are trained on multiple sessions, and within each session are trained on multiple trials. Then trial is nested in session, which is nested in animal/subject. Inclusion of animal- and session-specific random intercepts, a common random-effect specification for such experimental designs, will exhibit the challenges above.

In order to make the issue concrete, we describe perhaps the simplest case where the above problem presents. Specifically, we show how the original estimator requires extension in a simple *FLMM* model that is analogous to the functional analogue of the paired samples t-test: a model that includes a single binary covariate such as a treatment indicator (i.e., $X_{i,j} \in \{0, 1\}$), a random subject-specific intercept, and a random subject-specific slope. Suppose we observe a collection of trials indexed by j , on a collection of animals indexed by i . Then the above model can be expressed as

$$\mathbb{E}[\mathbf{Y}_{i,j}(s) \mid \mathbf{X}_{i,j}, \mathbf{Z}_{i,j}] = \beta_0(s) + \gamma_{0,i}(s) + X_{i,j} [\beta_1(s) + \gamma_{1,i}(s)]. \quad (7)$$

This model yields the vector of random-effect covariates $\mathbf{Z}_{i,j}^T = [1 \ X_{i,j}]$, and the associated matrix $\tilde{\mathbb{Z}}_i$ is

$$\tilde{\mathbb{Z}}_i = \begin{bmatrix} 1 & 2X_{1,1} & X_{1,1}^2 \\ 1 & 2X_{1,2} & X_{1,2}^2 \\ \vdots & \vdots & \vdots \\ 1 & 2X_{1,J_1} & X_{1,J_1}^2 \end{bmatrix} = \begin{bmatrix} 1 & 2X_{1,1} & X_{1,1} \\ 1 & 2X_{1,2} & X_{1,2} \\ \vdots & \vdots & \vdots \\ 1 & 2X_{1,J_1} & X_{1,J_1} \end{bmatrix}.$$

The above equality follows trivially because $X_{i,j}^2 = X_{i,j}$ since $X_{i,j}$ is binary. The columns therefore exhibit the linear dependence $2\tilde{\mathbb{Z}}_{\cdot,2} = \tilde{\mathbb{Z}}_{\cdot,3}$ and thus $\tilde{\mathbb{Z}}$ is not full rank. To our knowledge, no re-coding of the binary variable $X_{i,j}$ resolves the above linear dependence problem without changing the contrast (e.g., one could fit a model without a random intercept and include the covariates $[X_{i,j} \ (1 - X_{i,j})]$).

More generally, the reduced rank problem presents whenever a subject-specific random intercept is included (because it is encoded by a column of ones, $\mathbb{1}$, in $\tilde{\mathbb{Z}}$) alongside any binary covariate in $\mathbb{Z}$. As shown in the example above, this is because the column of ones used for random intercepts, when multiplied by binary covariates to construct the matrix $\tilde{\mathbb{Z}}$, will create linear dependence in the columns of $\tilde{\mathbb{Z}}$. This is problematic because binary covariates are needed to encode dichotomous and factor covariates (e.g., in the functional analogue of t-tests and ANOVAs), as well as, for example, session-specific random intercepts in nested designs. As such, resolving this issue is critical to properly model experiments that regularly arise in neuroscience.

Before deriving the estimator, we provide the necessary, but admittedly tedious, notation. Since a single random-effect can require many columns of $\mathbb{Z}$ to encode the corresponding covariate (e.g., if there are many session-specific random intercepts), separate indices are needed to distinguish between a single random-effect distribution and the (potentially many) columns of the random-effect design matrix associated with draws from that random-effect distribution (similar to how one distinguishes between a single factor covariate and the collection of binary indicator variables needed to encode each level of that factor).

Denote $\tilde{\mathbb{Z}} \in \mathbb{R}^{n^* \times \tilde{q}}$ to be a block diagonal matrix where the i^{th} block entry is $\tilde{\mathbb{Z}}_i$. The number of columns, $\tilde{q}$, will vary depending on the random-effect specification. For example, if one specifies a model that includes session-specific random-intercepts, this would require one column in $\mathbb{Z}$ (and thus multiple columns in $\tilde{\mathbb{Z}}$) for each animal and session to encode the corresponding animal- and session-specific “covariates” associated with these random-intercepts.

Let q^* be the number of unique random-effects distributions for a given model, where $q^* \leq q$. Take as an example a model that includes: 1) a subject-specific random intercept, $\gamma_{0,i}(s)$, and 2) subject- and session-specific random intercepts, $\gamma_{1,i,w}(s)$ for participant i on session w . If we assume that $\gamma_{1,i,w}(s) \stackrel{iid}{\sim} \mathcal{N}(0, \tilde{\sigma}^2) \forall i, w$, then $q^* = 2$, but $q \gg 2$ since encoding the subject- and session-specific random intercepts, $\gamma_{1,i,w}(s)$ would require many indicator variables (entered as columns of $\mathbb{Z}$). More generally, for random-effect covariate, r , denote $\mathcal{J}_r$ the set of column indices of $\mathbb{Z}$ that encode covariate r . Then denote q_r to be the cardinality of that set (i.e., $q_r = |\mathcal{J}_r|$). Note that we describe $\gamma_{r,i,w}(s)$ as drawn from a univariate Gaussian for explanatory purposes, but neither the estimator nor the modeling software assume independence between random-effects from *different* covariates.

Expanding Section 3.1, equation (4) from Cui et al. (2021) to be in terms of the column indices of the design matrix $\mathbb{Z}$

$$\begin{aligned} & \sum_{t=1}^{q_i} \sum_{v=1}^{q_i} Z_{i,k,t} Z_{i,j,v} \text{Cov}(\gamma_{i,t}(s_1), \gamma_{i,v}(s_2)) \\ &= \sum_{r=1}^{q^*} \sum_{m=1}^{q^*} \sum_{l \in \mathcal{J}_r} \sum_{b \in \mathcal{J}_m} Z_{i,k,l} Z_{i,j,b} \text{Cov}(\gamma_{i,l}(s_1), \gamma_{i,b}(s_2)) \\ &= \sum_{r=1}^{q^*} \sum_{m=1}^{q^*} \left\{ \sum_{l \in \mathcal{J}_r} Z_{i,k,l} \left[\sum_{b \in \mathcal{J}_m} Z_{i,j,b} \text{Cov}(\gamma_{i,l}(s_1), \gamma_{i,b}(s_2)) \right] \right\} \end{aligned}$$

Now recall that $\gamma_{i,l}(s) \stackrel{iid}{\sim} \mathcal{N}(0, \tilde{\sigma}_r^2)$ for all $l \in \mathcal{J}_r$. Thus, $\text{Cov}(\gamma_{i,l}(s_1), \gamma_{i,b}(s_2)) = \tilde{\rho}_{r,m}$ for $l \in \mathcal{J}_r$ and $b \in \mathcal{J}_m$. It then follows that we can simplify the above as,

$$\begin{aligned} & \sum_{r=1}^{q^*} \sum_{m=1}^{q^*} \left\{ \sum_{l \in \mathcal{J}_r} Z_{i,k,l} \left[\sum_{b \in \mathcal{J}_m} Z_{i,j,b} \text{Cov}(\gamma_{i,l}(s_1), \gamma_{i,b}(s_2)) \right] \right\} \\ &= \sum_{r=1}^{q^*} \sum_{m=1}^{q^*} \left[\sum_{l \in \mathcal{J}_r} Z_{i,k,l} \right] \left[\sum_{b \in \mathcal{J}_m} Z_{i,j,b} \right] \text{Cov}(\gamma_{i,r}(s_1), \gamma_{i,m}(s_2)) \end{aligned}$$

This expression suggests two potential estimation approaches. The first approach is to specify the products $A_{i,k,r} B_{i,j,m}$ as “covariates” in the OLS-based estimator described in expression (5), where $A_{i,k,r} = \sum_{l \in \mathcal{J}_r} Z_{i,k,l}$ and $B_{i,j,m} = \sum_{b \in \mathcal{J}_m} Z_{i,j,b}$. However, we anticipate this would exacerbate the co-linearity problems in some settings, since, for example, $A_{i,k,r} = 1$ for any random slope associated with a factor covariate.

Instead, we propose to follow another approach by first regressing the residual products onto $\{Z_{i,k,v}Z_{i,j,t} \mid v, t = 1, q\}$. Denote $\hat{\alpha}_{v,t}$ as the coefficient estimate associated with the “covariate” $Z_{i,j,v}Z_{i,k,t}$ in this regression. We exploit the fact that $\text{Cov}(\gamma_{i,l}(s_1), \gamma_{i,b}(s_2)) = \tilde{\rho}_{r,m}$ for all $l \in \mathcal{I}_r$ and $b \in \mathcal{I}_m$. Solving for the desired quantity then yields the estimator

$$\widehat{\text{Cov}}(\gamma_{i,r}(s_1), \gamma_{i,m}(s_2)) = \frac{1}{|\mathcal{I}_r||\mathcal{I}_m|} \sum_{l \in \mathcal{I}_r} \sum_{b \in \mathcal{I}_m} \hat{\alpha}_{l,b}. \quad (8)$$

While this partially resolves co-linearity problems, it does not ensure the design matrix, $\tilde{\mathbf{Z}}$, is of full rank for all random-effect specifications. In such cases, the solution to problem (6) is not unique. We solve for the “ridgeless regression” solution to problem (6) because it yields the minimum ℓ_2 norm estimator and exhibits desirable statistical properties (Hastie et al., 2019). This can be expressed as the solution to the optimization problem,

$$\begin{aligned} \min_{\boldsymbol{\alpha}(s_1, s_2)} \quad & \|\boldsymbol{\alpha}(s_1, s_2)\|_2^2 \\ \text{s.t.} \quad & \boldsymbol{\alpha}(s_1, s_2) = \underset{\boldsymbol{\alpha}(s_1, s_2)}{\text{argmin}} \left\| \hat{\boldsymbol{\epsilon}}(s_1, s_2) - \tilde{\mathbf{Z}}\boldsymbol{\alpha}(s_1, s_2) \right\|_2^2, \end{aligned} \quad (9)$$

where $\hat{\boldsymbol{\epsilon}}(s_1, s_2)$ are the residual products (i.e., with entries $\hat{\epsilon}_{i,j,k}(s_1, s_2) = [Y_{i,k}(s_1) - \mathbf{X}_{i,k}^T \boldsymbol{\beta}(s_1)][Y_{i,j}(s_2) - \mathbf{X}_{i,j}^T \boldsymbol{\beta}(s_2)]$) and $\boldsymbol{\alpha}(s_1, s_2) \in \mathbb{R}^{\tilde{q}}$. We can estimate $\hat{\mathbb{G}}(s_1, s_2)$ through reorganizing the elements of $\hat{\boldsymbol{\alpha}}(s_1, s_2)$ into the matrix $\hat{\mathbb{G}}(s_1, s_2)$. In practice the dimension of $\hat{\boldsymbol{\alpha}}(s_1, s_2)$ may differ from the number of columns of $\hat{\mathbb{G}}(s_1, s_2)$ because $\hat{\mathbb{G}}(s_1, s_2)$ may contain many repeated entries of functions of the element of $\hat{\boldsymbol{\alpha}}(s_1, s_2)$ (as suggested by expression (8)).

We can estimate $\boldsymbol{\alpha}(s_1, s_2)$ through the closed form expression,

$$\hat{\boldsymbol{\alpha}}(s_1, s_2) = \left(\tilde{\mathbf{Z}}^T \tilde{\mathbf{Z}} \right)^+ \tilde{\mathbf{Z}}^T \tilde{\mathbf{Y}}(s_1, s_2) = \mathbb{M} \tilde{\mathbf{Y}}(s_1, s_2),$$

where $(\cdot)^+$ denotes the Moore-Penrose pseudoinverse (Hastie et al., 2019) and $\hat{\boldsymbol{\alpha}}(s_1, s_2) \in \mathbb{R}^{\tilde{q}}$. Thus, while the estimator requires one to solve $L(L+1)/2$ problems of the form above (i.e., one for each unique $\{s_1, s_2\}$ pair), in practice we only need to calculate $\mathbb{M}$ once for all $\{s_1, s_2\}$ pairs. This allows for estimation of $\mathbb{G}$ (i.e., for all unique s_1, s_2 pairs) for datasets with many observations, complex random-effects specifications, and large L within a couple seconds in total.

We also explored the performance of two estimators for $\mathbb{G}$ based on mathematical programs that enforce non-negativity of variance components (before eigenvalue trimming). The first mathematical program was

$$\begin{aligned} \min_{\boldsymbol{\alpha}(s_1, s_2)} \quad & \|\boldsymbol{\alpha}(s_1, s_2)\|_2^2 \\ \text{s.t.} \quad & \boldsymbol{\alpha}(s_1, s_2) = \underset{\boldsymbol{\alpha}(s_1, s_2)}{\text{argmin}} \left\| \hat{\boldsymbol{\epsilon}} - \tilde{\mathbf{Z}}\boldsymbol{\alpha}(s_1, s_2) \right\|_2^2 \\ & \alpha_m \geq 0 \quad \text{for } m \in \mathcal{M}, \end{aligned} \quad (10)$$

where $\mathcal{M}$ denotes the set of indices corresponding to variance elements.

Expression (8) suggests that one could alternatively constrain the sum, as opposed to each element in the summation $\sum_{l \in \mathcal{I}_r} \sum_{b \in \mathcal{I}_m} \hat{\alpha}_{l,b}$. Thus we also explored performance of the estimator based on the mathematical program,

$$\begin{aligned} \min_{\alpha(s_1, s_2)} \quad & \|\alpha(s_1, s_2)\|_2^2 \\ \text{s.t.} \quad & \alpha(s_1, s_2) = \underset{\alpha(s_1, s_2)}{\operatorname{argmin}} \left\| \hat{\epsilon} - \tilde{\mathbb{Z}}\alpha(s_1, s_2) \right\|_2^2 \\ & \sum_{l \in \mathcal{I}_m} \alpha_l \geq 0 \quad \text{for } m \in \mathcal{M}. \end{aligned} \tag{11}$$

We found that, in practice, the performance of the estimators based on the mathematical programs (9), (10), and (11), performed comparably in simulations. This may be because the eigenvalue trimming applied to the solutions to the above optimization problems yields similar final estimates. Our software automatically implements estimators based on all three mathematical programs, but defaults to the estimator based on the solution to mathematical program (9). All results presented in the main text apply to this version.

In simulations, we observed that the regression coefficient estimate 95% CIs calculated using our proposed $\mathbb{G}$ estimator achieved the nominal *joint* coverage even with model specifications that yield reduced rank $\tilde{\mathbb{Z}}$. Therefore, the above estimator allows for a fast, flexible extension of the work in (Cui et al., 2021), thereby allowing for 95% CI calculation for general random-effects specifications. This statistical contribution is critical for neuroscience since studies often exhibit nested experimental designs that require sophisticated random-effect models to properly capture the rich information contained in photometry signals.

6.3 Appendix: Random-Effects Structures and Interpretations for Neuroscience Experiments

Functional random intercepts that are unique to each trial and/or sessions can be used to account for animal-to-animal, session-to-session, and trial-to-trial heterogeneity in the signal and is one example where one might include a covariate in the random-effects (that is, included in $\mathbb{Z}_i$) but not in the main effects terms, $\mathbb{X}_i$. Inclusion of these session- and trial-level random-effects is an example of one way to account for correlation within an animal across trials and sessions (“photometry signal on trial j for animal i is likely similar to the signal on trial $j + 1$). Indeed, many neuroscience studies involving behavioral paradigms that involve multiple sessions and trials (and potentially multiple groups) may require complicated random-effects structures to account for animal-to-animal, session-to-session and trial-to-trial heterogeneity in the photometry signal. For this reason, we explored the performance of many models to learn how to best model variability in the data.

FLMM yields interpretable regression coefficient estimates. Conveniently, they can be interpreted conditional on the functional random-effects or marginally with respect to the random-effects since both interpretations are numerically equivalent in the linear setting. For an intuitive introduction to this equivalence for linear mixed models, we recommend sections 2.2 and 7.4 of Crowder (2009). We present the marginal interpretation here because it is often more intuitive for photometry analyses. However, the conditional version has its own advantages.

6.4 Functional Regression Model Classes

7 Appendix Reanalysis Figures and Methods

In the following sections, we provide additional details for the reanalyses (Jeong et al., 2022) presented in the main text. In cases where it is helpful, we provide quotes from the original paper (Jeong et al., 2022) that informed our modeling decisions. These were taken from the main text of the paper, figure captions, and the supplement. In some places, we omitted minor portions of text to make the quotations more concise (e.g., figure references or citation numbers).

7.1 Section 2.3 Reanalyses

7.2 Appendix: Section Lick Bout Correlation

We present further analyses that illustrate how *FLMM* reveals effects obscured by standard methods. The authors note that the Reward Number–DA association, reported in their analyses, could arise from Lick Rate increases that also correlate with Reward Number. They tested this alternative explanation by applying a Pearson correlation between Lick Rate and DA. While they reported no significant association, the plot for the Lick Rate covariate from *FLMM* (Figure 12), shows Lick Rate exhibits 1) a significant positive association with DA before lick-bout onset, and 2) a negative association after lick-bout onset (reaching *joint* significance at ~ 1 s). These results suggest that the correlation analysis of reward period AUC in the paper missed this effect because the AUC summarized a time-window that contained opposing effects. The underlying association was likely diluted by averaging over time-points when Lick Rate is both positively correlated with DA (the first 0.5s of the reward period, visible in positive *FLMM* coefficient estimates) and negatively correlated with DA (the final 1s of the 1.5s time-window, with negative *FLMM* coefficient estimates). Identifying this effect with a summary measure would have required selecting these time intervals perfectly a priori; this is not necessary with *FLMM*.

Lick Rate Model We next repeated the analyses from Figure S8 C-D in (Jeong et al., 2022). The figure caption notes that “The consummatory lick rate is not correlated with dopamine reward response.” We used the methods described in the quoted paragraphs above, and provide further details in Appendix Section 7.2.

We fit the model for lick rate ($LR_{i,j,l}$):

$$\mathbb{E}[Y_{i,j,l}(s) | X_{i,j,l}, Z_{i,j,l}] = \beta_0(s) + \beta_1(s)LR_{i,j,l} + \gamma_{0,i}(s) + \gamma_{1,i,l}(s) + \text{trial}_{i,j,l} [\gamma_{2,i}(s) + \gamma_{3,i,l}(s)].$$

Consummatory lick-bout extraction

As stated in “Data Analysis: Experiment 1” of the Supplement (p.3-4), consummatory lick-bouts were defined as follows:

“To test whether lick rate affects dopamine reward response, we first classified licks into consummatory and non-consummatory licks. A group of licks with less than 1 s interval was defined as a lick bout. This criterion was conservative. Every lick in the first lick bout after reward delivery was defined as a consummatory lick, and all other licks were defined as nonconsummatory licks. Reward-by-reward correlation between consummatory lick rate and dopamine response was measured. To avoid any influence of the previous reward on the dopamine response or behavior to the current reward, we excluded rewards with less than 3 s IRI from the previous reward ($22.6 \pm 2.4\%$) for the above analyses. Rewards without lick until the next reward ($5.7 \pm 0.1\%$) were also excluded from analyses.”

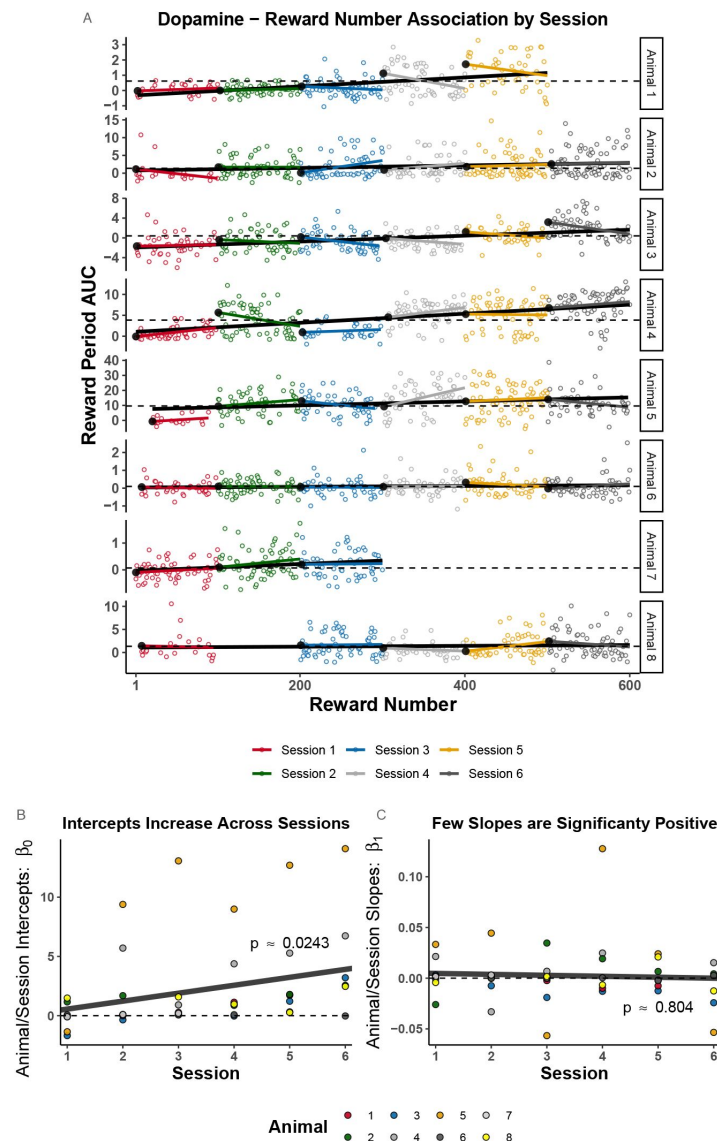


Figure 10.

Reward Number-AUC correlation within-session and across-session.

(A) Color indicates session number, rows denote animal number. Trial Number is the within-session Reward Number and ranges from 1-100 for each session. The black line that spans across sessions is a Reward Number-AUC linear regression fit, while the session color lines indicate a within-session Trial Number-AUC linear regression fit. The large black circles on the left side of each session-specific fit is the intercept, parameterized to yield the interpretation as the “expected AUC on the first trial of the corresponding session.” Dotted horizontal lines are set at the median of the intercepts to facilitate comparison. The intercepts tend to rise across sessions, while few slopes are significantly positive. (B)-(C) Each dot indicates the estimated intercept value (B) or slope (C) from the fits shown in (A). Lines and p-values were calculated in an LMM that was fit to the session-specific linear regression slopes, $\hat{\beta}_1$, and intercepts, $\hat{\beta}_0$, shown in (A). The LMM included animal-specific random intercepts and slopes. These plots quantify the trend observed in (A): the estimated intercepts significantly increase across sessions, but the slopes are mostly negative.

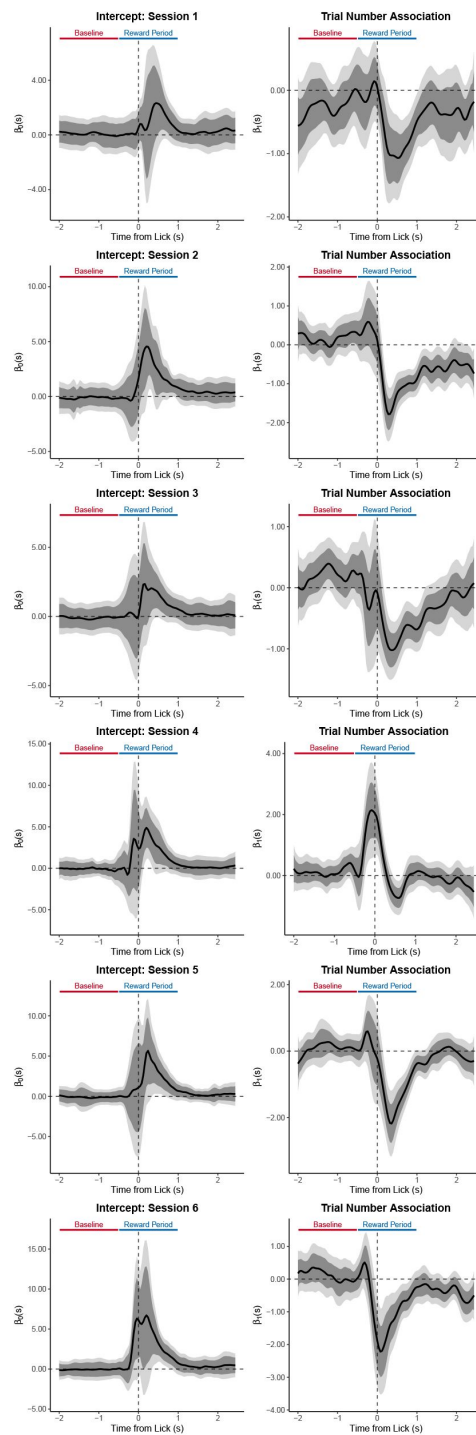


Figure 11.

Trial Number-DA correlation within-session on the random IRI task.

Row indicates session number. The intercept is parameterized to yield the interpretation as the “expected signal magnitude on the first trial of the corresponding session.” Effects are aligned to the first lick after reward delivery. This is the session-by-session version of the analysis presented in [Figure 5J-K](#).

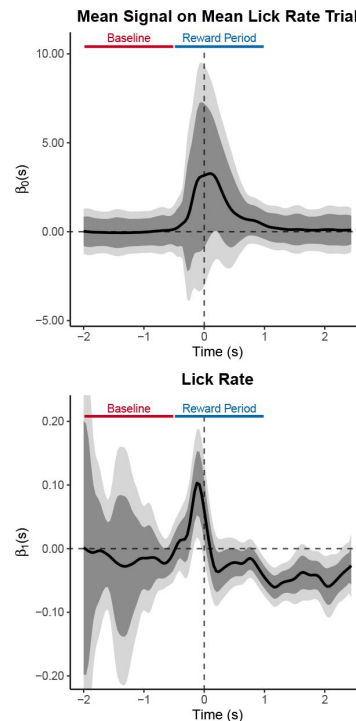


Figure 12.

***FLMM* reveals details occluded by summary measure analyses.**

Coefficient estimates from an *FLMM* analysis of the random inter-trial interval (IRI) reward delivery experiment. The top row contains the intercept term plots where the title provides interpretation of the intercept: the average dopamine (DA) signal on trials when Lick Rate is at its average value. The bottom row shows the coefficient estimate plot of the covariate in the model. The “Baseline” and “Reward Period” bars show the trial period that the original authors used to calculate the summary measure (AUC). Specifically, they quantified DA by a measure of normalized AUC of $\Delta F/F$ during a window 0.5 s before to 1 s after the first lick following reward delivery. All plots are aligned to this first lick after reward delivery. The interpretation of the y-value of the bottom plot at any time-point s : the mean change in the dopamine signal at s for a one unit change in Lick Rate. Association between DA and Lick Rate aligned to lick bout onset. Time-points when Lick Rate was negatively associated with DA (negative coefficient estimates in the final 1s of the 1.5s window) may have diluted time-points when they were positively associated (positive coefficient estimates in the first 0.5s of the reward period).

For an IRI that was less than 3s, we excluded the trial before and after that IRI so as to avoid any influence on either trial's signal. Our code implementations of the above methods can be found on the github page: https://github.com/gloewing/photometry_fGLMM

The reward period time-window and methods were described in the Supplement of Jeong et al. (2022):

“Dopamine response to reward was defined as the normalized area under curve (AUC) of $\Delta F/F$ during reward period. Reward period was defined as -0.5 to 1 s from the first lick after reward delivery. We defined the window with reference to the first lick time, not reward delivery time, because the response latency to reward differs across trials. Also, we used a window starting from 0.5 s ahead the first lick because dopamine response started to increase even before an animal made the first lick (as they get better at sensing reward delivery in late sessions). The AUC during 1.5 s time window before reward period was subtracted from AUC during reward period to normalize baseline activity.”

7.3 Photobleaching

In main text Section 2.3, we conducted analyses that showed DA decreased within-session during the post-lick interval of the reward period ([0, 1] sec from lick-onset): Trial Number was negatively correlated with signal magnitude during the post-lick period. We include a portion of main **Figure 5** here to assist in the photobleaching discussion.

Because photobleaching is thought to depend on light intensity in a non-linear fashion (Serra and Terentjev, 2008), we reasoned that if photobleaching caused the within-session decrease post-lick, indications of photobleaching would also be evident in other time-windows in the trial when the average signal was high (e.g., the pre-lick period). The signal–Trial Number association is, however, actually slightly positive pre-lick (albeit non-significant), indicating that the signal does not decrease on average within-session pre-lick. Moreover, if the degree of photobleaching scales with signal magnitude, then we would expect to see a negative signal–Trial Number association only during the [0,0.5] sec time-window when the mean signal is higher than it was during the pre-lick period (since the pre-lick period does not exhibit any negative signal–Trial Number correlation). Instead, the negative correlation is evident in a time-window when the mean signal has returned to close to zero [0, 2.5] sec. In Appendix Figure 11, we show the same analyses conducted on each session separately. *FLMM* estimates a large positive Trial Number effect pre-lick and post-lick on session 4, despite the fact that the average signal was higher than on sessions 1-3 when the Trial Number is negative. Since there are sessions on which it is possible to detect positive Trial Number effects, we reasoned that photobleaching would not have occluded any true DA increases within-session. Finally, we repeated the above analyses while adjusting for lick frequency and found that controlling for various behavioral engagement summaries did not impact the Trial Number effects.

In Appendix Figure 14, we conducted a similar analysis but aligned the signal to reward-delivery. If photobleaching were the main cause of the within-session reductions described above, one might expect the Trial Number coefficient to be most negative when the average signal is most positive. Instead, these analyses show that the within-session signal decrease (i.e., the negative Trial Number effect) and the average signal exhibit distinct temporal dynamics. For example, the peak average signal occurs around 0.75 sec post-reward, while the Trial Number coefficient is most negative around 1.25-1.5 sec. Moreover, while the average signal magnitude has returned to nearly 0 around 1.75 sec, the Trial Number remains significantly negative until at least 2.5 sec after reward-delivery.

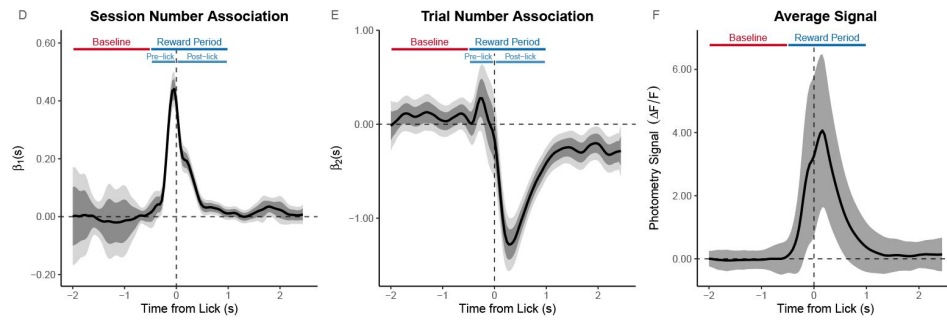


Figure 13.

Random IRI experiment aligned to lick-onset: Reward Number-DA association analyzed as within-session (Trial Number) and between-session (Session Number) linear effects.

The average reward signal shows the average trace with standard error of the mean indicated by the shaded region. The Trial Number Session Number effects are *FLMM* coefficient estimate plots.

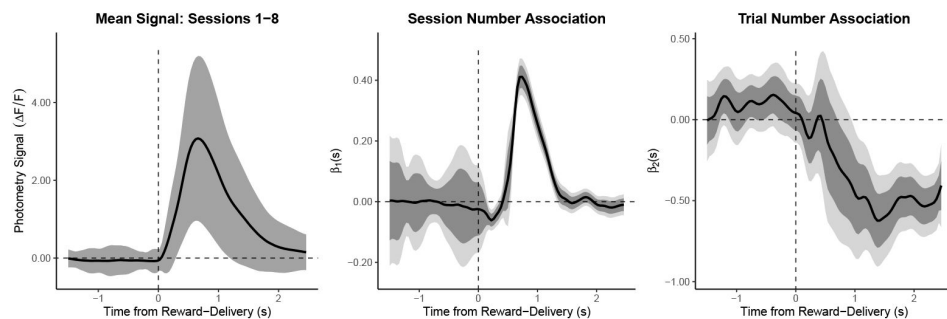


Figure 14.

Random IRI experiment aligned to reward-delivery: Reward Number-DA association analyzed as within-session (Trial Number) and between-session (Session Number) linear effects.

The average reward signal shows the average trace with standard error of the mean indicated by the shaded region. The Trial Number Session Number effects are *FLMM* coefficient estimate plots.

7.3.1 Photobleaching: Delay-length Experiment

We next sought to determine whether within-session decreases occurred in a different reward learning task collected from the same mice. We analyzed cue responses on the data from the delay-length experiments (presented in main text Section 2.4). We used data from the final short-delay session because the animals were well trained on the Pavlovian task at that stage. During personal communications, Jeong et al. (2022) [\[1\]](#) suggested these analyses because it provided an opportunity to analyze stabilized event-triggered DA responses in a different task from the same animals. **Figure 15** [\[1\]](#) shows the peak magnitude of the average signal at cue-onset (of the delay length data) is about 4.5 $\Delta F/F$ units (see the Intercept plot), which is about 15% *higher* than the peak magnitude of the average signal during the post-lick period of the experiment described above (5 shows that the peak magnitude of the average signal was about 4 $\Delta F/F$ units). Because the signal is higher and photobleaching is thought to exert a greater effect on larger signals, one would expect photobleaching to have a *larger* effect in this experiment (i.e., a more negative Trial Number effect). However, **Figure 15** [\[1\]](#) shows that the within-session signal *increases* significantly over trials during the cue period (i.e., a positive Trial Number effect). **Figure 15** [\[1\]](#) also shows that the signal decreases within-session across trials at reward-delivery (3s after cue-onset) despite exhibiting a substantially lower average signal than during the Cue Period. This echoes the within-session reductions observed around reward-consumption on the random IRI task above. We note additional analyses that adjusted for anticipatory licking and other indications of behavioral engagement did not noticeably impact the Trial Number effects. Analyses that pooled together multiple short-delay sessions yielded similar results. Taken together, these analyses of data from the same animals provide additional evidence against a photobleaching explanation as the main contributor for the within-session reduction described in **Figure 5** [\[1\]](#). One potential caveat is that there may be additional unmeasured factors (e.g., change in motivation) that increase across trials within a session, thereby occluding an effect of photobleaching. However, we reason that the parsimonious explanation in this case is that photobleaching is not a significant concern.

7.3.2 Photobleaching: Background Reward Experiment

Next we show results from analyses suggested during personal communications with Jeong et al. (2022) [\[1\]](#) to further rule out the possibility that the within-session decrease was a result of photobleaching. We specifically reanalyzed “Test 7” in Jeong et al. (2022) [\[1\]](#) (original analysis results shown in **Figure 4M** [\[1\]](#)-P of their manuscript). They described the experiment as follows:

“To test whether the significant positive dopamine responses following extinction reflect a retrospective association between the cue and reward, we selectively reduced the retrospective association without reducing the prospective association. We maintained the fixed reward following the cue but added unpredictable rewards during the intertrial interval. In this experiment, not all rewards are preceded by the cue (i.e., retrospective association is weak), but all cues are followed by reward (i.e., prospective association is high). ANCCR predicts a rapid drop in dopamine cue response whereas RPE predicts no change in cue response if TDRL only considers the cue-reward “trial period” (Test 7, fig. S10). The dopamine cue response remained significantly positive but decayed across trials faster than during extinction.”

This experiment was like long-delay sessions in the delay-length experiment (i.e., a CS+ followed by reward-delivery 9 sec later), but it also included rewards delivered randomly without any predictive stimuli at any point in the session (e.g., between the CS+ and the predicted-reward). These “background rewards” were similar to the reward-delivery schedule in the random IRI experiment presented in **Figure 5** [\[1\]](#) and therefore provide a critical point of comparison. Similar to our random IRI experiment analyses, we analyzed the same “reward period” time window aligned to the first lick after reward-delivery. We removed background reward “trials” that were

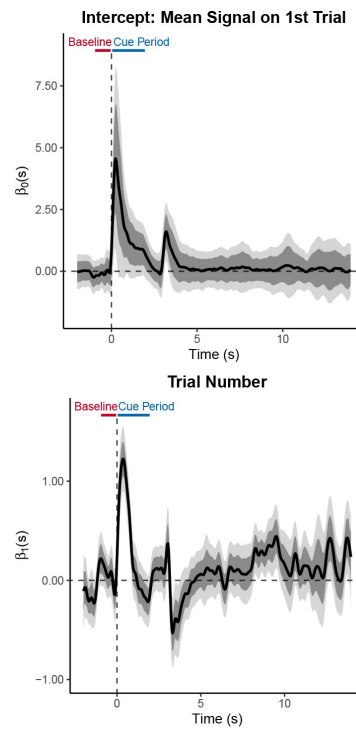


Figure 15.

***FLMM* identifies how the signal increases across trials during Cue Period and decreases across trials after reward-delivery (3 sec).**

too close to each other to avoid signal bleed-over from adjacent trials (e.g., trials with inter-reward intervals that were too short), and trials for which no licks occurred between two successive reward-deliveries (background or reward-predicted) to avoid “double-counting” a trial.

Figure 16 [↗](#) shows *FLMM* estimates a negative Trial Number effect. The magnitude of this negative coefficient is comparable to that seen in the random rewards experiments post-lick (i.e., after the first lick following reward-delivery). However, the within-session effect is most negative pre-lick, unlike what we observed in the random IRI task presented in **Figure 5** [↗](#). It seems unlikely that this would be solely explained through satiation since the animals cannot be consuming the reward in the pre-lick period. Finally, we note additional analyses that adjust for lick frequency and other indications of behavioral engagement did not impact the Trial Number effects. Thus, while these results might be expected if photobleaching was causing the within-session signal reductions shown in **Figure 5** [↗](#), we argue that the results suggest that it is unlikely that photobleaching is the sole contributor to these within-session reductions.

Appendix: Section 4.5.3 Reanalyses

We sought to evaluate the hypothesis described in the following paragraph in the main text of Jeong et al. (2022) [↗](#):

“Further, when a learned delay between cue onset and reward (3 seconds) is extended permanently to a new, longer delay (9 seconds), RPE predicts that as animals learn the longer delay and suppress anticipatory licking at the previous short delay, there will be a concomitant reduction in the dopamine cue response due to temporal discounting (46). By contrast, ANCCR predicts little to no change in the dopamine cue response as the structure of the task is largely unchanged (Test 4, **Fig. 4G** [↗](#) and figs. S9 and S10; intuitively, relative to the long intertrial interval, the cue-reward delay is still short). Experimentally, we observed that although animals learned the new delay rapidly, dopaminergic cue responses showed no significant change (**Fig. 4, G** [↗](#) to **I** [↗](#)). After the extension of the cue-reward delay, RPE predicts a suppression of dopamine after the old delay expires without reward. Because the increase in cue-reward delay is permanent [unlike in prior experiments (45)], ANCCR predicts that the delay representation in the internal causal model of the animal would be updated to reflect the new delay. This predicts no reward omission response at the old delay (3 seconds) after the increase in the delay (9 seconds). Thus, ANCCR predicts no negative omission response after the old delay expires without reward. (Test 5). Experimentally, we observed no suppression of dopamine response at 3 seconds in this experiment but did observe suppression in a separate experiment when the reward was indeed omitted (**Fig. 4J** [↗](#) and fig. S10).”

The authors described the results from their analyses in the **Figure 4** [↗](#) (main text) caption: “(H and I) Behavior is learned abruptly by all animals, but the dopaminergic cue response shows little to no change. The dashed vertical line is the trial at which the experimental condition transitions [in (H, K, and P)]. We tested for the lack of change by showing that the Akaike Information Criterion (AIC) is similar between a model assuming change and a model assuming no change. Paired t test for abruptness of change; $t(6) = 22.92$; two-tailed $P = 4.52 \times 10^{-7}$; one-sample t test for Δ AIC against a null of zero; $t(6) = 7.49$ for lick, -0.86 for dopamine; two-tailed $P = 2.9 \times 10^{-4}$ for lick, 0.4244 for dopamine ($n = 7$ animals). (J) The dopaminergic cue response of an example animal remains positive well after it learns extinction of the cue-reward association. (K to L) Across all animals, the dopaminergic cue response remains significantly positive despite abrupt behavioral learning of extinction [paired t test for abruptness of change; $t(6) = 5.67$; two-tailed $P = 0.0013$; paired t test for change trial; $t(6) = -2.40$; two-tailed $P = 0.0531$; $n = 7$ animals].”

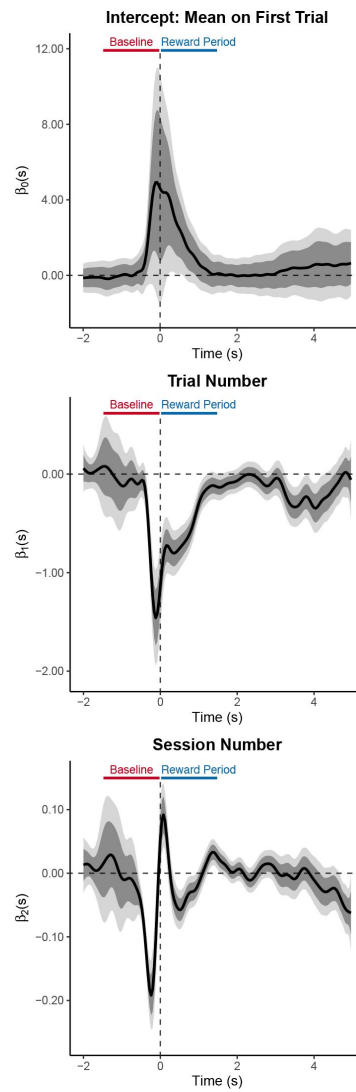


Figure 16.

Background Reward Experiment Analyses

7.5 Using *FLMM* to test signal changes within- and across-trials

We show how *FLMM* enables hypothesis testing of signal changes within- and across-trials. In the experiment in Section “Tests 9 to 11” of Jeong et al. (2022) [\[1\]](#), the authors test whether, across several sessions of Pavlovian learning, DA activity “backpropagates” from reward delivery (3s after cue-onset) to the presentation of reward-predictive cues. They analyzed a summary measure defined as the difference between the average signal during pre-reward-delivery (“Late”) and cue-onset (“Early”) time-windows.

We tested the “backpropagation” question with a *FLMM* model with session binary indicators as covariates, similar to a functional repeated-measures ANOVA. This yields estimates of mean signal changes, at each time-point, between pairs of sessions. We did not observe significant “Late” period changes, corroborating the authors’ findings (**Figure 17** [\[1\]](#); see Appendix Figure 18 for individual-animal fits). However, we identified an additional effect that would be hard to find with summaries: the average (peak) size of cue-elicited DA, exhibited later in training, is similar to the degree that reward-delivery DA decreased. This is evident by comparing the symmetry between the magnitudes of peak increases during the “Early” period and peak decreases during the post-“Late” period on the last session (**Figure 17D** [\[1\]](#)). This illustrates the capability in *FLMM* of directly testing effects visible in graphs, instead of having to perform hypothesis tests on a summary-of-summaries (e.g., a ratio or difference of AUCs).

7.5.1 Reanalysis Methods: Using *FLMM* to test signal changes within- and across-trials

We reanalyzed data presented in the section “Tests 9-11 (Backpropagation within a trial)” of (Jeong et al., 2022 [\[1\]](#)). We sought to evaluate the author’s hypothesis described in the following paragraph in the main text:

“[in the] TDRL RPE account (...) dopamine responses drive value learning of the immediately preceding state. We tested three predictions of this central postulate that are each inconsistent with ANCCR. The first is that during the acquisition of trace conditioning, dopamine response systematically backpropagates from the moment immediately prior to reward to the cue onset (50) (Test 9, **Fig. 6A** [\[1\]](#)). Unlike TDRL RPE, ANCCR does not make such a prediction since delay periods are not broken into states in ANCCR (...) Our observations were not consistent with a backpropagating bump of activity and were instead consistent with an increase in cue response over trials of learning (**Fig. 6B** [\[1\]](#))”

We fit a random slope model using indicator variables of the 1st, 2nd, 3rd, and subject-specific final sessions as covariates. We analyzed data from sessions 1-3, and the final session for each animal, as these were the sessions where we noticed the greatest changes. We discarded data from other sessions and thus the interpretation of the intercept is the mean signal on the first session. We specified the random slope model:

$$\mathbb{E}[\mathbf{Y}_{i,j,l}(s) \mid \mathbf{X}_{i,l}, \mathbf{Z}_{i,l}] = \beta_0(s) + \gamma_{0,i}(s) + \mathbb{1}(l = 2) [\beta_1(s) + \gamma_{1,i}(s)] \\ + \mathbb{1}(l = 3) [\beta_2(s) + \gamma_{2,i}(s)] + \mathbb{1}(l = S_i) [\beta_3(s) + \gamma_{3,i}(s)].$$

where l denotes the session number, S_i is subject i ’s final session (which can differ between animals), and $\mathbb{1}(l = s)$ is an indicator variable for session s .

As an example, we show the code from our package to fit the above model (note that the period variable below is a factor variable):

```
model_fit = fui(photometry ~ period + (period | id), data = photometry_data)
```

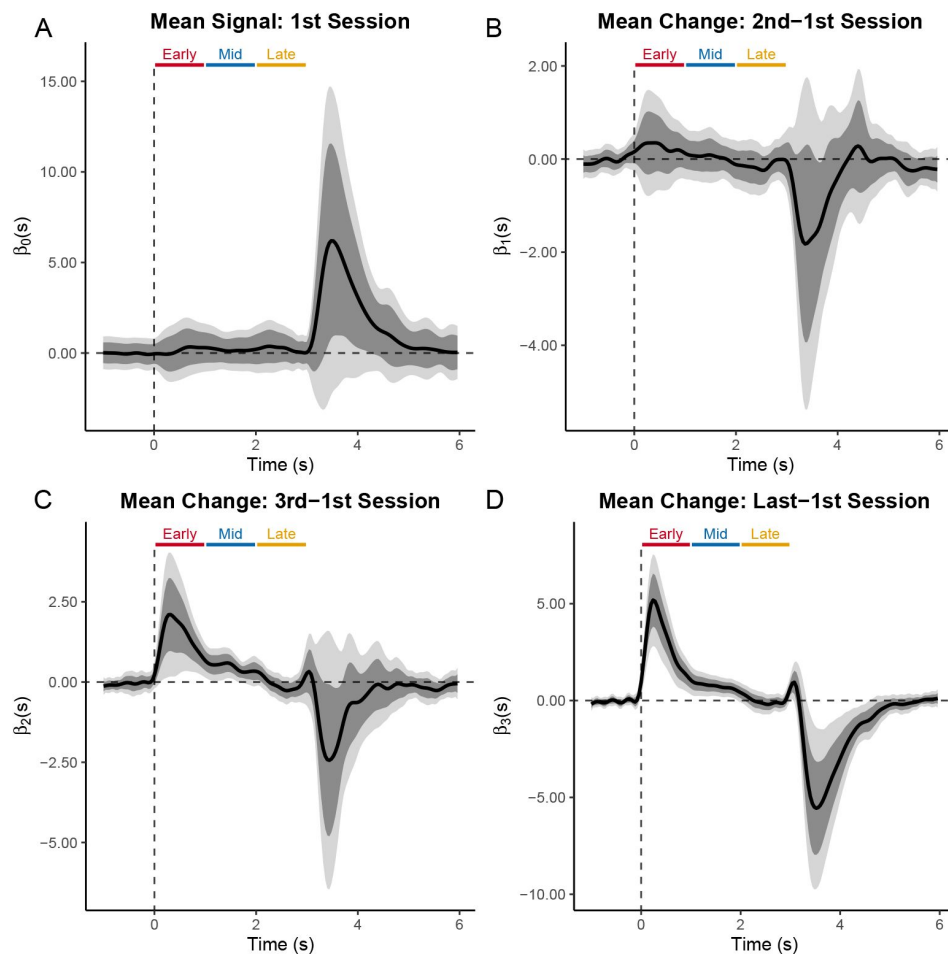


Figure 17.

FLMM identifies how the signal evolves across trial time-points, and how the temporal location of transients progresses across sessions in a statistically significant manner.

The panels show coefficient estimates from *FLMM* analyses of the “backpropagation” experiment. Panel (A) contains the intercept term plot corresponding to the average signal on the first session of training. The “Early” (0-1s), “Mid” (1-2s), and “Late” (2-3s) bars show the trial time-periods that the original authors used to calculate summary AUC measures. Trials are aligned to cue onset (cues lasted 2s) and rewards were delivered at 3s. Panels (B)-(D) show the coefficient estimates corresponding to the mean change in signal values from 2nd, 3rd or 4th sessions, respectively, compared to the first session (positive values indicate an increase from the first session). Plots are aligned to cue onset.

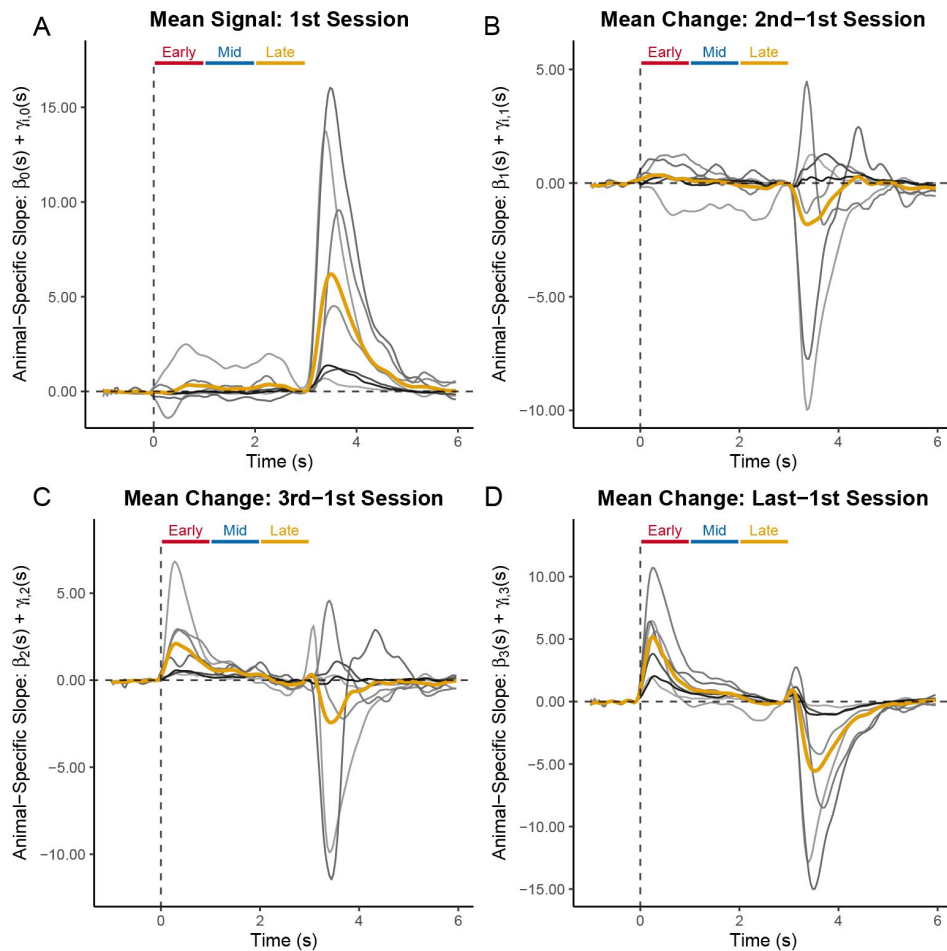


Figure 18.

Individual-level coefficient estimates from *FLMM* analyses of “backpropagation” experiment: gold lines indicate the fixed-effect estimates and grey lines indicate animal-specific functional random-effect estimates (Best Linear Unbiased Predictor). Panel (A) contains the intercept term plot where the title provides an interpretation: the average signal on the first session of training. The “Early” (0-1s), “Mid” (1-2s), and “Late” (2-3s) bars show the trial time-periods that the original authors used to calculate summary AUC measures. Trials are aligned to cue onset and rewards were delivered at 3s. Panels (B)–(D) show the coefficient estimates which are interpreted as the change in mean signal from 2nd, 3rd or 4th sessions, respectively, compared to the first session (positive values indicate an increase from the first session). Plots are aligned to cue onset.

8 Appendix Simulations

8.1 Appendix: Simulation Scheme

As described in the main text, we simulated data from the model,

$$Y_{i,j}(s) = \beta_0(s) + \gamma_{0,i}(s) + \text{Delay}_{i,j} [\beta_1(s) + \gamma_{1,i}(s)] + \epsilon_{i,j}(s), \quad (12)$$

where $\epsilon_{i,j}(s)$ is element-wise for all $s = 1, 2, S$ and $i = 1, 2, \dots, n$. $\epsilon_i \sim \mathcal{N}_{n_i}(\mathbf{0}, \hat{\Sigma}_i)$ where

$\hat{\Sigma}_i = \widehat{\text{Cov}}(\hat{\epsilon}_i)$ and $\hat{\epsilon}_i$ are the residuals from the model 12 fit to the data. We draw

$\gamma_{i,k} \sim \mathcal{N}_S(\mathbf{0}, \hat{\Sigma}_\gamma^{(k)})$ where $\hat{\Sigma}_\gamma^{(k)}$ is the $S \times S$ covariance matrix for the random-effects k , where

element (l, t) of the matrix contains $(\gamma_{i,k}(l), \gamma_{i,k}(t))$. $\gamma_{i,k}(l)\gamma_{i,k}(t)$ for $l, t = 1, 2, S, i = 1, 2, n$ and kk' .

Thus, the $S \times 1$ vector of random-effects k are drawn as a Gaussian Process. The covariance matrix is estimated from the model using the corresponding elements of $G(l, t)$ which we describe in the Appendix section 6.2. We take the simulated $\beta(s)$ in 4.7 to be equal to the estimated coefficients from model 12 fit to the real data in (Jeong et al., 2022).

To “simulate” the covariates (i.e., just the Delay indicator vector), we randomly draw a subset of animal IDs and concatenate all of their *observed* covariates from the (Jeong et al., 2022). That is for each simulation replicate, r , of sample size $n^* \in \{4, 5, n\}$ we randomly draw a sample of n^* animal subject IDs, denoted as N_r from the set $\{1, 2, n\}$ with uniform probability without replacement. For each animal ID in the sample, N_r , we concatenate the covariates (i.e., the design matrix) of the corresponding subjects in the *observed* data to be the “simulated” covariates. That is, the design matrix $\mathbb{X}_r$ for simulation replicate r , is the row concatenation of each $\mathbb{X}_i$ for $i \in N_r$.

8.2 Appendix: Additional Simulation Results

To explore how analyzing summary measures can drown-out effects, we compared method performances on the same analyses across different cue period lengths (2s, 2.5s, or 3s from cue onset), which we visualize in Figure 19A. These relatively small adjustments substantially influenced estimation performance (see Figure 19B), and statistical power (see Figures 19D and 20). The pointwise 95% CI coverage (see Figure 19C) of the summary measure approaches (t-test and LMM) is also very sensitive to the specified length of the cue period because they analyze a summary that averages over a time-window that contains heterogeneous effects. However, the *FLMM* and *Perm* CI coverages (averaged over the cue period) remain stable to differing time-window lengths, because they evaluate each time-point. Notably, *Perm* exhibits low coverage at smaller sample sizes and the t-test yields poor coverage in all settings tested, likely because of the animal-to-animal variability in the signal magnitude and delay-length change effect. Finally, Appendix Figure 21 shows *FLMM* fits in around 10 seconds for datasets with about 800 trials (pooled across animals). Taken together, these data-driven simulations demonstrate that at low sample sizes, and in the presence of individual-differences, the *FLMM* consistently 1) achieves the nominal *pointwise* and *joint* coverage, 2) improves statistical power, and 3) can be fit quickly.

Here we define the estimation error (RMSE) of statistical methods for the average difference in photometry signal amplitude during the cue period as a function of how long that cue period was (2s, 2.5s, 3s). The error was defined as $\frac{1}{\sqrt{R}} \left\| \bar{\beta}_1(s) - \hat{\beta}_1(s) \right\|_2$ where $\bar{\beta}_1(s) = \frac{1}{|S|} \sum_{s \in S} \beta_1(s)$. That is, the average of the coefficients for covariate k across time-points s in a fixed interval (indexed by S). Since the model only contains a single binary covariate, we can compare the average of functional coefficients for the slope parameter in a *FLMM* model with the slope coefficient

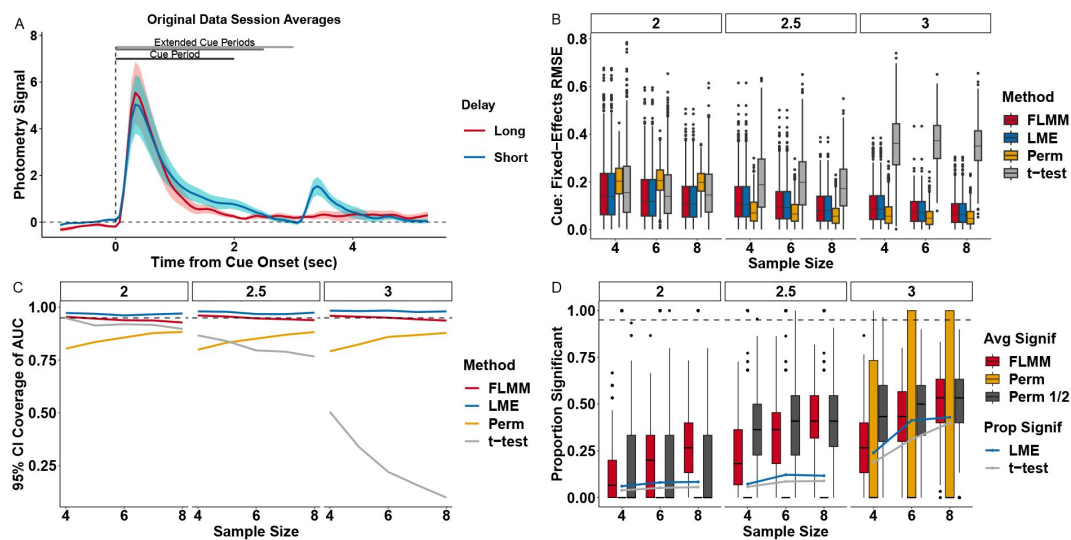


Figure 19.

Summary measure analyses are highly sensitive to minor changes in the summary time-window. (A) Average Short/Long delay data. Bars show cue period length used in (2s) and “extended” delays analyzed in additional simulations. (B) Estimation error (RMSE), $\frac{1}{\sqrt{R}} \|\hat{\beta}_1(s) - \tilde{\beta}_1(s)\|_2$ where $\tilde{\beta}_1(s) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \beta_k(s)$, associated with mean difference during the cue periods (panels) and n on the x-axis. Lower numbers indicate more accurate estimates. (C) pointwise 95% CI coverage associated with mean difference during cue period (panels) and n on the x-axis. Higher values indicate better CI coverage. (D) Statistical power during cue period. The LMM and t-test were fit on the signal averaged over the cue period and thus each simulation replicate yields a single indicator of CI inclusion or statistical significance, which we represent with a line plot. For other methods, estimates are provided at each time-point and performance is averaged across the time-points. We summarize these simulation replicate-specific averages with a boxplot.

estimate in a scalar LMM applied on the outcome: $\bar{Y}_{i,j} = \frac{1}{|S|} \sum_{s \in S} Y_{i,j}(s)$ and a paired-samples t-test using the outcome: $\bar{Y}_i = \frac{1}{|J_i|} \frac{1}{|S|} \sum_{s \in S} \sum_{j \in J_i} Y_{i,j}(s)$ where J_i is the set of trials observed for subject i .

9 Appendix Additional Reanalyses

We include analyses here conducted on a second recent article [Coddington et al. \(2023\)](#); [Dudman \(2023\)](#) proposing a new reinforcement learning model for the role of mesolimbic dopamine in learning.

9.1 Appendix: Additional Reanalyses Results

9.1.1 Functional methods allow for testing how signal “dynamics” early in training predict behavior later in training

We next analyze data from a second paper focused on the role of mesolimbic DA in reward learning ([Coddington et al., 2023](#)). We first examine how between-animal differences in behavior correlate with average nucleus accumbens dopamine neuron calcium changes (NAc-DA)⁹. In this experiment, mice were exposed to a 0.5s stimulus, followed by reward 1s after cue-offset. The authors identified significant correlations between average Reward period NAc-DA (trial-averaged on *early* training sessions) with measures of average behavior (trial-averaged on *late* sessions).

We fit two analogous univariate models with the average behavioral measures as the covariate.¹⁰

Our results reveal details missed in the summary measure analyses. Consistent with the author’s findings, we found Reward period NAc-DA was associated with Final Latency. However, our analyses also revealed significant associations during the Cue period, an effect hard to see on average traces: while the average signal increases substantially during the Reward period, it exhibits comparatively little change during the Cue period. This demonstrates the difficulty of constructing summary measures: the time-windows when the signal is associated with covariates may not align with time-periods when the average signal exhibits noticeable changes. Remarkably, the association sign differs from the direction of average signal change: Final Latency is *positively* associated with NAc-DA during both Cue and Reward periods, yet the mean signal *decreases* during the Cue Period and *increases* during the Reward period.

9.1.2 FLMM provides test of how signal differences between trial-types change across training

Our final example demonstrates how questions that might otherwise require analysis of summaries of summary measures (e.g., ratios of average AUCs), can be precisely specified in a functional model so as to provide greater detail. Specifically, the authors sought to investigate how the difference in NAc-DA signals between trials with (Lick+) and without (Lick-) preparatory licking changed with learning. They reported a significant correlation between reward collection latency (Final Latency) and a “ratio of NAc-DA reward signals [AUCs] on lick-vs lick+ trials.” We fit an FLMM model with an interaction between Final Latency and Lick State (a Lick-/Lick+ indicator). The interaction term provides a hypothesis test of their question at *each trial time-point*. The longest portion of joint significance occurs *between* Cue and Reward periods. Our analysis confirms the authors’ results and adds detail obscured by standard methods: the interaction does not reach joint statistical significance until a couple seconds after reward-delivery, well-beyond the time when the average signal has fallen from its peak (shown by the intercept). This example highlights the challenge with constructing an adequate summary measure for this analysis: the

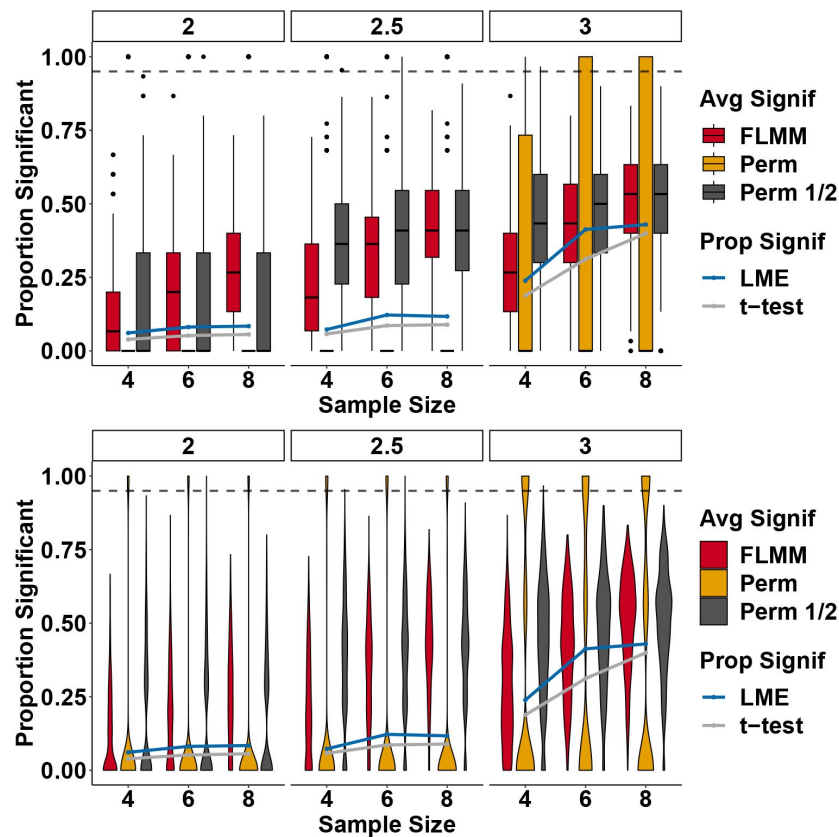


Figure 20.

Statistical power associated with mean difference defining the cue period as 2s, 2.5s, and 3s (panels) and sample sizes (numbers of animals) on the x-axis. The two panels presents the same data in either violin or boxplot forms. Higher numbers indicate better power. For *FLMM* and *Perm*, power is averaged across the time-points in the cue period whereas the others assess the power using the average signal (across the cue period) as the outcome. Since each simulation replicate takes the proportion of significant time-points in the cue period for *FLMM* and *Perm*, these are presented as boxplots (or violin plots), whereas the rest are simply presented as the proportion of simulation replicates that identified the mean signal during the cue as statistically significant (either 0 or 1 for each replicate).

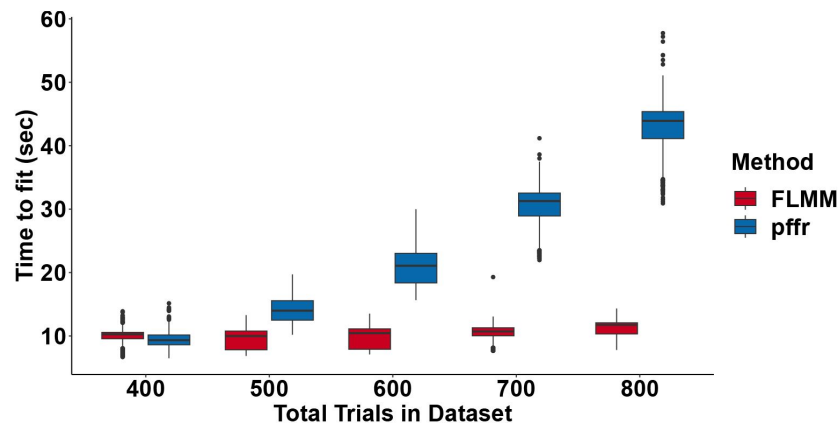


Figure 21.

Time to fit *FLMM* fit with our software (using a closed-form variance calculation) on simulated data (each datapoint represents one replicate). *pffr* shows the time to fit the functional linear mixed model (with the same model specification) with the *pffr()* function in the *refund* package. Number of animals in simulations shown in plots ranges from 4-8 (i.e., $n = \text{trials}/100$).

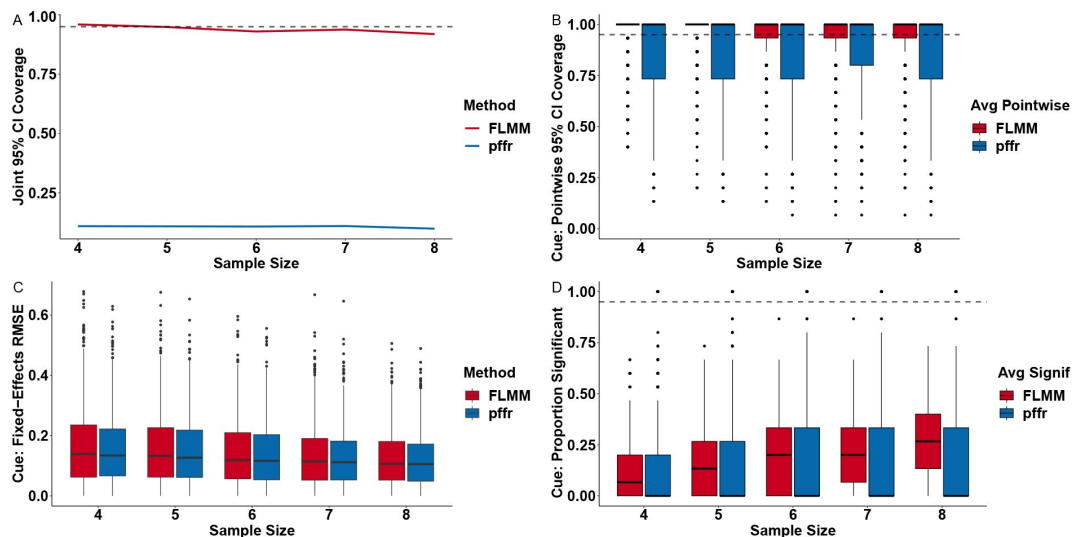


Figure 22.

FLMM fit with our software achieves comparable or superior performance to the functional linear mixed model (with the same model specification) fit with the *pffr()* function in the *refund* package. (A) *FLMM* achieves *joint* 95% CI coverage at roughly the nominal level. *pffr* does not provide *joint* 95% CIs and thus the *pointwise* 95% CIs that it does provide achieve low *joint* coverage. (B) *FLMM* achieves *pointwise* 95% CI coverage at or above the nominal level. The *pointwise* 95% CI coverage of *pffr* is close to but below the nominal level. *pointwise* 95% CI coverage associated with mean difference during cue period (panels) and n on the x-axis. Higher values indicate better CI coverage. (C) *FLMM* and *pffr* exhibit comparable fixed-effects estimation performance. Estimation error (RMSE), $\frac{1}{\sqrt{R}} \|\hat{\beta}_1(s) - \tilde{\beta}_1(s)\|_2$ where $\tilde{\beta}_1(s) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \beta_k(s)$, associated with mean difference during the cue periods (panels) and n on the x-axis. Lower numbers indicate more accurate estimates. (D) *FLMM* exhibits superior statistical power compared to *pffr* during the cue period. (B)-(D) Since estimates are provided at each time-point for both methods, pointwise performance is averaged across the time-points. We summarize these simulation replicate-specific averages as one point in a boxplot.

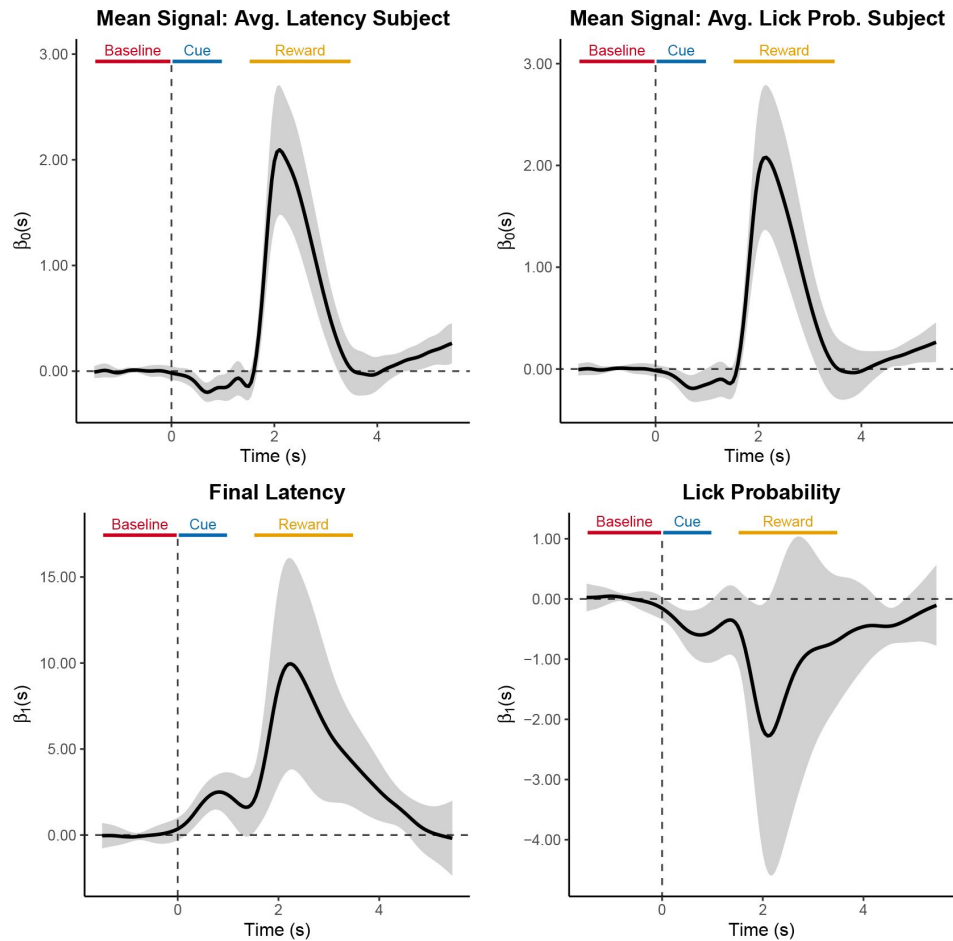


Figure 23.

Coefficient estimates from *FLMM* analyses of Final Latency and Lick Probability models. The top row contains intercept term plots where the title provides interpretation of the intercept: the average NAc-DA signal on trials when the covariates in the model are at their average value. The bottom row shows the coefficient estimate plot of the covariate in the model. (Left) Association between average NAc-DA (averaged over first 100 trials) and latency to lick Final Latency (averaged over trials 700-800). (Right) Association between average NAc-DA (averaged over first 100 trials) lick probability (averaged over trials 700-800).

period of clearest effect of the Lick State is before the average signal is largest. This may explain why the authors did not construct a summary measure during the interval between the Cue and Reward periods, when the Lick State effect was strongest. In sum, this example demonstrates how summary-of-summary measure analyses can be translated into simple *FLMM* regressions that provide greater detail about the time-course and magnitude of the effects throughout the trial.

9.2 Appendix: Additional Reanalyses Methods

9.2.1 Methods for Appendix Section 9.1.1

This paper measured photometry signals in which DAT-Cre::*ai32* transgenic mice were injected with a Cre-dependent jRCaMP1b virus across the ventral midbrain enabling the measurement of calcium dynamics in mesolimbic DA cells specifically. We sought to conduct an analysis based upon the following quote:

“Unexpectedly, initial NAc–DA reward signals were negatively correlated with the amount of preparatory behaviour at the end of training (NAc–DA reward trials 1-100 versus preparatory index trials 700-800, $r=-0.85$, $P=0.004$), as well as the speed of reward collection (NAc–DA reward trials 1-100 versus reward collection latency trials 700-800, $r=0.81$, $P=0.008$).”

Preparatory licking and latency-to-lick were used as indicators of learning.

Since there is no way to meaningfully pair the neural activity of one trial with behavior on a separate trial, analyzing trial-level data is not appropriate and we therefore modeled the *average* photometry signal (averaged across trials 1-100) as a function of *average* behavior (averaged across trials 700-800). We removed trials with latencies over 1 second as they constituted behavioral outliers comprising less than 1% of trials.

To conduct a functional regression most analogous to the Pearson correlations the authors conducted, we fit the univariate linear regression models for final latency, FL_i ,

$$\mathbb{E}[\bar{Y}_i(s) | \mathbf{X}_i] = \beta_0(s) + FL_i \beta_1(s).$$

Similarly, the we fit the lick probability, LP_i , functional regression model,

$$\mathbb{E}[\bar{Y}_i(s) | \mathbf{X}_i] = \beta_0(s) + LP_i \beta_1(s).$$

These models were fit with the R package *refund* with the `fosr()` function (where *fosr* abbreviates function-on-scalar regression used for a functional outcome and scalar covariate). Note that this package function only provides *pointwise* 95% CIs and thus the plots in the main text only have one shade of grey for the 95% CIs. Thus the interpretation for this analysis is confined to a *pointwise* interpretation.

9.2.2 Methods for Appendix Section 9.1.2

We sought to conduct an analysis based upon the following quote: “[the proposed] scheme also predicts that [mesolimbic DA] reward signals should reflect the evolution of reward collection policy across learning...Indeed,...mouse data exhibited differential reward responses on trials with (‘lick+’) or without (‘lick-’) preparatory licking as learning progressed” (Coddington et al., 2023 [C](#)). The authors reported (see the **Figure 8d** [C](#) caption (Coddington et al., 2023 [C](#))) a significant Pearson correlation between the final reward collection latency (Final Latency) and a “ratio of NAc–DA reward signals on lick-vs lick+ trials.” They constructed this measure by first summarizing the reward-period NAc–DA (with AUC), averaged across Lick+ and Lick-trials separately, and then calculating a ratio for each animal. Similar to the analysis we presented in the

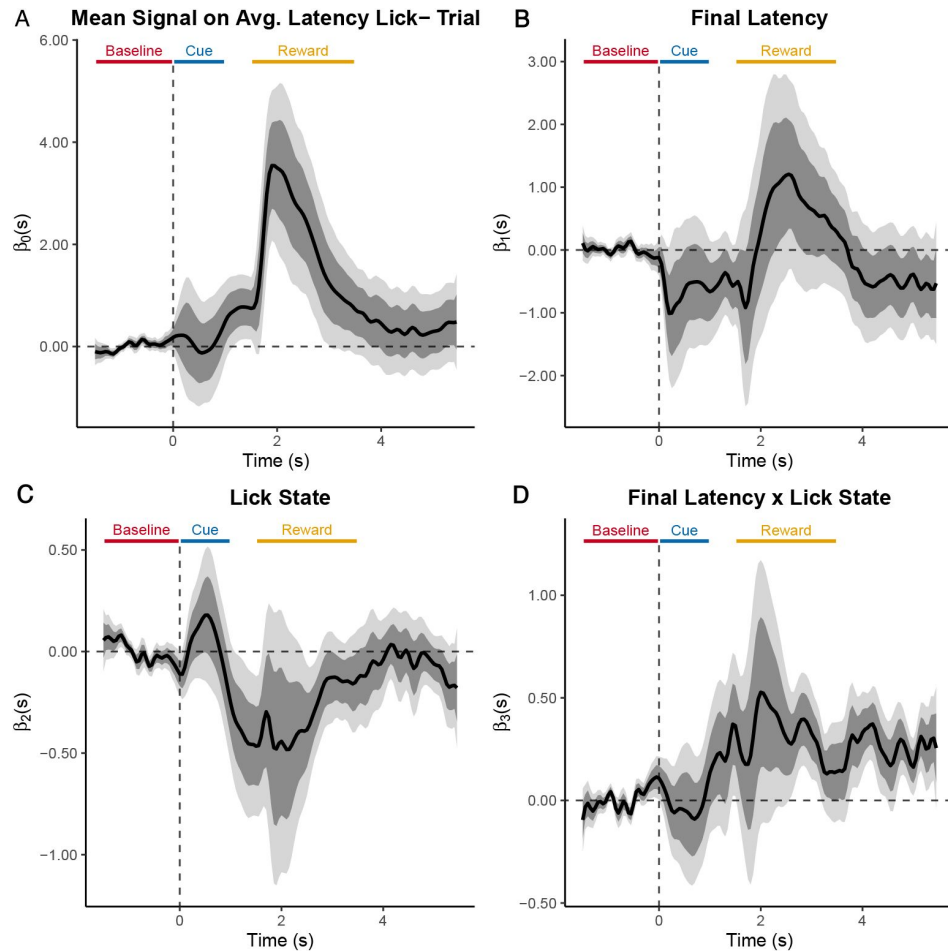


Figure 24.

Coefficient estimates from a single *FLMM* analysis of Final Latency $\times$ Lick State interaction model (including main effects). A simple *FLMM* model enables characterization of how the association between photometry signals and behavioral responding (Final Latency) differs between conditions (Lick+/Lick-) at each time-point in the trial. Panel (A) contains the intercept term plot where the title provides an interpretation: the average photometry signal on Lick-trials for animals that exhibit average Final Latency values. Panels (B)-(D) show the three covariates of main effects and interaction of Lick State and Final Latency. The Final Latency functional coefficient is interpreted as the effect of Final Latency on Lick-trials. The Lick State main effect is interpreted as the difference in average NAc-DA between Lick+ and Lick-trials for an animal with an average Final Latency value. The interaction is interpreted as a difference in differences: during the Reward period, a 1 standard deviation increase in average Final Latency is associated with pointwise significantly higher NAc-DA signals on Lick+ than on Lick-trials (with a portion of joint significance) during most of the Reward period.

previous section, the authors compared behavior on one set of trials to dopamine activity in a separate set of trials. Because there is no way to meaningfully pair the neural activity of one trial with behavior on a separate trial, we analyzed the *average* photometry signal (Lick+/Lick- separately averaged over the trials specified in the paper) as a function of *average* behavior. Specifically, we modeled the trial-averaged signal denoted as $\bar{Y}_{i,v}(s)$ for animal i and lick state v . The signal was trial-averaged separately across lick+ and lick-trials. We both indicate lick state with the subscript v on $\bar{Y}_{i,v}(s)$ and also as a covariate in the model where $LS_{i,v} = 1$ for lick+ trials and $LS_{i,v} = 0$ for lick-trials. We denote the final latency below as FL_i . We fit the random slope *FLMM* model,

$$\mathbb{E}[\bar{Y}_{i,v}(s) | \mathbf{X}_{i,v}, \mathbf{Z}_i] = \beta_0(s) + FL_i [\beta_1(s) + \gamma_{1,i}(s)] + LS_{i,v}\beta_2(s) + FL_i \times LS_{i,v}\beta_3(s).$$

Given that the outcome was trial-averaged, the number of observations of the out-come was only twice the number of animals (i.e., one observation of the functional outcome for lick+ and one for lick-). We were thus limited in the number of random-effects we could include and still retain an identifiable model. For that reason, our model did not include a random intercept. As for all other analyses in our manuscript, we compared multiple candidate models that we specified based upon the authors' analyses. We then selected the model with the best AIC/BIC model fit criteria.

We normalized the (average) Final Latency variable (across trials) to have mean 0 and unit variance. The intercept therefore shows that mean NAc-DA activity for a mouse with average Final Latency on Lick-trials is *jointly* significantly elevated during portions of the Reward period. The Final Latency functional coefficient is interpreted as the effect of Final Latency on Lick-trials and is only briefly pointwise significant in the middle of the Reward period. The Lick State main effect is interpreted as the difference in average NAc-DA between Lick+ and Lick-trials for an animal with an average Final Latency value. The interaction is interpreted as a difference in differences: during the Reward period, a 1 standard deviation increase in average Final Latency is associated with pointwise significantly higher NAc-DA signals on Lick+ than on Lick-trials (with a portion of joint significance) during most of the Reward period.

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Reviewer #1 (Public Review):

Summary:

Fiber photometry has become a very popular tool in recording neuronal activity in freely behaving animals. Despite the number of papers published with the method, as the authors rightly note, there are currently no standardized ways to analyze the data produced. Moreover, most of the data analyses confine to simple measurements of averaged activity and by doing so, erase valuable information encoded in the data. The authors offer an approach based on functional linear mixed modeling, where beyond changes in overall activity various functions of the data can also be analyzed. More in-depth analysis, more variables taken into account, and better statistical power all lead to higher quality science.

Strengths:

The framework the authors present is solid and well-explained. By reanalyzing formerly published data, the authors also further increase the significance of the proposed tool opening new avenues for reinterpreting already collected data.

Weaknesses:

However, this also leads to several questions. The normalization method employed for raw fiber photometry data is different from lab to lab. This imposes a significant challenge to applying a single tool of analysis. Does the method that the authors propose work similarly efficiently whether the data are normalized in a running average dF/F as it is described in the cited papers? For example, trace smoothing using running averages (Jeong et al. 2022) in itself may lead to pattern dilution. The same question applies if the z-score is calculated based on various responses or even baselines. How reliable the method is if the data are non-stationary and the baselines undergo major changes between separate trials?

Finally, what is the rationale for not using non-linear analysis methods? Following the paper's logic, non-linear analysis can capture more information that is diluted by linear methods.

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Reviewer #2 (Public Review):

Summary:

This work describes a statistical framework that combines functional linear mixed modeling with joint 95% confidence intervals, which improves statistical power and provides less conservative statistical inferences than in previous studies. As recently reviewed by Simpson

et al. (2023), linear regression analysis has been used extensively to analyze time series signals from a wide range of neuroscience recording techniques, with recent studies applying them to photometry data. The novelty of this study lies in 1) the introduction of joint 95% confidence intervals for statistical testing of functional mixed models with nested random-effects, and 2) providing an open-source R package implementing this framework. This study also highlights how summary statistics as opposed to trial-by-trial analysis can obscure or even change the direction of statistical results by reanalyzing two other studies.

Strengths:

The open-source package in R using a similar syntax as the lme4 package for the implementation of this framework on photometry data enhances the accessibility, and usage by other researchers. Moreover, the decreased fitting time of the model in comparison with a similar package on simulated data, has the potential to be more easily adopted.

The reanalysis of two studies using summary statistics on photometry data (Jeong et al., 2022; Coddington et al., 2023) highlights how trial-by-trial analysis at each time-point on the trial can reveal information obscured by averaging across trials. Furthermore, this work also exemplifies how session and subject variability can lead to opposite conclusions when not considered.

Weaknesses:

Although this work has reanalyzed previous work that used summary statistics, it does not compare with other studies that use trial-by-trial photometry data across time-points in a trial.

As described by the authors, fitting pointwise linear mixed models and performing t-test and Benjamini-Hochberg correction as performed in Lee et al. (2019) has some caveats. Using joint confidence intervals has the potential to improve statistical robustness, however, this is not directly shown with temporal data in this work. Furthermore, it is unclear how FLMM differs from the pointwise linear mixed modeling used in this work.

In this work, FLMM usages included only one or two covariates. However, in complex behavioral experiments, where variables are correlated, more than two may be needed (see Simpson et al. (2023), Engelhard et al. (2019); Blanco-Pozo et al. (2024)). It is not clear from this work, how feasible computationally would be to fit such complex models, which would also include more complex random effects.

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Reviewer #3 (Public Review):

Summary:

Loewinger et al., extend a previously described framework (Cui et al., 2021) to provide new methods for statistical analysis of fiber photometry data. The methodology combines functional regression with linear mixed models, allowing inference on complex study designs that are common in photometry studies. To demonstrate its utility, they reanalyze datasets from two recent fiber photometry studies into mesolimbic dopamine. Then, through simulation, they demonstrate the superiority of their approach compared to other common methods.

Strengths:

The statistical framework described provides a powerful way to analyze photometry data and potentially other similar signals. The provided package makes this methodology easy to implement and the extensively worked examples of reanalysis provide a useful guide to others on how to correctly specify models.

Modeling the entire trial (function regression) removes the need to choose appropriate summary statistics, removing the opportunity to introduce bias, for example in searching for optimal windows in which to calculate the AUC. This is demonstrated in the re-analysis of Jeong et al., 2022, in which the AUC measures presented masked important details about how the photometry signal was changing.

Meanwhile, using linear mixed methods allows for the estimation of random effects, which are an important consideration given the repeated-measures design of most photometry studies.

Weaknesses:

While the availability of the software package (fastFMM), the provided code, and worked examples used in the paper are undoubtedly helpful to those wanting to use these methods, some concepts could be explained more thoroughly for a general neuroscience audience.

While the methodology is sound and the discussion of its benefits is good, the interpretation and discussion of the re-analyzed results are poor:

In section 2.3, the authors use FLMM to identify an instance of Simpson's Paradox in the analysis of Jeong et al. (2022). While this phenomenon is evident in the original authors' metrics (replotted in Figure 5A), FLMM provides a convenient method to identify these effects while illustrating the deficiencies of the original authors' approach of concatenating a different number of sessions for each animal and ignoring potential within-session effects. The discussion of this result is muddled. Having identified the paradox, there is some appropriate speculation as to what is causing these opposing effects, particularly the decrease in sessions. In the discussion and appendices, the authors identify (1) changes in satiation/habitation/motivation, (2) the predictability of the rewards (presumably by the click of a solenoid valve) and (3) photobleaching as potential explanations of the decrease within days. Having identified these effects, but without strong evidence to rule all three out, the discussion of whether RPE or ANCCR matches these results is probably moot. In particular, the hypotheses developed by Jeong et al., were for a random (unpredictable) rewards experiment, whereas the evidence points to the rewards being sometimes predictable. The learning of that predictability (e.g. over sessions) and variation in predictability (e.g. by attention level to sounds of each mouse) significantly complicate the analysis. The FLMM analysis reveals the complexity of analyzing what is apparently a straightforward task design. If this paper is not trying to arbitrate between RPE and ANCCR, as stated in the text, the post hoc reasoning of the authors of Jeong et al 2022 provided in the discussion is not germane. Arbitrating between the models likely requires new experimental designs (removing the sound of the solenoid, satiety controls) or more complex models (e.g. with session effects, measures of predictability) that address the identified issues.

Of the three potential causes of within-session decreases, the photobleaching arguments advanced in the discussion and expanded greatly in the appendices are not convincing. The data being modeled is a processed signal ($\Delta F/F$) with smoothing and baseline correction and this does not seem to have been considered in the argument. Furthermore, the photometry readout is also a convolution of the actual concentration changes over time, influenced by the on-off kinetics of the sensor, which makes the interpretation of timing effects of photobleaching less obvious than presented here and more complex than the dyes considered in the cited reference used as a foundation for this line of reasoning.

Within this discussion of photobleaching, the characterization of the background reward experiments used in part to consider photobleaching (appendix 7.3.2) is incorrect. In this experiment (Jeong et al., 2022), background rewards were only delivered in the inter-trial-interval (i.e. not between the CS+ and predicted reward as stated in the text). Both in the authors' description and in the data, there is a 6s before cue onset where rewards are not

delivered and while not described in the text, the data suggests there is a period after a predicted reward when background rewards are not delivered. This complicates the comparison of this data to the random reward experiment.

The discussion of the lack of evidence for backpropagation, taken as evidence for ANCCR over RPE, is also weak. A more useful exercise than comparing FLMM to the methods and data of Jeong et al., 2022, would be to compare against the approach of Amo et al., 2022, which identifies backpropagation (data publicly available: DOI: 10.5061/dryad.hhmgqknkjw). The replication of a positive result would be more convincing of the sensitivity of the methodology than the replication of a negative result, which could be a result of many factors in the experimental design. Given that the Amo et al. analysis relies on identifying systematic changes in the timing of a signal over time, this would be particularly useful in understanding if the smoothing steps in FLMM obscure such changes.

<https://doi.org/10.7554/eLife.95802.1.sa0>

Author Response

This important work presents a new methodology for the statistical analysis of fiber photometry data, improving statistical power while avoiding the bias inherent in the choices that are necessarily made when summarizing photometry data. The reanalysis of two recent photometry data sets, the simulations, and the mathematical detail provide convincing evidence for the utility of the method and the main conclusions, however, the discussion of the re-analyzed data is incomplete and would be improved by a deeper consideration of the limitations of the original data. In addition, consideration of other data sets and photometry methodologies including non-linear analysis tools, as well as a discussion of the importance of the data normalization are needed.

Thank you for the thorough and positive review of our work! We will incorporate this feedback to strengthen the manuscript. Specifically, we plan to revise the Discussion section to include a deeper consideration of the limitations of the original data, a description of the capacities of our method for conducting non-linear analyses, and the role data normalization plays in applicability of our tool.

Reviewer 1:

Strengths:

The framework the authors present is solid and well-explained. By reanalyzing formerly published data, the authors also further increase the significance of the proposed tool opening new avenues for reinterpreting already collected data.

Weaknesses:

However, this also leads to several questions. The normalization method employed for raw fiber photometry data is different from lab to lab. This imposes a significant challenge to applying a single tool of analysis.

Thank you for the positive feedback, we will address your comments in our revision. We agree that any data pre-processing steps will have down-stream consequences on the statistical inference from our method. Note, though, that this would also be the case with standard analysis approaches (e.g., t-tests, correlations) applied to summary measures like AUCs. For that reason, we do not believe that variability in pre-processing is an impediment to widespread adoption of a standard analysis procedure. Rather, we argue that the sensitivity of analysis results to pre-processing choices underscores the need for establishing

statistical techniques that reduce the need for pre-processing, and properly account for structure in the data arising from experimental designs. The reviewer brings up an excellent point that we can further elaborate on how our methods actually reduce the need for such pre-processing steps. Indeed, our method provides smooth estimation results along the functional domain (i.e., across trial timepoints), has the ability to adjust for between-trial and -animal heterogeneity, and provides a valid statistical inference framework that quantifies the resulting uncertainty. For example, adjustment for session-to-session variability in signal magnitudes or dynamics could be accounted for, at least in part, through the inclusion of session-level random effects. This heterogeneity would then influence the width of the confidence intervals. This stands in contrast to “sweeping it under the rug” with a pre-processing step that may have an unknown impact on the final statistical inferences. Similarly, the level of smoothing is at least in part selected as a function of the data, and again is accounted for directly in the equations used to construct confidence intervals. In sum, our method provides both a tool to account for challenges in the data, and a systematic framework to quantify the additional uncertainty that accompanies accounting for those data characteristics.

Does the method that the authors propose work similarly efficiently whether the data are normalized in a running average dF/F as it is described in the cited papers? For example, trace smoothing using running averages (Jeong et al. 2022) in itself may lead to pattern dilution. The same question applies if the z-score is calculated based on various responses or even baselines.

This is an important question given how common this practice is in the field. Briefly, application of pre-processing steps will change the interpretation of the results from our analysis method. For example, if one subtracts off a pre-trial baseline average from each trial timepoint, then the “definition of 0”, and the interpretation of coefficients and their statistical significance, changes. Similarly, if one scales the signal (e.g., divides the signal magnitude by a trial- or animal-specific baseline), then this changes the interpretation of the FLMM regression coefficients to be in terms of an animal-specific signal unit as opposed to a raw dF/F . This is, however, not specific to our technique, and pre-processing would have a similar influence on, for example, linear regression (and thus t-tests, ANOVAs and Pearson correlations) applied to summary measures. We agree with the reviewer that explicitly discussing this point will strengthen the paper.

While it is difficult to make general claims about the anticipated performance of the method under all the potential pre-processing steps taken in the field, we believe that most common pre-processing strategies will not negatively influence the method’s performance or validity; they would, instead, change the interpretation of the results. We are releasing a series of vignettes to guide analysts through using our method and, to address your comment, we will add a section on interpretation after pre-processing.

How reliable the method is if the data are non-stationary and the baselines undergo major changes between separate trials?

This is an excellent question. We believe the statistical inferences will be valid and will properly quantify the uncertainty from non-stationarities, since our framework does not impose stationarity assumptions on the underlying process. It is worth mentioning that non-stationarity and high trial-to-trial variability may increase variance estimates if the model does not include a rich enough set of covariates to capture the source of the heterogeneity across trial baselines. However, this is a feature of our framework, rather than a bug, as it properly conveys to the analyst that high unaccounted for variability in the signal may result in high model uncertainty. Finally, mixed effects modeling provides a transparent, statistically reasonable, and flexible approach to account for between-session, and between-

trial variability, a type of non-stationarity. We agree with the reviewer that this should be more explicitly discussed in the paper, and will do so.

Finally, what is the rationale for not using non-linear analysis methods? Following the paper's logic, non-linear analysis can capture more information that is diluted by linear methods.

Functional data analysis assumes that the function varies smoothly along the functional domain (i.e., across trial timepoints). It is a type of non-linear modeling technique over the functional domain since we do not assume a linear model (straight line). Therefore, our functional data analysis approach is able to capture more information that is diluted by linear models. While the basic form of our model assumes a linear change in the signal at a fixed trial timepoint, across trials/sessions, our package allows one to easily model changes with non-linear functions of covariates using splines or other basis functions. One must consider, however, the tradeoff between flexibility and interpretability when specifying potentially complex models.

Reviewer 2

Strengths:

The open-source package in R using a similar syntax as the lme4 package for the implementation of this framework on photometry data enhances the accessibility, and usage by other researchers. Moreover, the decreased fitting time of the model in comparison with a similar package on simulated data, has the potential to be more easily adopted.

The reanalysis of two studies using summary statistics on photometry data (Jeong et al., 2022; Coddington et al., 2023) highlights how trial-by-trial analysis at each time-point on the trial can reveal information obscured by averaging across trials. Furthermore, this work also exemplifies how session and subject variability can lead to opposite conclusions when not considered.

Thank you for the positive assessment of our work!

Weaknesses:

Although this work has reanalyzed previous work that used summary statistics, it does not compare with other studies that use trial-by-trial photometry data across time-points in a trial.

As described by the authors, fitting pointwise linear mixed models and performing t-test and Benjamini-Hochberg correction as performed in Lee et al. (2019) has some caveats. Using joint confidence intervals has the potential to improve statistical robustness, however, this is not directly shown with temporal data in this work. Furthermore, it is unclear how FLMM differs from the pointwise linear mixed modeling used in this work.

We agree with the reviewers that providing more detail about the drawbacks of the approach applied in Lee et al., 2019 will strengthen the paper. We will add an example analysis applying the method proposed by Lee et al., 2019 to show how the set of timepoints at which coefficient estimates reach statistical significance can vary dramatically depending on the sampling rate one subsamples their data at, a highly undesirable property of this strategy. Our approach is robust to this, and still provides a multiple comparisons correction through the joint confidence intervals.

In this work, FLMM usages included only one or two covariates. However, in complex behavioral experiments, where variables are correlated, more than two may be needed (see Simpson et al. (2023), Engelhard et al. (2019); Blanco-Pozo et al. (2024)). It is not clear from this work, how feasible computationally would be to fit such complex models, which would also include more complex random effects.

This is a good point. In our experience, the code is still quite fast (often taking seconds to tens of seconds in our experience) on a standard laptop when fitting complex models that include, for example, 10 covariates, or complex random effect specifications on dataset sizes common in fiber photometry. In the manuscript, we included results from simpler models with few covariates in an attempt to show results from the FLMM versions of the standard analyses (e.g., correlations, t-tests) applied in Jeong et al., 2022. Our goal was to show that our method reveals effects obscured by standard analyses even in simple cases. Some of our models did, however, include complex nested random effects (e.g., the models described in Section 4.5.2).

Like other mixed-model based analyses, our method becomes slower when the number of observations in the dataset is on the order of tens of thousands of observations. However, we coded the methods to be memory efficient so that even these larger analyses can be run on standard laptops. We thank the reviewer for this point, as we worked extremely hard to scale the method to be able to efficiently fit models commonly applied in neuroscience. Indeed, challenges with scalability were one of the main motivations for applying the estimation procedure that we did; in the appendix we show that the fit time of our approach is much faster than existing FLMM software such as the refund package function `pffr()`, especially for large sample sizes. While `pffr()` appears to scale exponentially with the number of clusters (e.g., animals), our method appears to scale linearly. We will more explicitly emphasize the scalability in the revision, since we agree this will strengthen the final manuscript.

Reviewer #3

Strengths:

The statistical framework described provides a powerful way to analyze photometry data and potentially other similar signals. The provided package makes this methodology easy to implement and the extensively worked examples of reanalysis provide a useful guide to others on how to correctly specify models.

Modeling the entire trial (function regression) removes the need to choose appropriate summary statistics, removing the opportunity to introduce bias, for example in searching for optimal windows in which to calculate the AUC. This is demonstrated in the re-analysis of Jeong et al., 2022, in which the AUC measures presented masked important details about how the photometry signal was changing.

Meanwhile, using linear mixed methods allows for the estimation of random effects, which are an important consideration given the repeated-measures design of most photometry studies.

Thank you for the positive assessment of our work!

Weaknesses:

While the availability of the software package (fastFMM), the provided code, and worked examples used in the paper are undoubtedly helpful to those wanting to use these methods, some concepts could be explained more thoroughly for a general neuroscience audience.

We appreciate this and, to address your and other reviewers' comments, we are creating a series of vignettes walking users through how to analyze photometry data with our package. We will include algebraic illustrations to help users gain familiarity with the regression modeling here.

While the methodology is sound and the discussion of its benefits is good, the interpretation and discussion of the re-analyzed results are poor:

In section 2.3, the authors use FLMM to identify an instance of Simpson's Paradox in the analysis of Jeong et al. (2022). While this phenomenon is evident in the original authors' metrics (replotted in Figure 5A), FLMM provides a convenient method to identify these effects while illustrating the deficiencies of the original authors' approach of concatenating a different number of sessions for each animal and ignoring potential within-session effects. The discussion of this result is muddled. Having identified the paradox, there is some appropriate speculation as to what is causing these opposing effects, particularly the decrease in sessions. In the discussion and appendices, the authors identify (1) changes in satiation/habitation/motivation, (2) the predictability of the rewards (presumably by the click of a solenoid valve) and (3) photobleaching as potential explanations of the decrease within days. Having identified these effects, but without strong evidence to rule all three out, the discussion of whether RPE or ANCCR matches these results is probably moot. In particular, the hypotheses developed by Jeong et al., were for a random (unpredictable) rewards experiment, whereas the evidence points to the rewards being sometimes predictable. The learning of that predictability (e.g. over sessions) and variation in predictability (e.g. by attention level to sounds of each mouse) significantly complicate the analysis. The FLMM analysis reveals the complexity of analyzing what is apparently a straightforward task design.

While we are disappointed to hear the reviewer felt our initial interpretations and discussion were poor, the reviewer brings up an excellent point that we had not considered. They have convinced us that acknowledging and elaborating on this alternative perspective will strengthen the paper. We agree that the ANCCR/RPE model predictions were made for unpredictable rewards and, as the reviewer rightly points out, there is evidence that the animals sense the reward delivery. Regardless of the learning theory one adopts (RPE, ANCCR or others), we agree that this (potentially) learned predictability alone could account for the increase in signal magnitude across sessions.

After reading the reviewer's comments, we consulted with a number of researchers in this area, and several felt that a CS+ can serve as a reward, within itself. From this perspective, the rewards in the Jeong et al., 2022 experiment might still be considered unexpected. After discussing extensively with the authors of Jeong et al., 2022, it is clear that they went to enormous trouble to prevent the inadvertent generation of a CS+, and it is likely changes in pressure from the solenoid (rather than a sound) that served as a cue. This underscores the difficulty of preventing perception of reward delivery in practice. As this paper is focused on analysis approaches, we feel that we can contribute most thoughtfully to the dopamine-learning theory conversation by presenting both sides.

Overall, we agree with the reviewer that future experiments will be needed for testing the accuracy of the models' predictions for random (unpredicted) rewards. While we understand that our attempt to document our conversations with the Jeong et al., 2022 authors may have room for improvement, we hope the reviewer can appreciate that this was done with the best of intentions. We wish to emphasize that we also consulted with several other researchers in the field when crafting the discussion. The Jeong et al., 2022 authors could easily have avoided acknowledging the potential incompleteness of their theory, by claiming that our results do not invalidate their predictions for a random reward, as the reward was not

unpredicted in the experiment (as a result of the inadvertent solenoid CS+). Instead, they went out of their way to emphasize that their experiment did test a random reward, and that our results do present problems for their theory. We think that engagement with re-analyses of one's data, even when findings are inconvenient, is a good demonstration of open science practice. For that reason as well, we feel providing readers with a perspective on the entire discussion will contribute to the scientific discourse in this area.

Finally, we would like to reiterate that this conversation is happening because our method, by analyzing the signal at every trial timepoint, revealed a neural signal that appears to indicate that the animals sense reward delivery. Ultimately, this was what we set out to do: help researchers ask questions of their data that they could not ask before. We believe that having a demonstration that we can indeed do this for a “live” issue is the most appropriate way of demonstrating the usefulness of the method.

It is clear the reviewer put a lot of time into understanding what we did, and was very thoughtful about the feedback. We would like to thank the reviewer again for taking such care in reviewing our paper.

If this paper is not trying to arbitrate between RPE and ANCCR, as stated in the text, the post hoc reasoning of the authors of Jeong et al 2022 provided in the discussion is not germane.

While we appreciate that the post hoc reasoning of the authors of Jeong et al., 2022 may not seem germane, we would like to provide some context for its inclusion. As statisticians and computer scientists, our role is to create methods, and this often requires using open source data and recreating past analyses. This usually involves extensive conversation with authors about their data and analysis choices because, if we cannot reproduce their findings using their analysis methods, we cannot verify that results from our own methods are valid. As such, we prefer to conduct method development in a collaborative fashion, and we strive to constructively, and respectfully, discuss our results with the original authors. We feel that giving them the opportunity to suggest analyses, and express their point of view if our results conflict with their original conclusions, is important, and we do not want to discourage authors from making their datasets public. As such, we conducted numerous analyses at the suggestion of Jeong et al., 2022 and discussed the results over the course of many months. Indeed the analyses in the Appendix that the reviewer is referring to were conducted at the suggestion of the authors of Jeong et al., 2022, in an attempt to rule out alternative explanations. We nevertheless appreciate that our interpretations of these results can include some of the caveats suggested by the reviewer, and we will strive to improve these sections.

Arbitrating between the models likely requires new experimental designs (removing the sound of the solenoid, satiety controls) or more complex models (e.g. with session effects, measures of predictability) that address the identified issues.

We agree with the reviewer that the results suggest that new experimental designs will likely be necessary to adjudicate between models. It is our hope that, by weighing the different issues and interpretations, our paper might provide useful suggestions into what experimental designs would be most beneficial to rule out competing hypotheses in future data collection efforts. We believe that our methodology will strengthen our capacity to design new experiments and analyses. We will make the reviewer's suggestions more explicit in the discussion by emphasizing the limitations of the original data.

Of the three potential causes of within-session decreases, the photobleaching arguments advanced in the discussion and expanded greatly in the appendices are not convincing. The data being modeled is a processed signal ($\Delta F/F$) with smoothing and baseline correction and this does not seem to have been considered in the argument.

We are disappointed to hear that this extensive set of analyses, much of which was conducted at the suggestion of Jeong et al., 2022, was not convincing. We agree that acknowledging any pre-processing would provide useful context for the reader. We do wish to clarify that we analyzed the data that were made available online (raw data was not available). Moreover, for comparison with the authors' results, we felt it was important to maintain the same pre-processing steps as they did. These conditions were held constant across analysis approaches; therefore, we think that the changes within-trial are likely not influenced substantially by these pre-processing choices. While we cannot speak definitively to the impact any of the processing conducted by the authors had on the results, we believe that it was likely minor, given that the timing of signals at other points in the trial, and in other experiments, were as expected (e.g., the signal rose rapidly after cue onset in Pavlovian tasks).

Furthermore, the photometry readout is also a convolution of the actual concentration changes over time, influenced by the on-off kinetics of the sensor, which makes the interpretation of timing effects of photobleaching less obvious than presented here and more complex than the dyes considered in the cited reference used as a foundation for this line of reasoning.

We appreciate the nuance of this point, and we will add it to our discussion. In response to your criticism, we have consulted with more experts in the field regarding the potential for bleaching in this data, and it is not clear to us why photobleaching would be visible in one time-window of a trial, but not at another (less than a second away), despite high dF/F magnitudes in both time-windows. We do wish to point out that, at the request of the authors, we analyzed many experiments from the same animals and in most cases did not observe other indications of photobleaching. Hence, it is not clear to us why this particular set of experiments would garner additional skepticism regarding the potential for photobleaching to invalidate results. While the role of photobleaching may be more complicated with this sensor than others in the references, that citation was included, at the suggestion of Jeong et al., 2022 simply as a way of acknowledging that non-linearities in photobleaching can occur.

Within this discussion of photobleaching, the characterization of the background reward experiments used in part to consider photobleaching (appendix 7.3.2) is incorrect. In this experiment (Jeong et al., 2022), background rewards were only delivered in the inter-trial-interval (i.e. not between the CS+ and predicted reward as stated in the text). Both in the authors' description and in the data, there is a 6s before cue onset where rewards are not delivered and while not described in the text, the data suggests there is a period after a predicted reward when background rewards are not delivered. This complicates the comparison of this data to the random reward experiment.

Thank you for pointing this out!! We will remove the parenthetical on page 18 of the appendix that incorrectly stated that rewards can occur between the CS+ and the predicted reward.

The discussion of the lack of evidence for backpropagation, taken as evidence for ANCCR over RPE, is also weak.

This point was meant to acknowledge that, although our method yields results that conflict with the conclusions described by Jeong et al., 2022 on data from some experiments, on other experiments our method supports their results. Again, we believe that a critical part of open science is acknowledging both areas where analyses support and conflict with those of the original authors. We agree with the reviewer that qualifying our results so as not to emphasize support for/against RPE/ANCCR will strengthen our paper, and we will make these changes.

A more useful exercise than comparing FLMM to the methods and data of Jeong et al., 2022, would be to compare against the approach of Amo et al., 2022, which identifies backpropagation (data publicly available: DOI: 10.5061/dryad.hhmgqnkjw). The replication of a positive result would be more convincing of the sensitivity of the methodology than the replication of a negative result, which could be a result of many factors in the experimental design. Given that the Amo et al. analysis relies on identifying systematic changes in the timing of a signal over time, this would be particularly useful in understanding if the smoothing steps in FLMM obscure such changes.

Thank you for this suggestion, and we agree this could be a useful analysis for the field. Your thoughtful review has convinced us that focusing on our statistical contribution will strengthen the paper, and we will make changes to further emphasize that we are not seeking to adjudicate between RPE/ANCCR. We only had space in the manuscript to include a subset of the analyses conducted on Jeong et al., 2022, and had to relegate the results from the Coddington et al., data to an appendix. Realistically, it would be hard for us to justify analyzing a third dataset. As you may surmise from the one we presented, reanalyzing a new dataset is usually very time consuming, and invariably requires extensive communication with the original authors. We did include numerous examples in our manuscript where we already replicated positive results, in a way that we believe demonstrates the sensitivity of the methodology. We have also been working with five groups at NIH and elsewhere using our approach, in experiments targeting different scientific questions. In fact, one paper that extensively applies our method and compares the results from those yielded by standard analysis of AUCs is already accepted and in press. Hence there should soon be additional demonstrations of what the method can do in less controversial settings. Finally, our forthcoming vignettes include additional analyses, not included in the manuscript, that replicate positive results. We take your point that our description of the data supporting one theory or the other should be qualified, and we will correct that. Again, your review was very thorough, and we appreciate your taking so much time to help us improve our work.

Reviewer #2 (Recommendations For The Authors):

First, I would like to commend the authors for the clarity of the paper, and for creating an open-source package that will help researchers more easily adopt this type of analysis.

Thank you!

I would suggest the authors consider adding to the manuscript, either some evidence or some intuition on how feasible would be to use FLMM for very complex model specifications, in terms of computational cost and model convergence.

This is an excellent point and we will make this suggested change in the Methods and Discussion section in the next draft.

From my understanding, this package might potentially be useful not just for photometry data but also for two-photon recordings for example. If so, I would also suggest the authors add to the discussion this potential use.

We appreciate your thinking on this point, as it would definitely help expand use of the method. We included a brief point in the Discussion that this package would be useful for other techniques, but we will expand upon this.

Reviewer #3 (Recommendations For The Authors):

The authors should define 'function' in context, as well as provide greater detail of the alternate tests that FLMM is compared to in Figure 7. Given the novelty of estimating joint CIs, the authors should be clearer about how this should be reported and how this differs from pointwise CIs (and how this has been done in the past).

Thank you, this is a very good point and will be critical for helping analysts describe and interpret results. We will add more detail to the Methods section on this point.

The authors identify that many photometry studies are complex nested longitudinal designs, using the cohort of 8 animals used in five task designs of Jeong et al. 2022 as an example. The authors miss the opportunity to illustrate how FLMM might be useful in identifying the effects of subject characteristics (e.g. sex, CS+ cue identity).

This is a great suggestion and we will add this important point to the discussion, especially in light of the factorial designs common in neuroscience experiments.

In discussing the delay-length change experiment, it would be more accurate to say that proposed versions of RPE and ANCCR do not predict the specific change.

We will make this change and agree this is a better phrasing.